Supplementary Materials

This document contains the supplementary figures and a numbered register of the supplementary tables accompanying the thesis. Complete tables are supplied as CSV files in the tables folder; the displayed extracts are provided for readability and verification.

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Figure S19. Within-condition semantic dispersion
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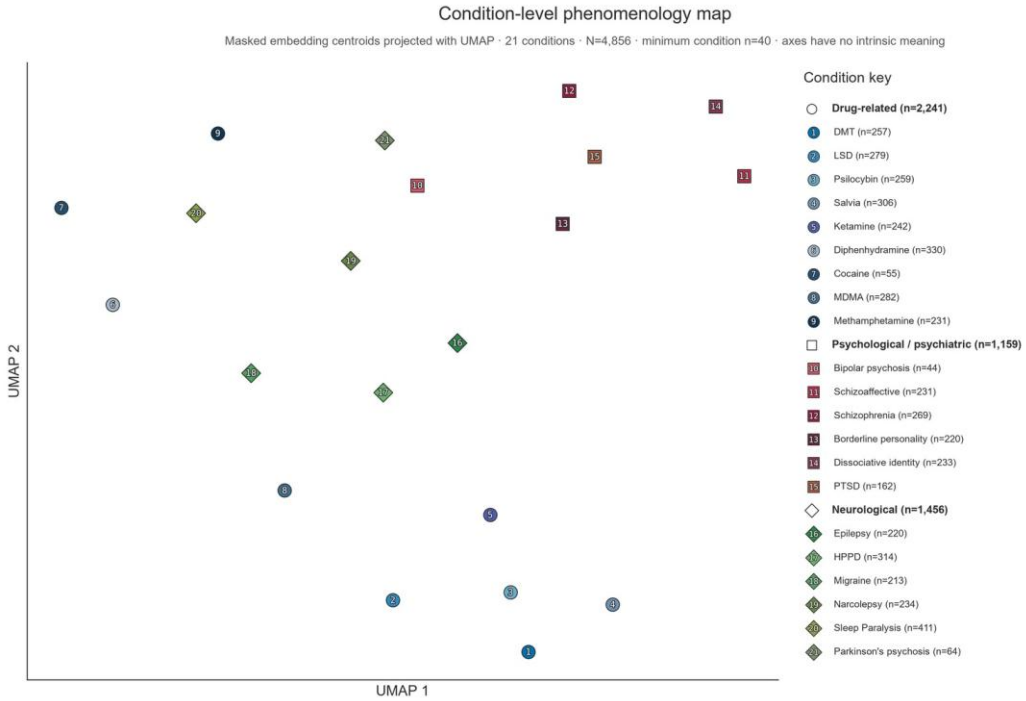
Supplementary tables

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Supplementary Figures

Figures S1-S13 report the principal supplementary analyses. Figure Series S14A-S14U and S15A-S15U contain the independent condition-specific NMF summaries and topic maps. Figure Series S16A-S16C provides category-restricted UMAPs, and Figures S17-S20 report category similarity, family similarity, semantic dispersion, and topic breadth.

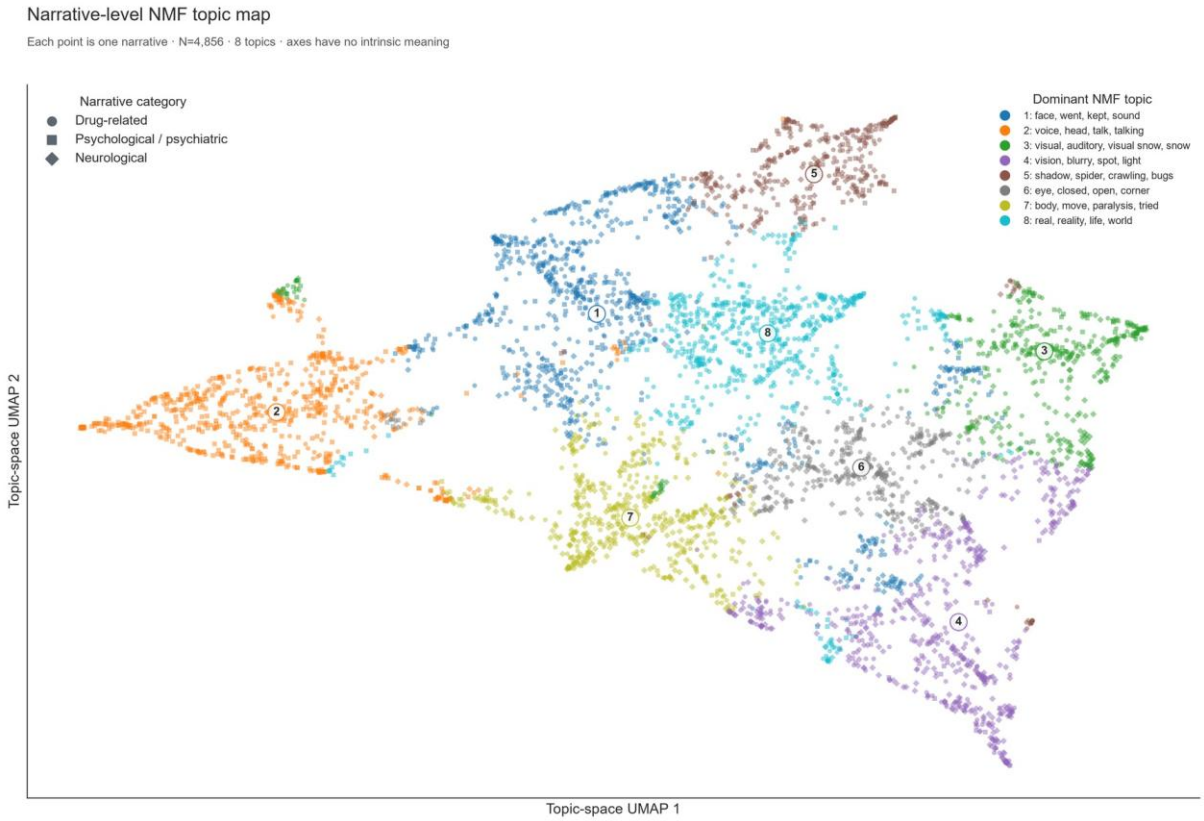
Supplementary Figure S1



Supplementary Figure S1. *Condition centroid UMAP*

Centroids are shown for the 21 conditions included in condition-level comparisons. Marker shapes denote broad category; axes have no intrinsic meaning.

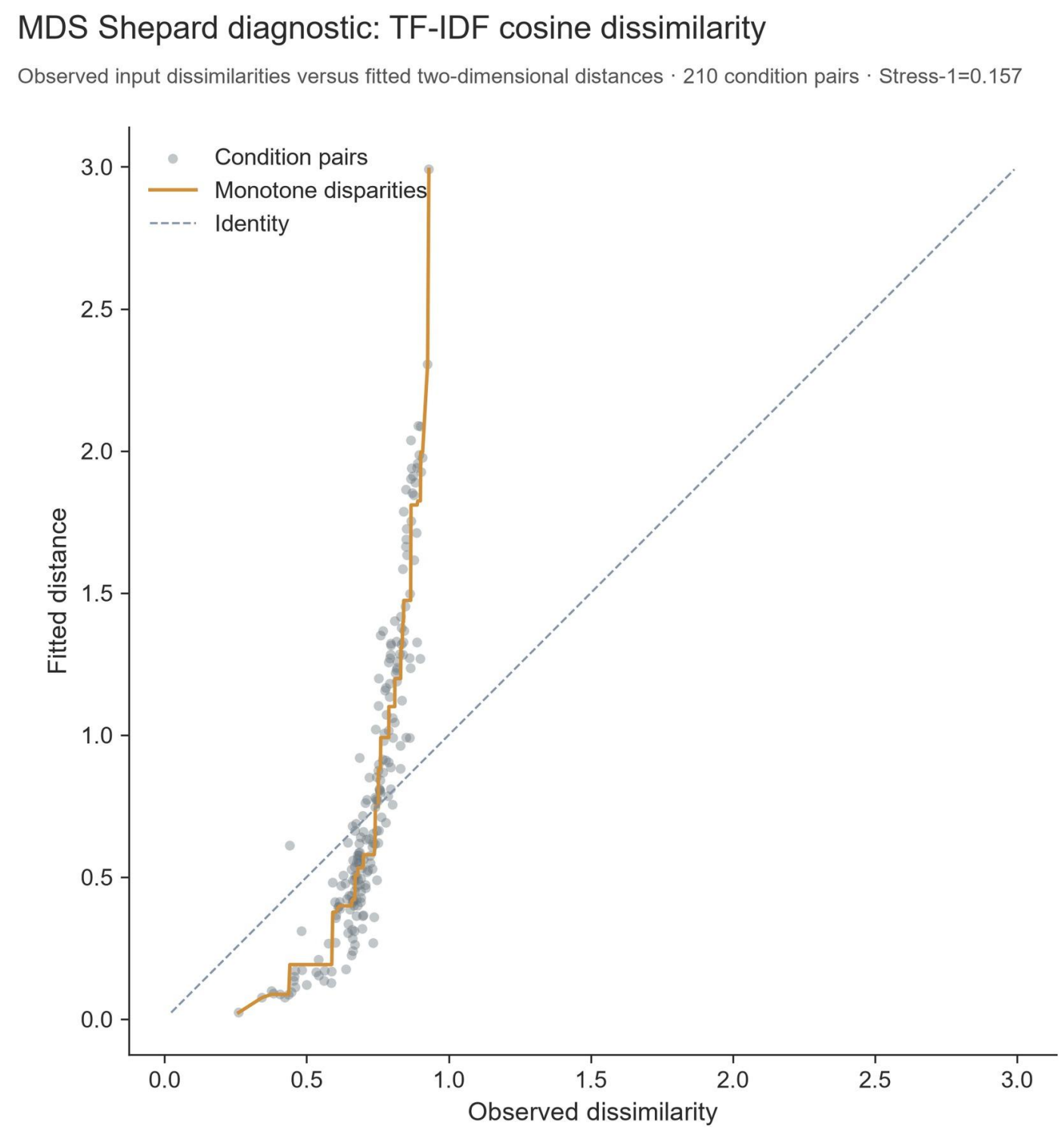
Supplementary Figure S2



Supplementary Figure S2. *Narrative-level NMF topic map*

Each point represents one reconstructed narrative. Colour denotes the narrative's highest-weight NMF topic and shape denotes broad category; axes have no intrinsic meaning.

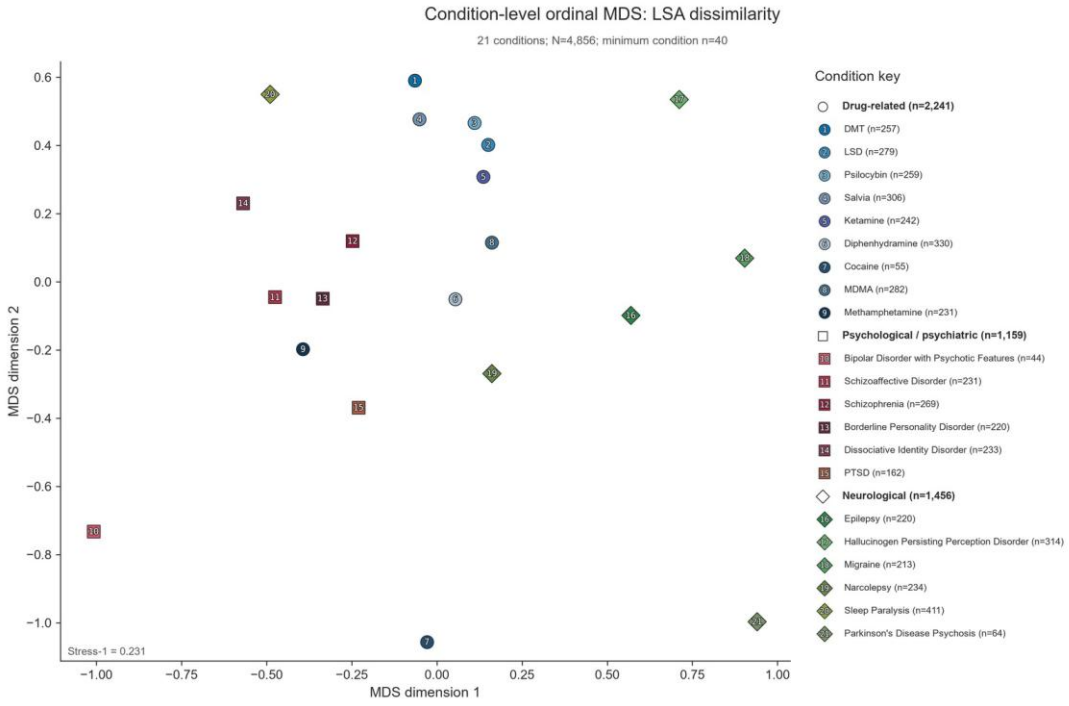
Supplementary Figure S3



Supplementary Figure S3. *TF-IDF MDS Shepard diagnostic*

Observed TF-IDF cosine dissimilarities are compared with fitted two-dimensional distances. The monotone disparity curve is the ordinal MDS fit.

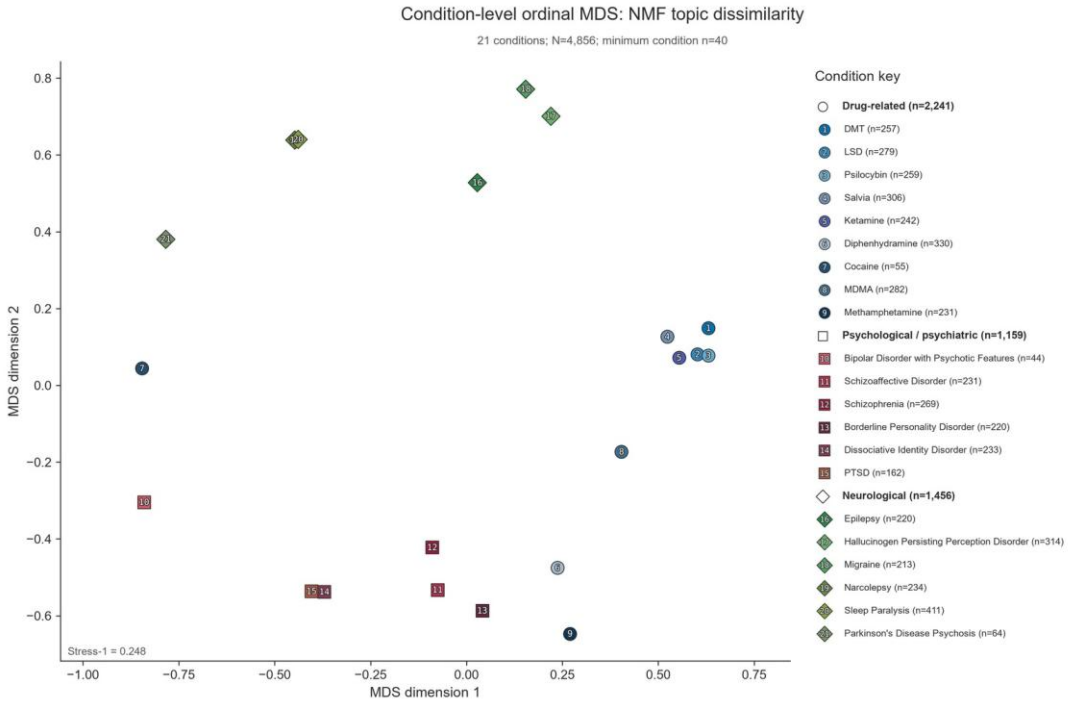
Supplementary Figure S4



Supplementary Figure S4. *LSA semantic MDS*

Ordinal MDS of condition-level latent semantic analysis dissimilarities. Marker numbers correspond to the condition key; axes have no intrinsic meaning.

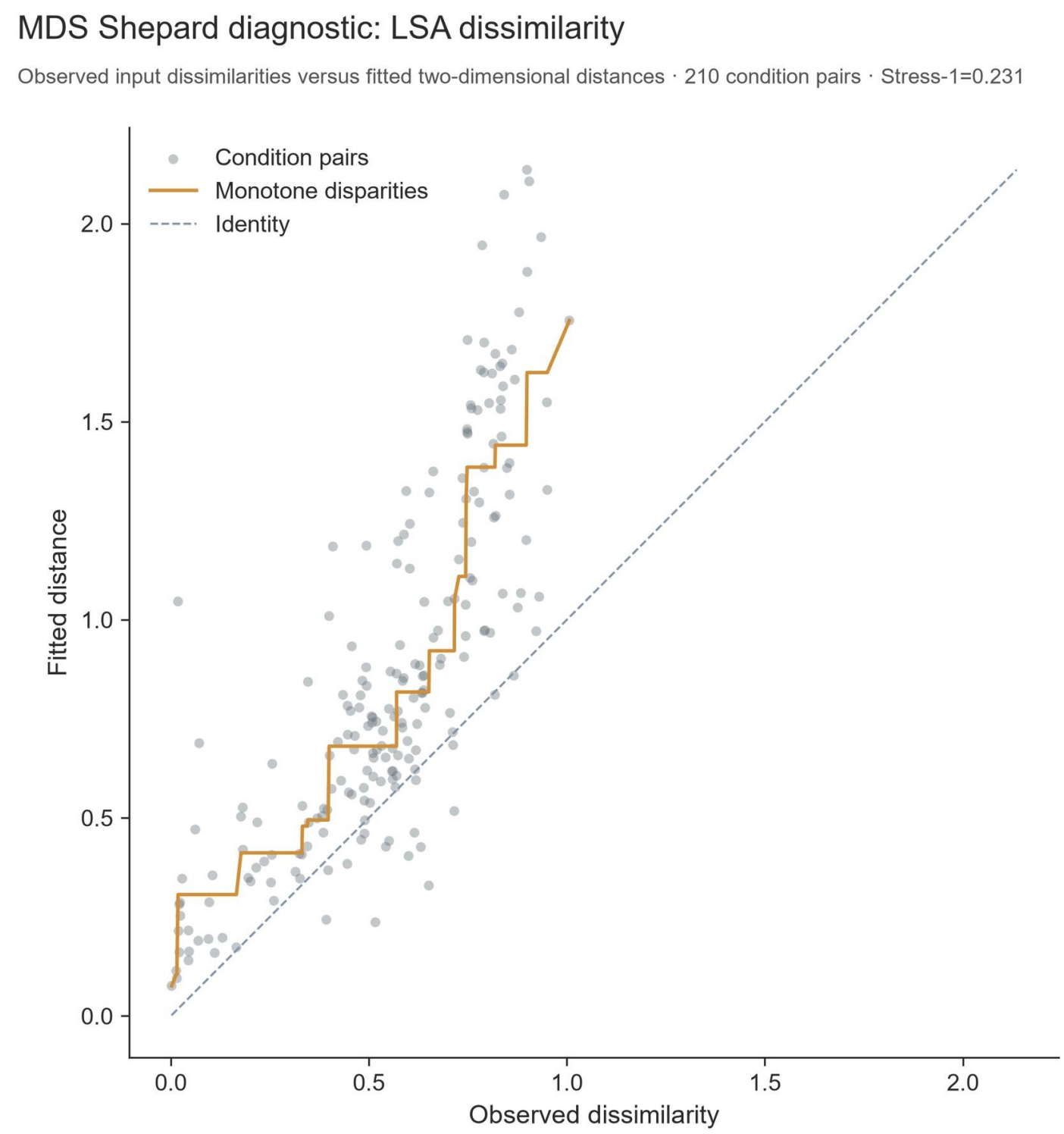
Supplementary Figure S5



Supplementary Figure S5. *NMF topic-profile MDS*

Ordinal MDS of condition-level NMF topic-profile dissimilarities. Marker numbers correspond to the condition key; axes have no intrinsic meaning.

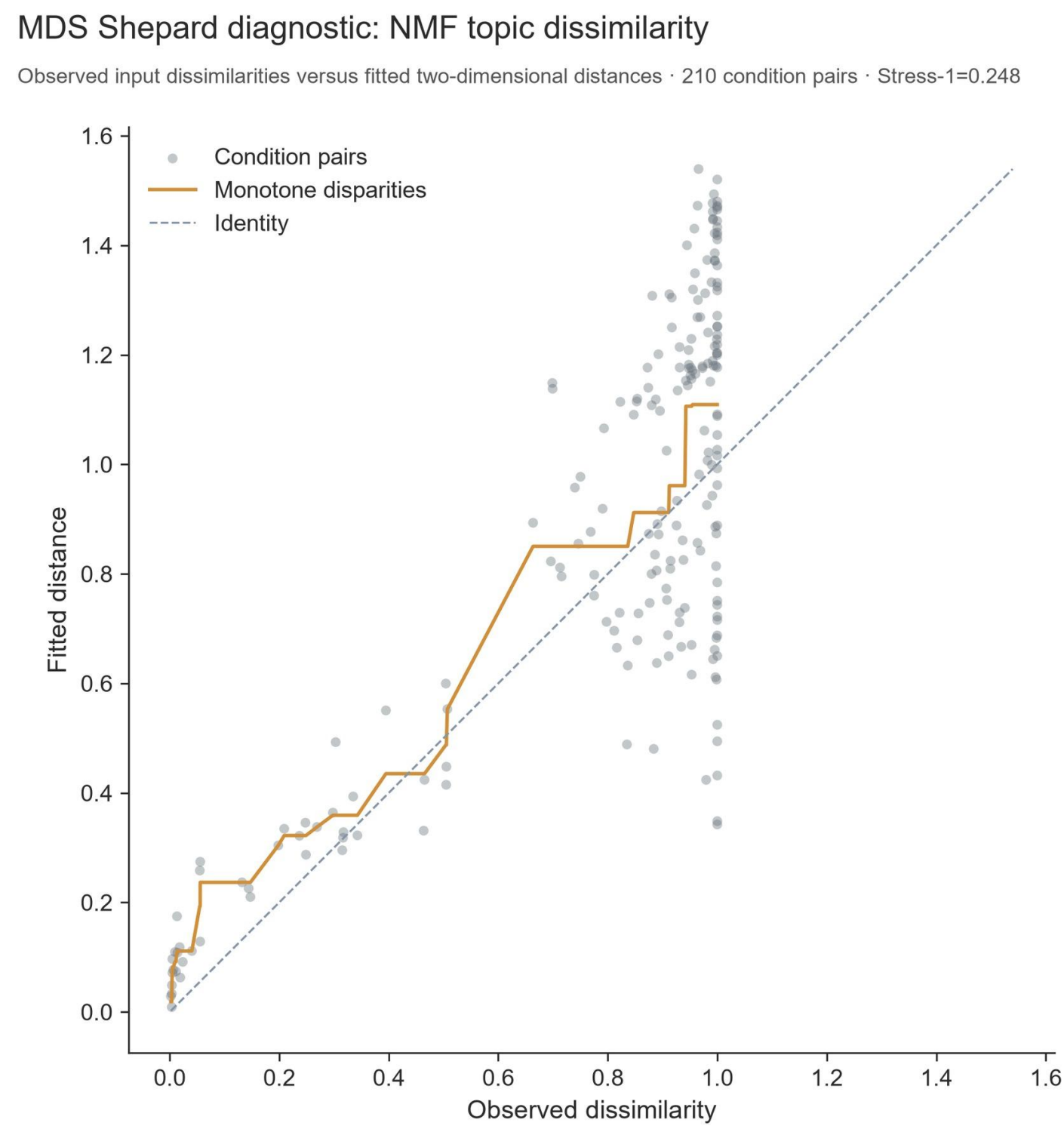
Supplementary Figure S6



Supplementary Figure S6. *LSA MDS Shepard diagnostic*

Observed LSA dissimilarities are compared with fitted two-dimensional distances. The monotone disparity curve is the ordinal MDS fit.

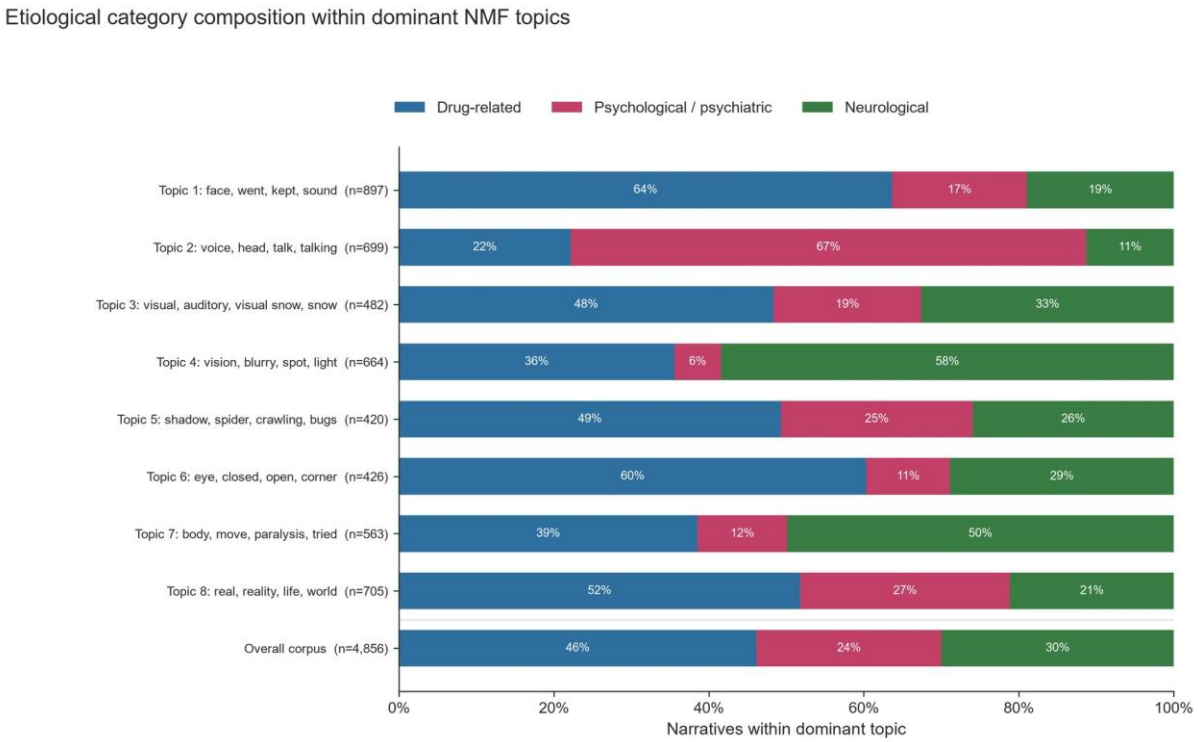
Supplementary Figure S7



Supplementary Figure S7. *NMF MDS Shepard diagnostic*

Observed NMF topic-profile dissimilarities are compared with fitted two-dimensional distances. The monotone disparity curve is the ordinal MDS fit.

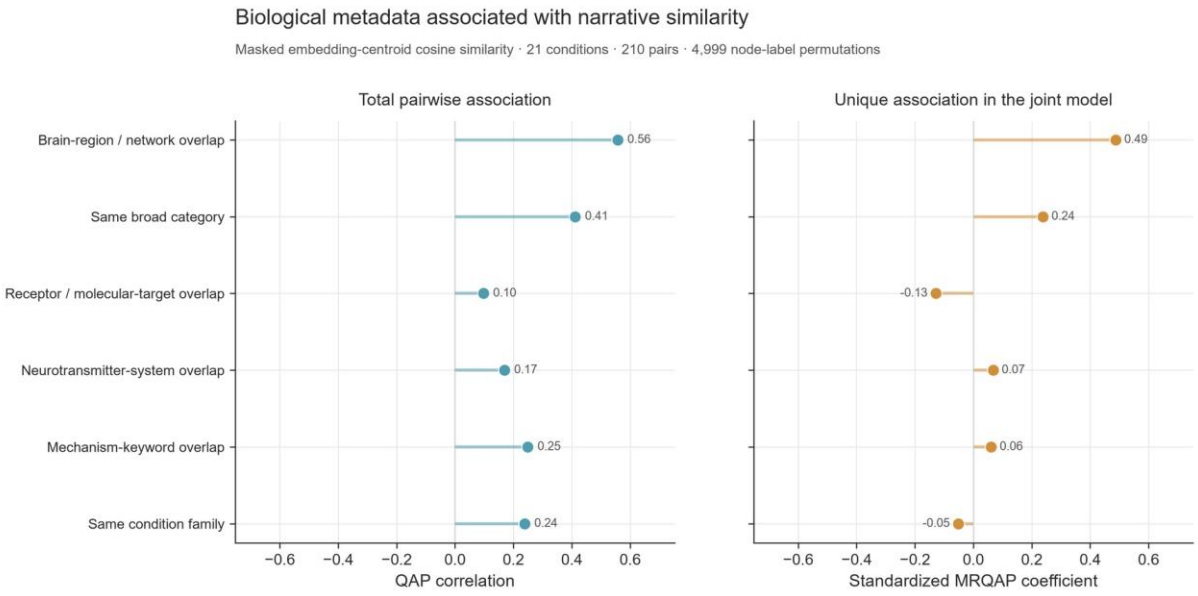
Supplementary Figure S8



Supplementary Figure S8. *NMF topic composition by broad category*

Bars show the broad-category composition of narratives assigned to each highest-weight NMF topic. Topic assignment is used for summary only; topic weights remain graded.

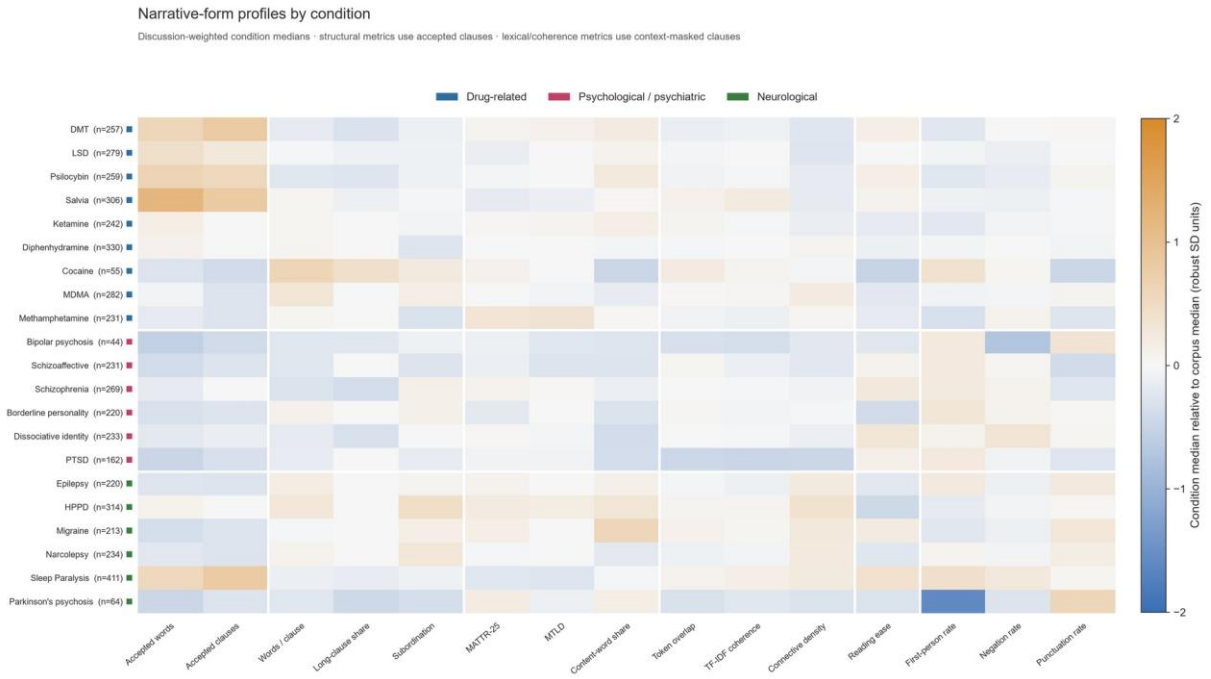
Supplementary Figure S9



Supplementary Figure S9. *Biological predictors of condition similarity*

Total pairwise associations and unique associations from the joint model are shown for the curated biological predictors. These exploratory results do not establish mechanisms.

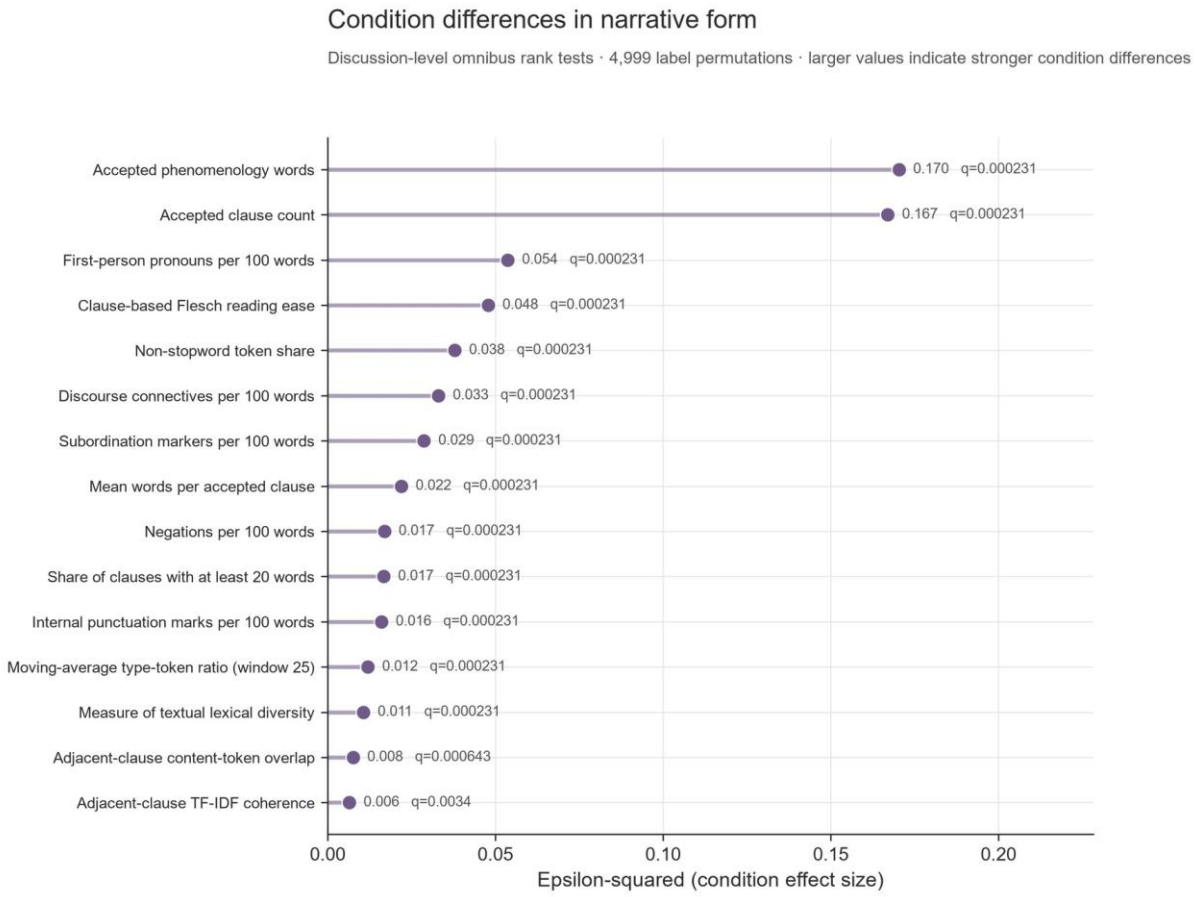
Supplementary Figure S10



Supplementary Figure S10. *Condition profiles of narrative form*

Discussion-weighted linguistic-form summaries are standardized within each measure to show relative condition profiles.

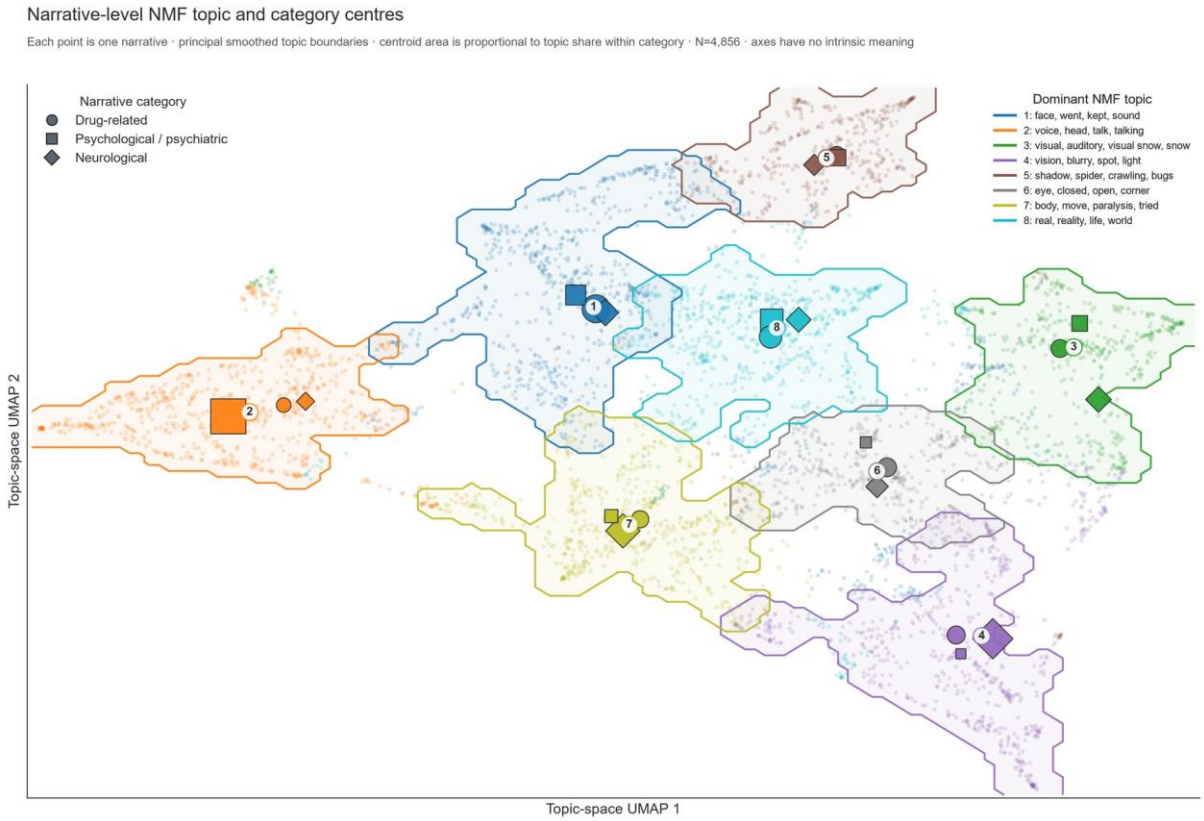
Supplementary Figure S11



Supplementary Figure S11. *Condition effects on narrative form*

Epsilon-squared values summarize the magnitude of condition differences in each narrative-form measure. Adjusted q values are printed beside the estimates.

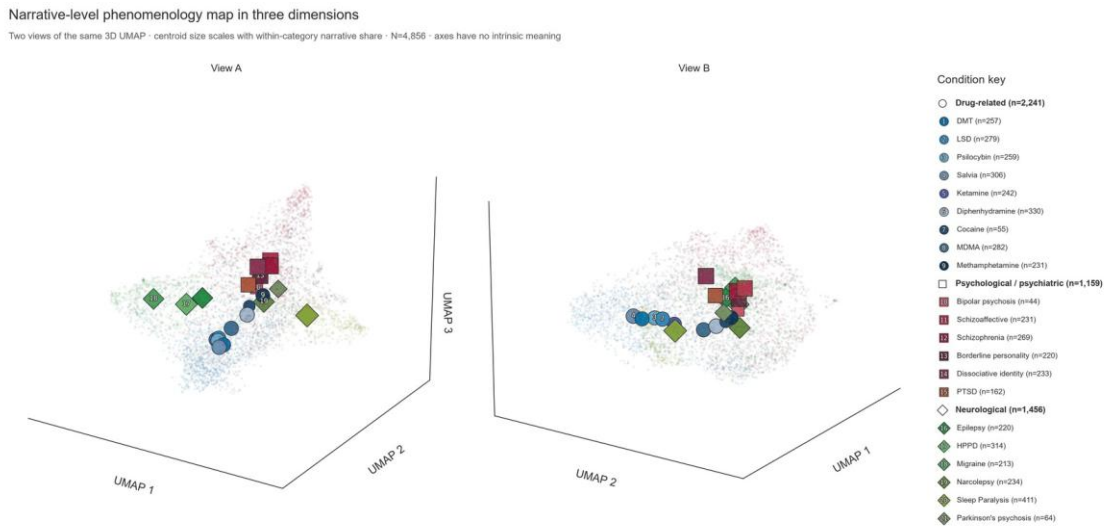
Supplementary Figure S12



Supplementary Figure S12. *Narrative NMF topic and category centres*

Topic boundaries summarize principal regions of the narrative topic map. Category centres are plotted within each topic and sized by within-category topic share.

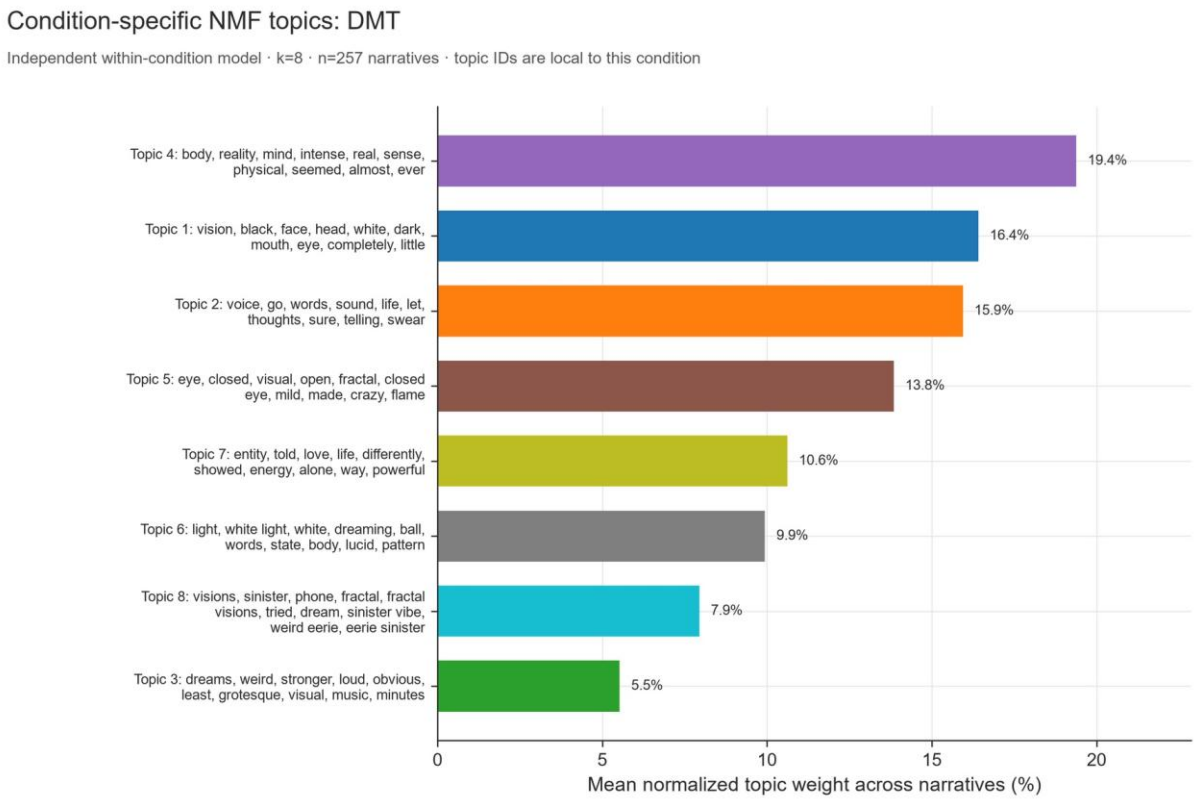
Supplementary Figure S13



Supplementary Figure S13. *Three-dimensional condition UMAP*

Static view of the three-dimensional narrative-level condition UMAP. The interactive HTML companion is included in the interactive folder; axes have no intrinsic meaning.

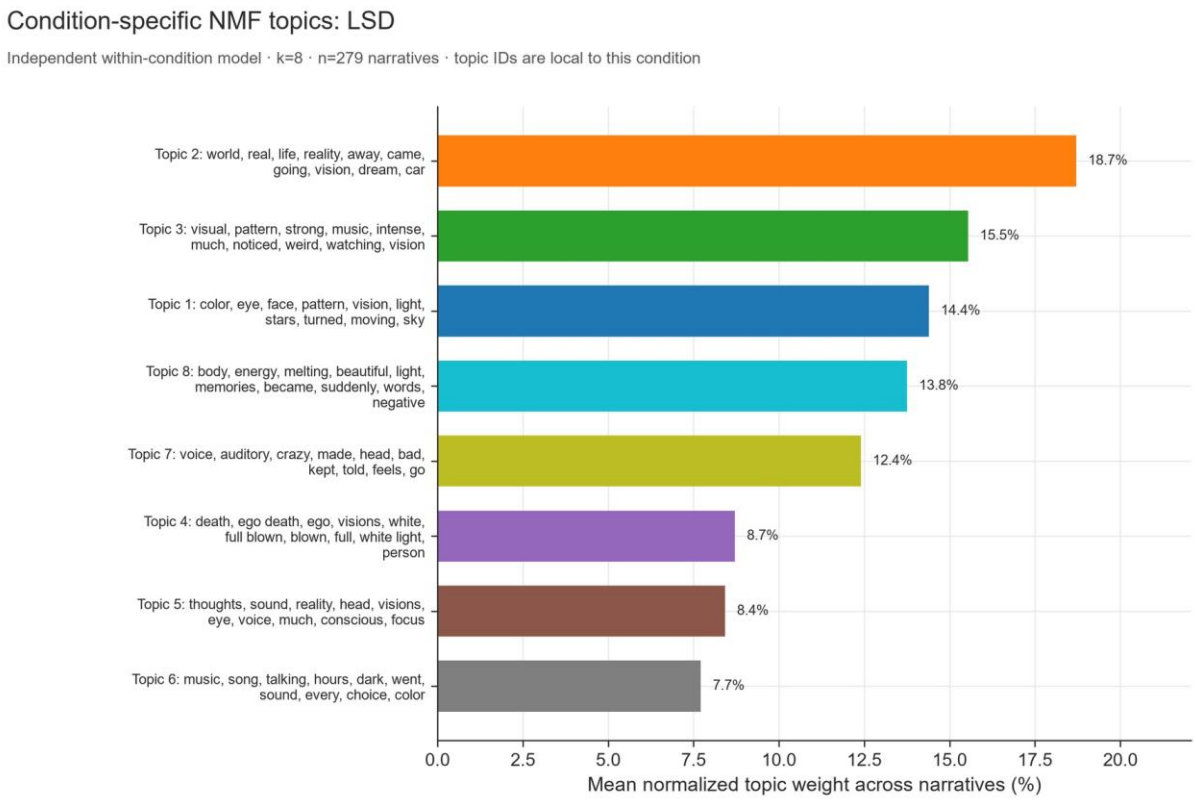
Supplementary Figure S14A



Supplementary Figure S14A. *Condition-specific NMF topic summary for DMT*

Eight-topic NMF model fitted independently within DMT. Topic numbers are local to this condition and are not directly comparable across conditions.

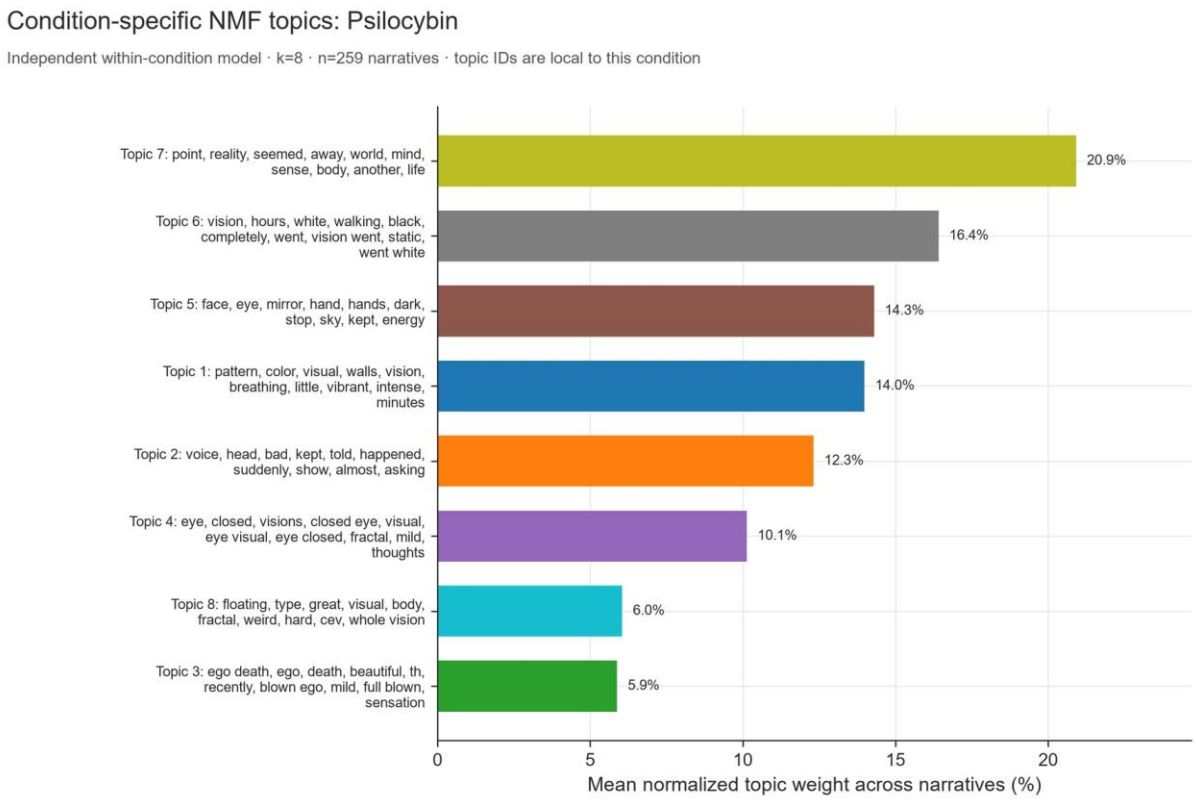
Supplementary Figure S14B



Supplementary Figure S14B. *Condition-specific NMF topic summary for LSD*

Eight-topic NMF model fitted independently within LSD. Topic numbers are local to this condition and are not directly comparable across conditions.

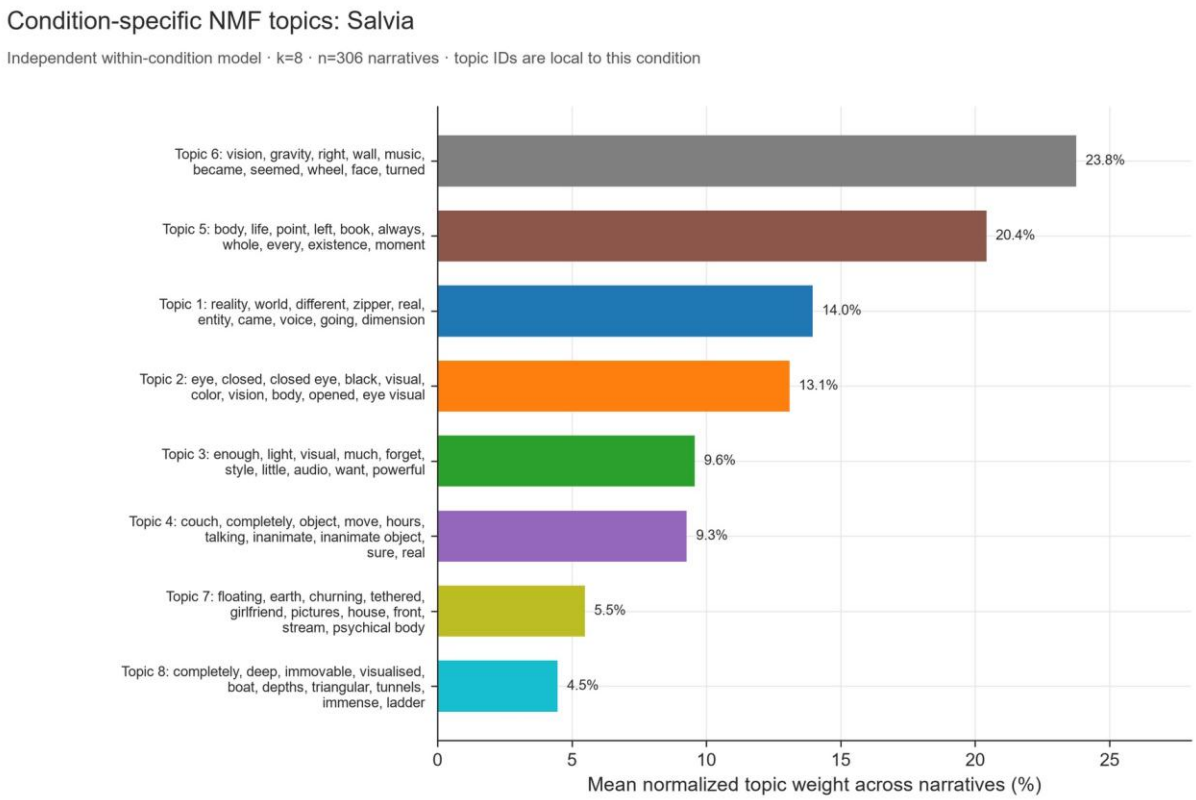
Supplementary Figure S14C



Supplementary Figure S14C. *Condition-specific NMF topic summary for Psilocybin*

Eight-topic NMF model fitted independently within Psilocybin. Topic numbers are local to this condition and are not directly comparable across conditions.

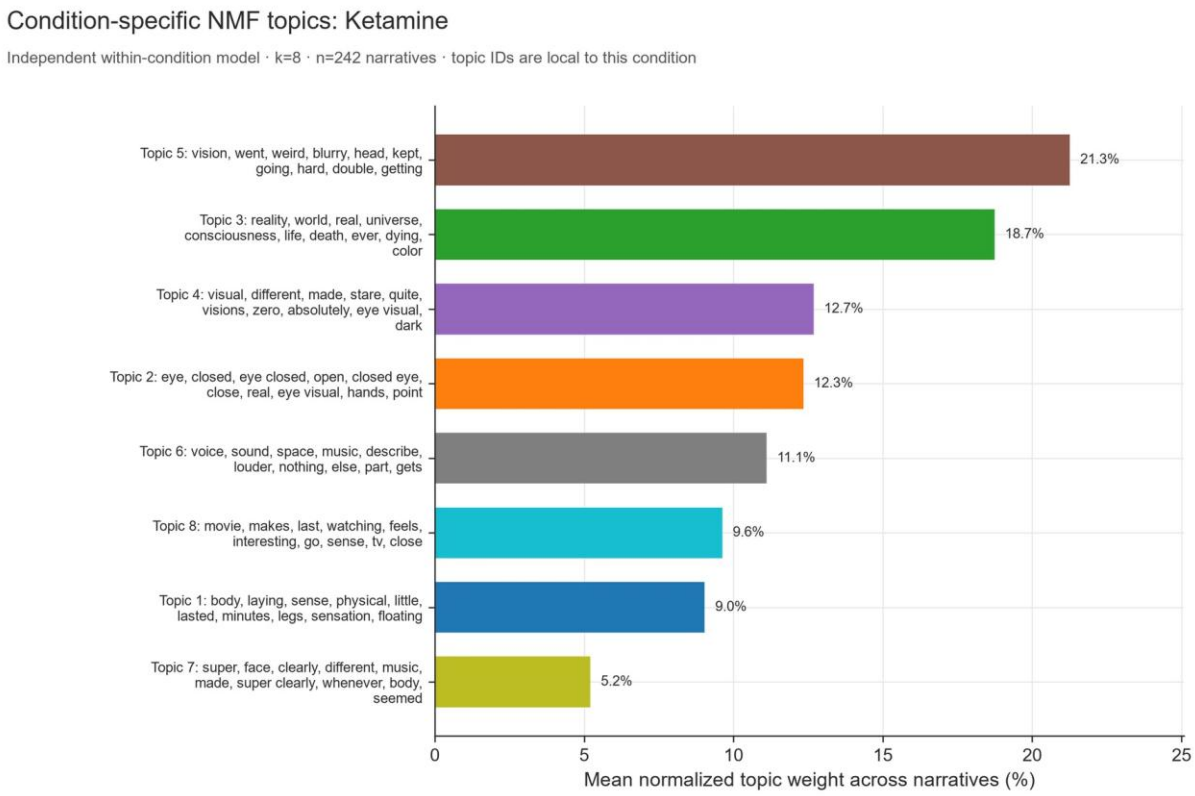
Supplementary Figure S14D



Supplementary Figure S14D. *Condition-specific NMF topic summary for Salvia*

Eight-topic NMF model fitted independently within Salvia. Topic numbers are local to this condition and are not directly comparable across conditions.

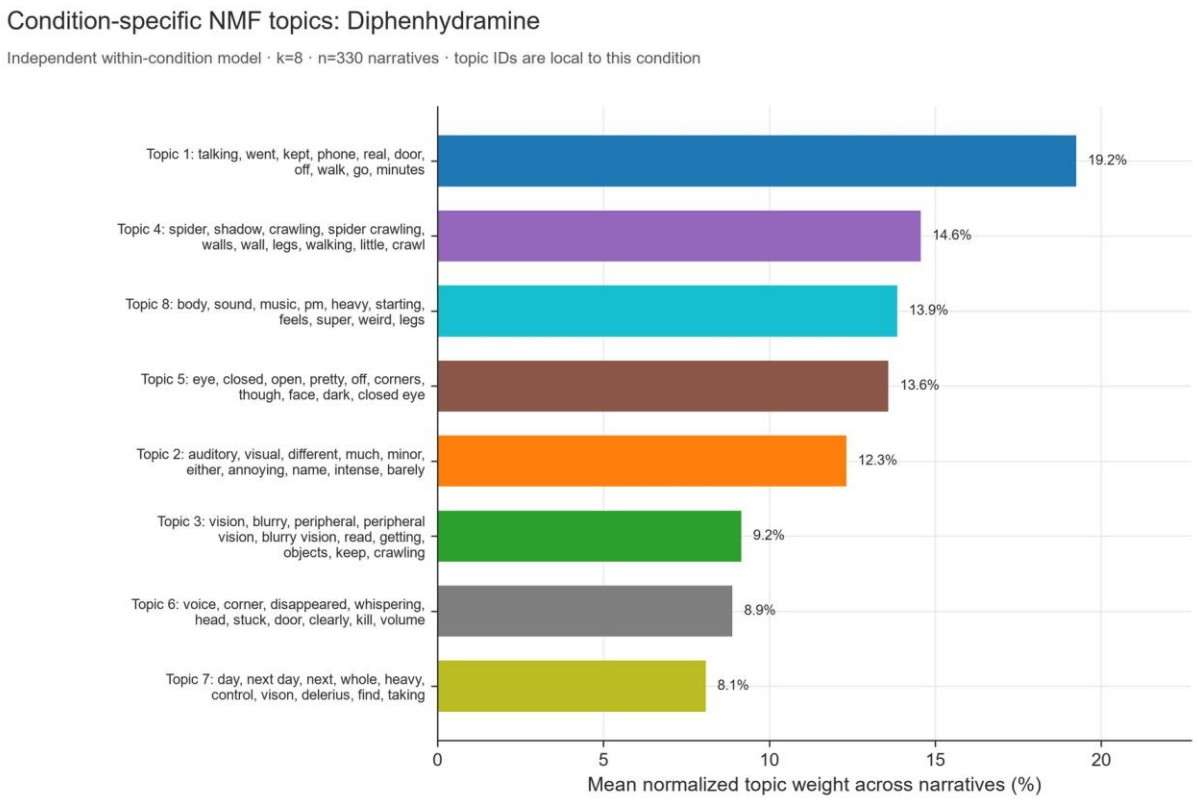
Supplementary Figure S14E



Supplementary Figure S14E. *Condition-specific NMF topic summary for Ketamine*

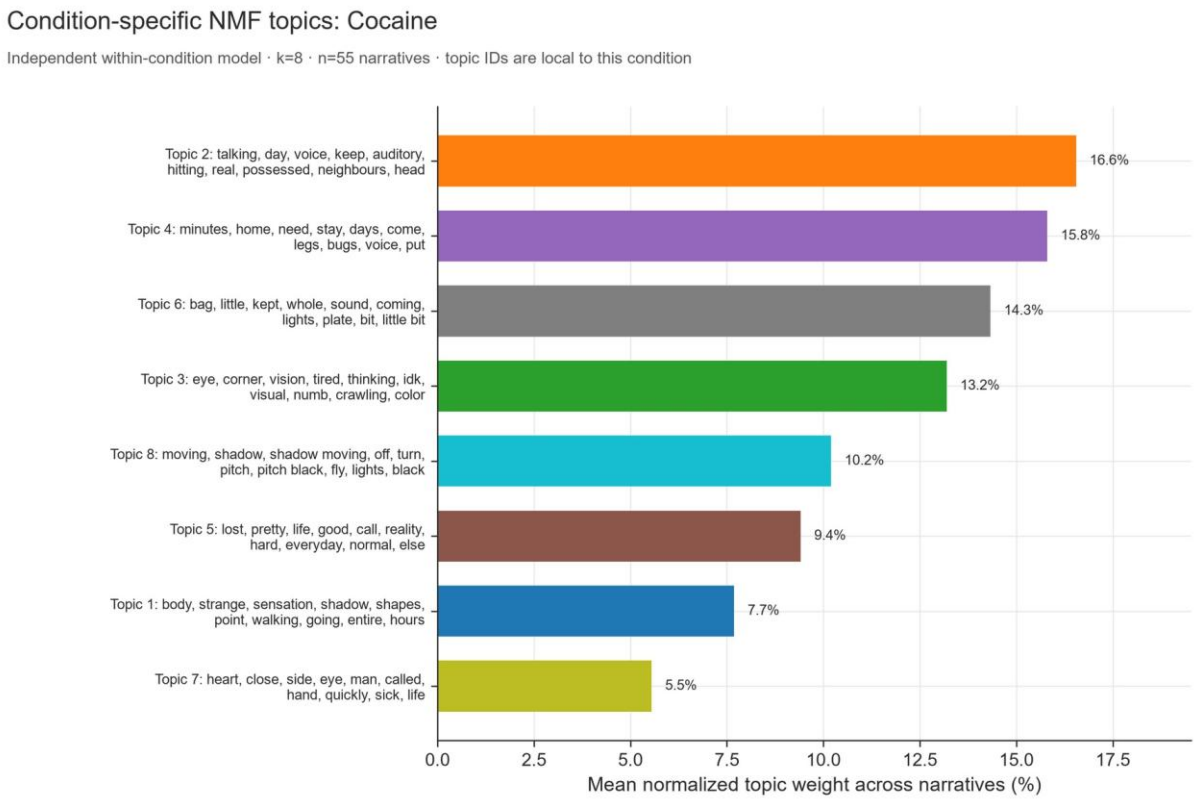
Eight-topic NMF model fitted independently within Ketamine. Topic numbers are local to this condition and are not directly comparable across conditions.

Supplementary Figure S14F



Supplementary Figure S14F. *Condition-specific NMF topic summary for Diphenhydramine*
Eight-topic NMF model fitted independently within Diphenhydramine. Topic numbers are local to this condition and are not directly comparable across conditions.

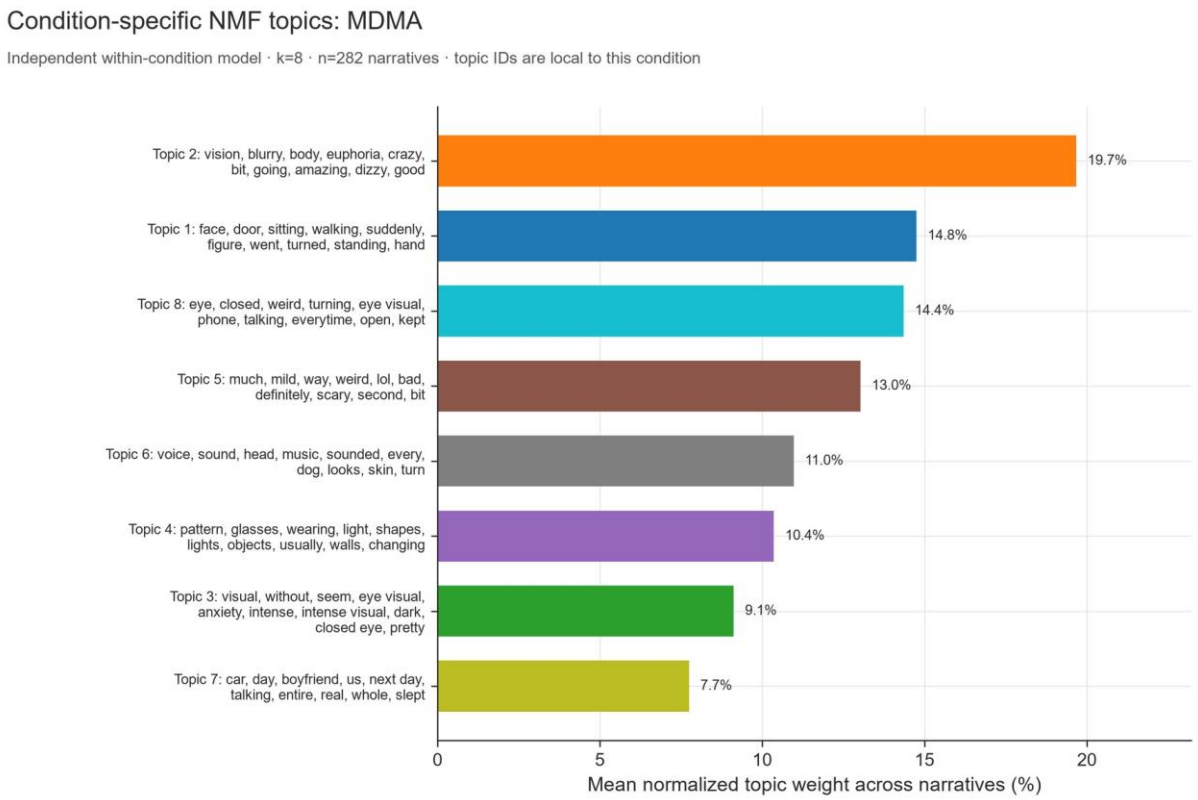
Supplementary Figure S14G



Supplementary Figure S14G. *Condition-specific NMF topic summary for Cocaine*

Eight-topic NMF model fitted independently within Cocaine. Topic numbers are local to this condition and are not directly comparable across conditions.

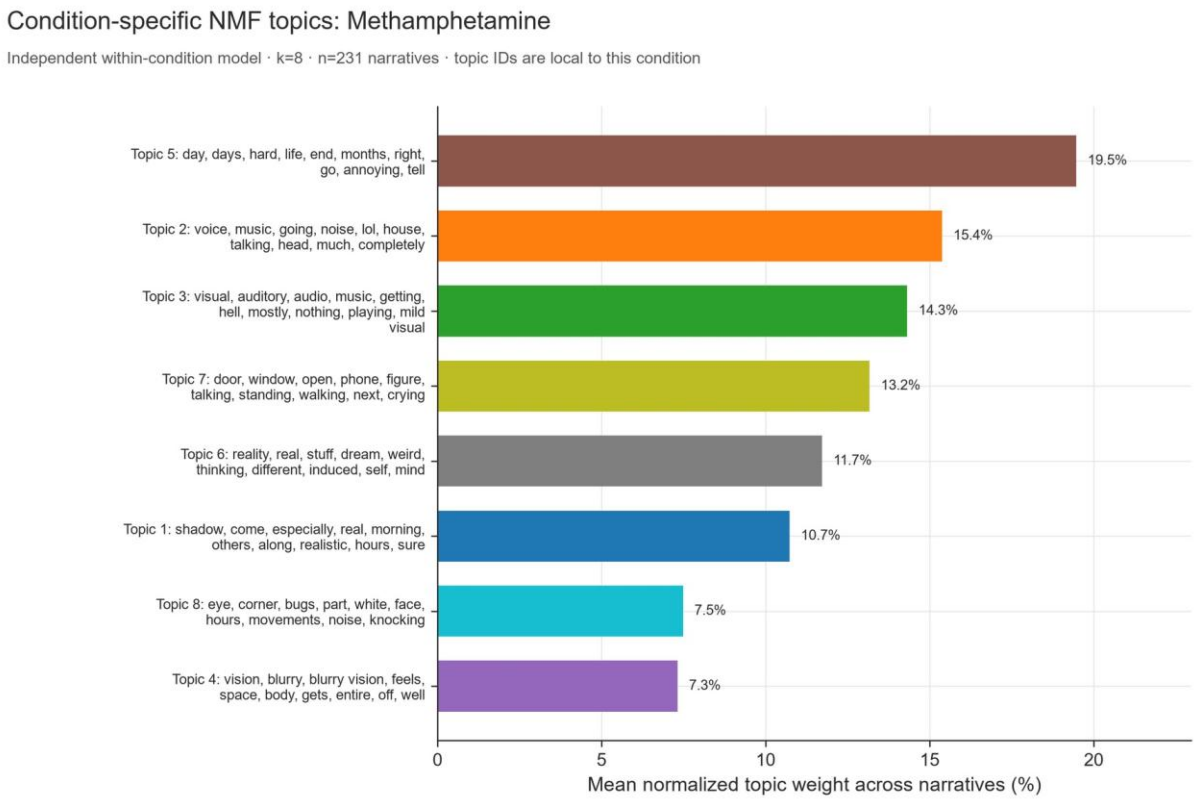
Supplementary Figure S14H



Supplementary Figure S14H. *Condition-specific NMF topic summary for MDMA*

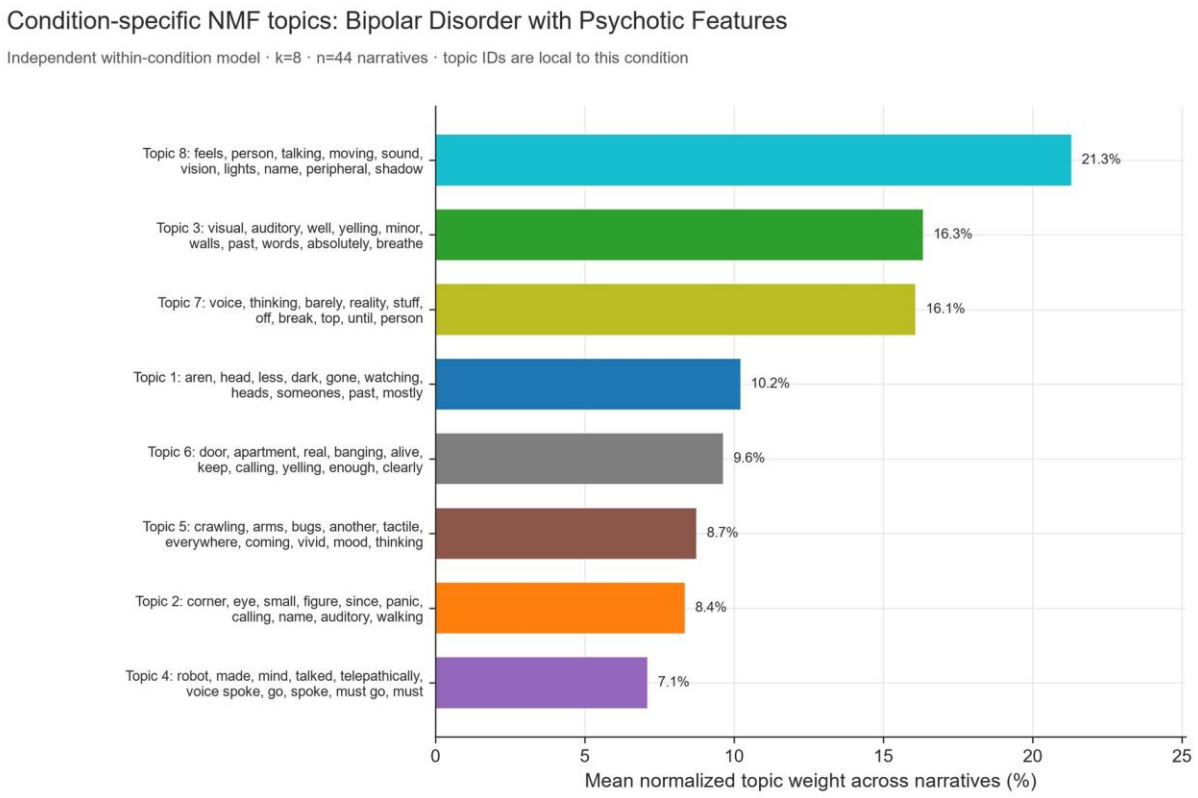
Eight-topic NMF model fitted independently within MDMA. Topic numbers are local to this condition and are not directly comparable across conditions.

Supplementary Figure S14I



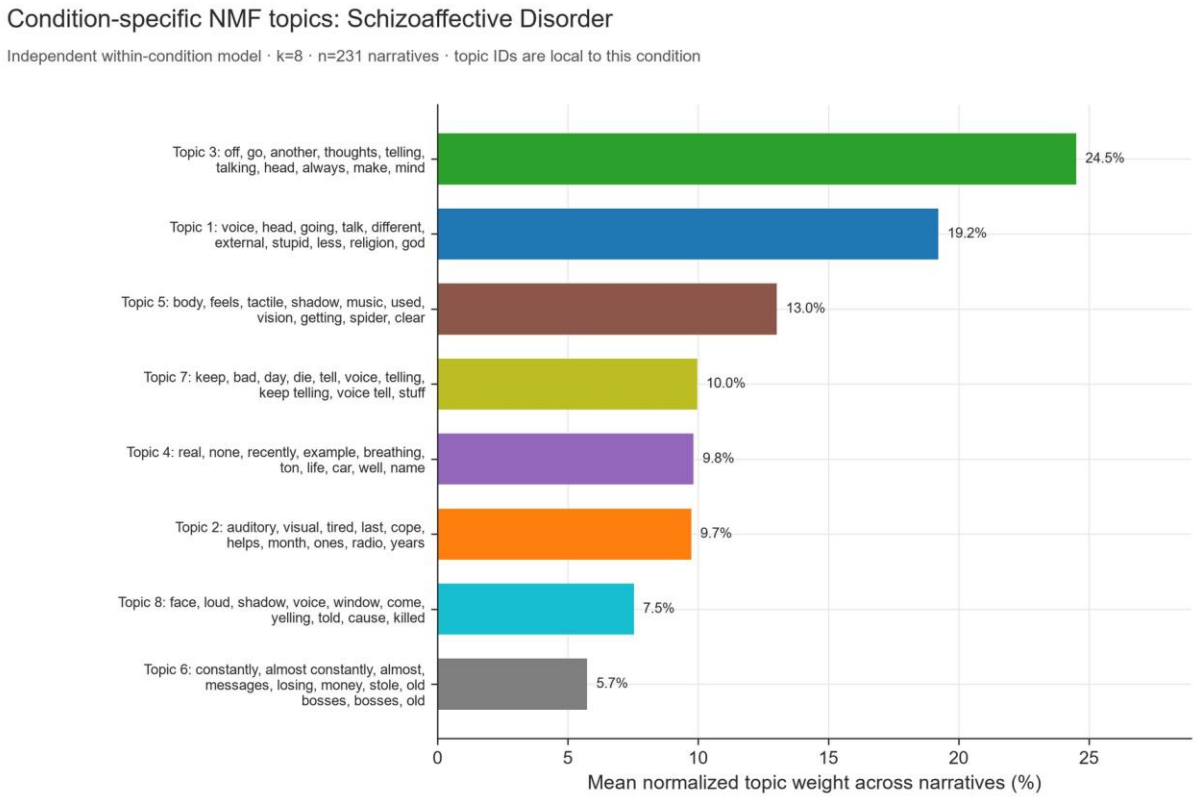
Supplementary Figure S14I. *Condition-specific NMF topic summary for Methamphetamine*
Eight-topic NMF model fitted independently within Methamphetamine. Topic numbers are local to this condition and are not directly comparable across conditions.

Supplementary Figure S14J



Supplementary Figure S14J. *Condition-specific NMF topic summary for Bipolar disorder with psychotic features*
Eight-topic NMF model fitted independently within Bipolar disorder with psychotic features. Topic numbers are local to this condition and are not directly comparable across conditions.

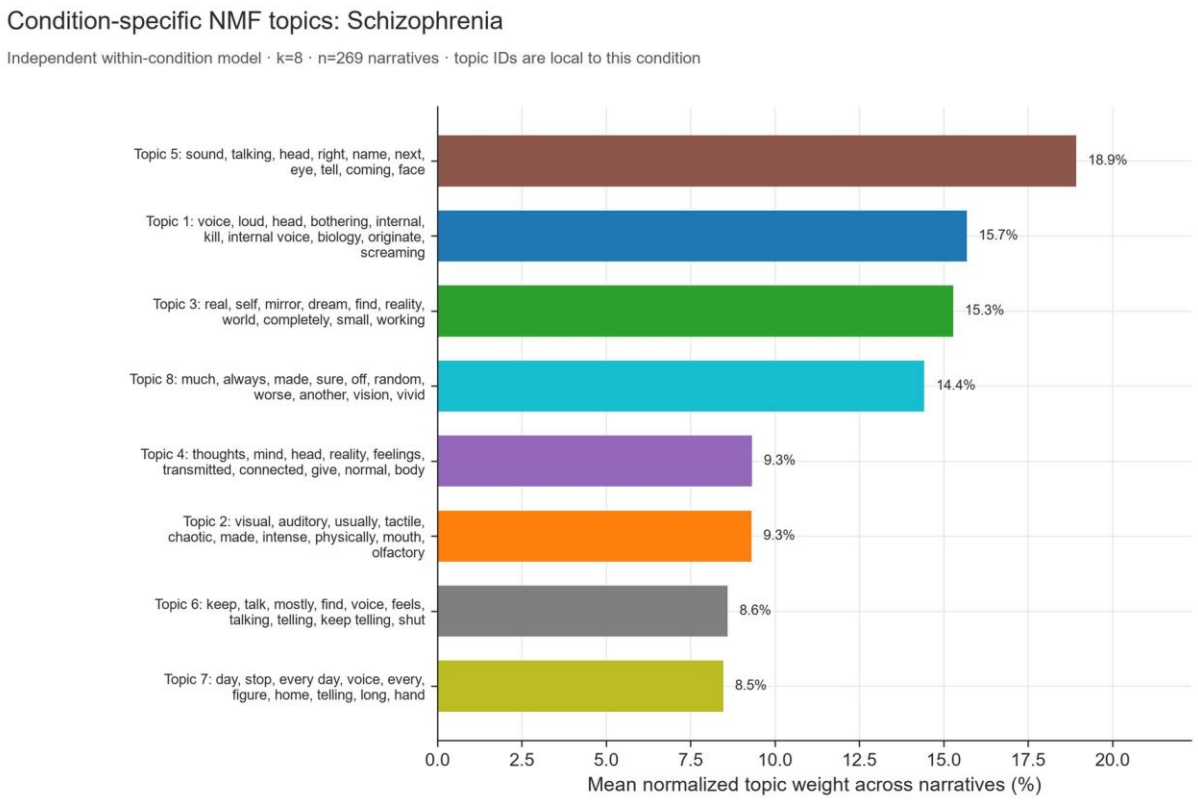
Supplementary Figure S14K



Supplementary Figure S14K. *Condition-specific NMF topic summary for Schizoaffective disorder*

Eight-topic NMF model fitted independently within Schizoaffective disorder. Topic numbers are local to this condition and are not directly comparable across conditions.

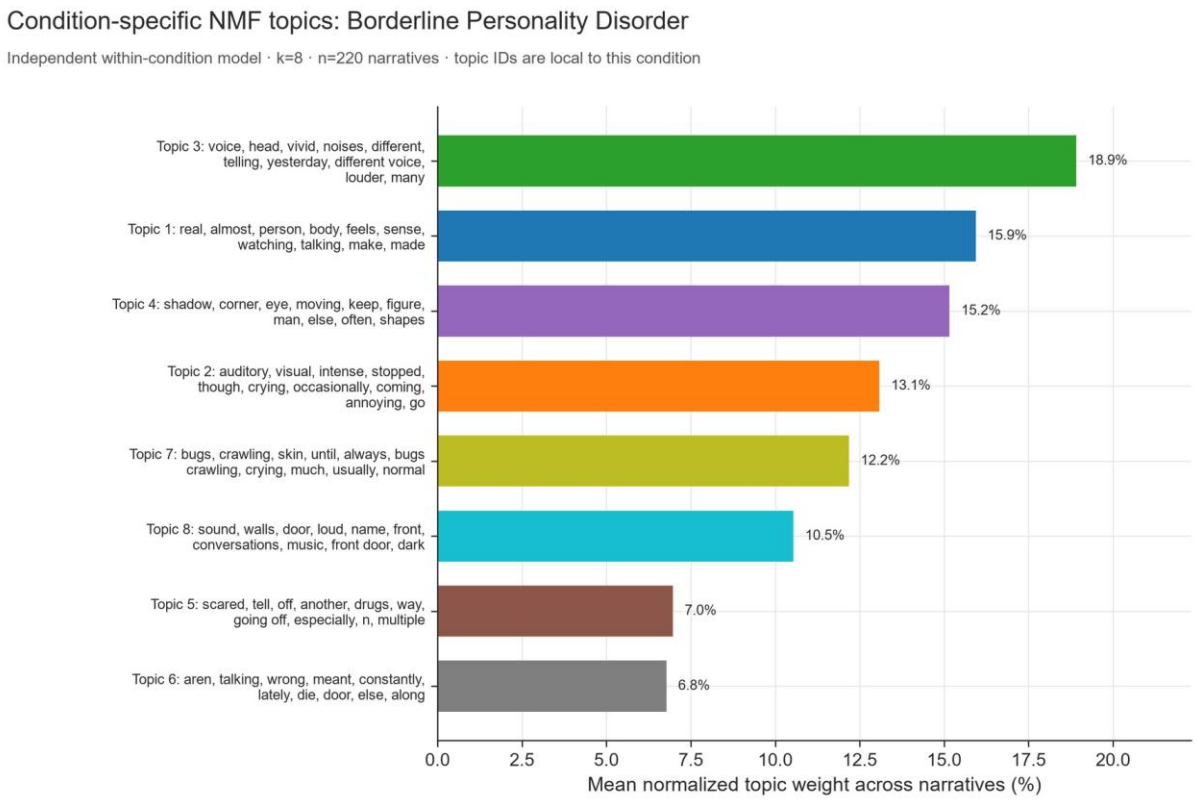
Supplementary Figure S14L



Supplementary Figure S14L. *Condition-specific NMF topic summary for Schizophrenia*

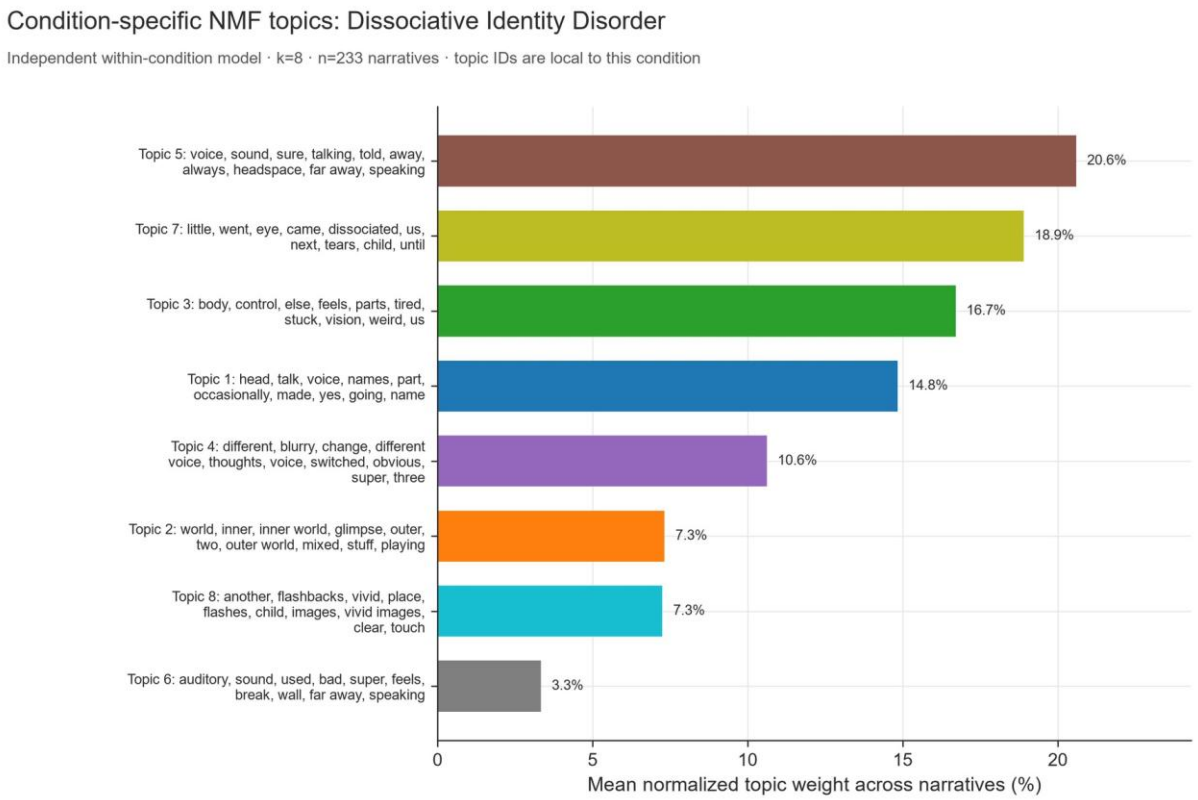
Eight-topic NMF model fitted independently within Schizophrenia. Topic numbers are local to this condition and are not directly comparable across conditions.

Supplementary Figure S14M



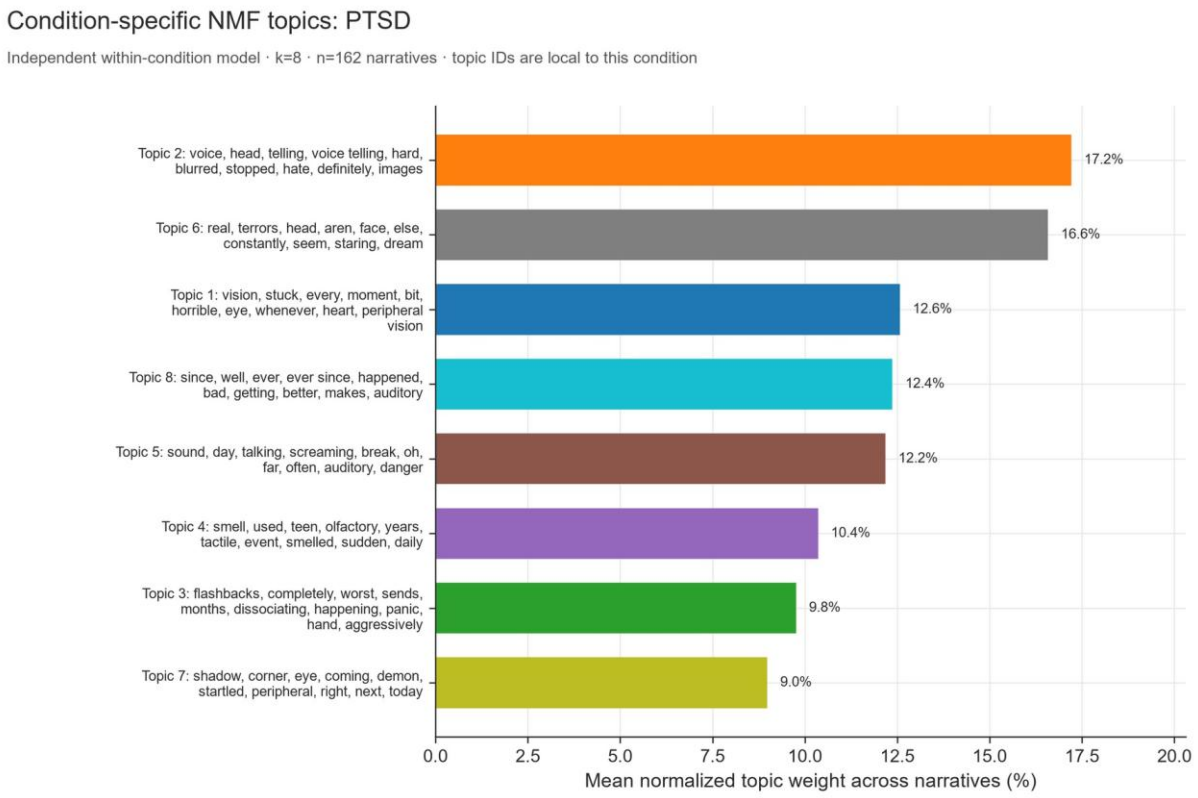
Supplementary Figure S14M. *Condition-specific NMF topic summary for Borderline personality disorder*
Eight-topic NMF model fitted independently within Borderline personality disorder. Topic numbers are local to this condition and are not directly comparable across conditions.

Supplementary Figure S14N



Supplementary Figure S14N. *Condition-specific NMF topic summary for Dissociative identity disorder*
Eight-topic NMF model fitted independently within Dissociative identity disorder. Topic numbers are local to this condition and are not directly comparable across conditions.

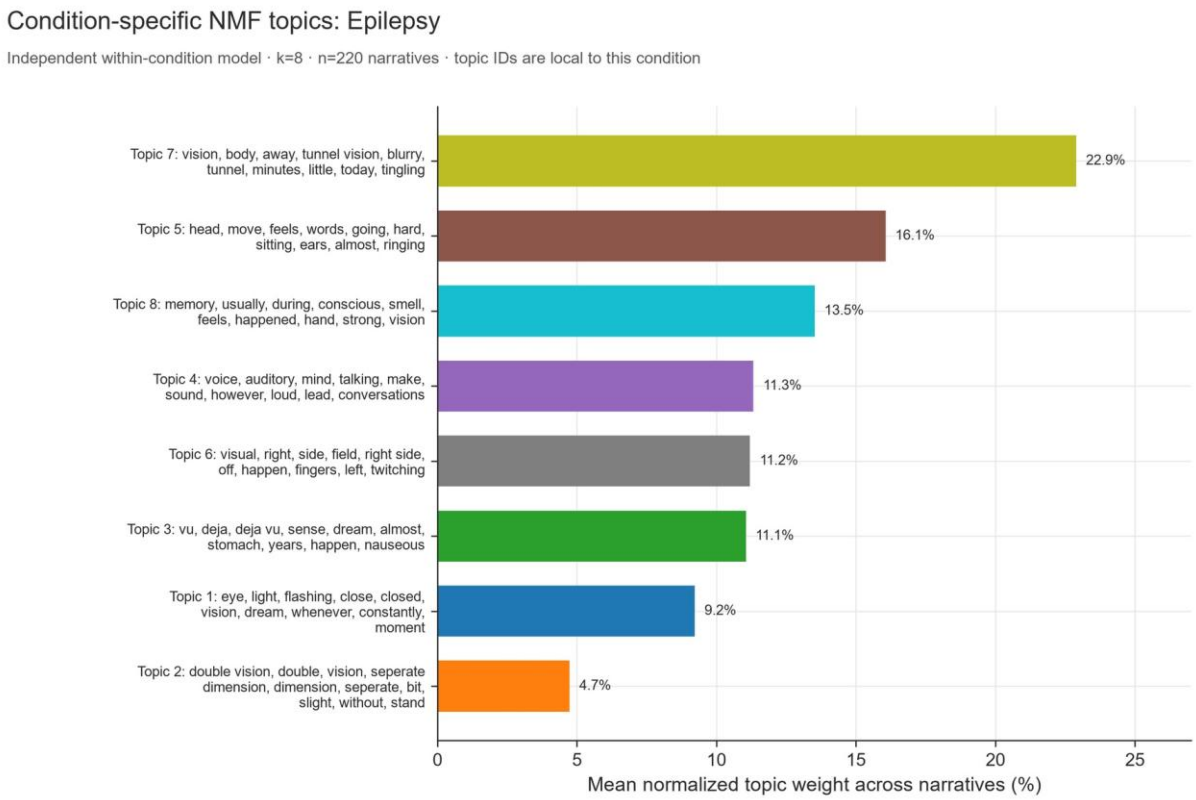
Supplementary Figure S14O



Supplementary Figure S14O. *Condition-specific NMF topic summary for PTSD*

Eight-topic NMF model fitted independently within PTSD. Topic numbers are local to this condition and are not directly comparable across conditions.

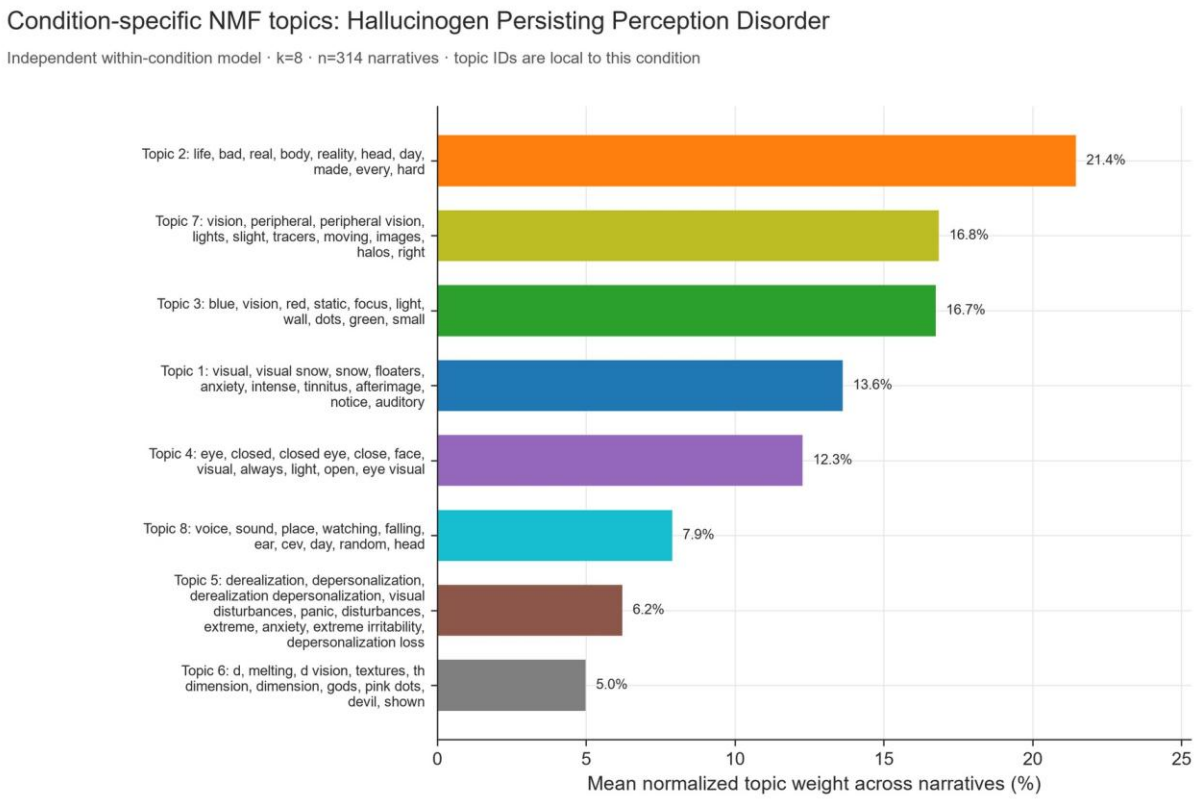
Supplementary Figure S14P



Supplementary Figure S14P. *Condition-specific NMF topic summary for Epilepsy*

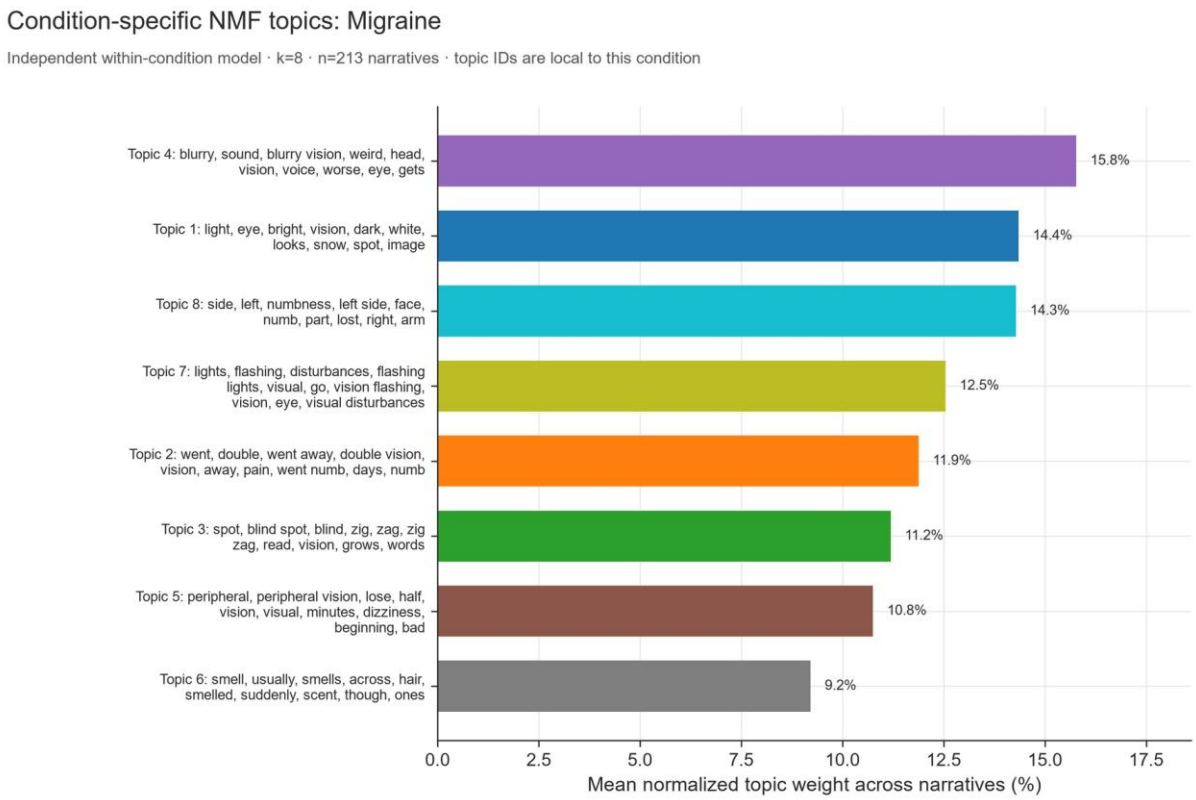
Eight-topic NMF model fitted independently within Epilepsy. Topic numbers are local to this condition and are not directly comparable across conditions.

Supplementary Figure S14Q



Supplementary Figure S14Q. *Condition-specific NMF topic summary for Hallucinogen persisting perception disorder*
Eight-topic NMF model fitted independently within Hallucinogen persisting perception disorder. Topic numbers are local to this condition and are not directly comparable across conditions.

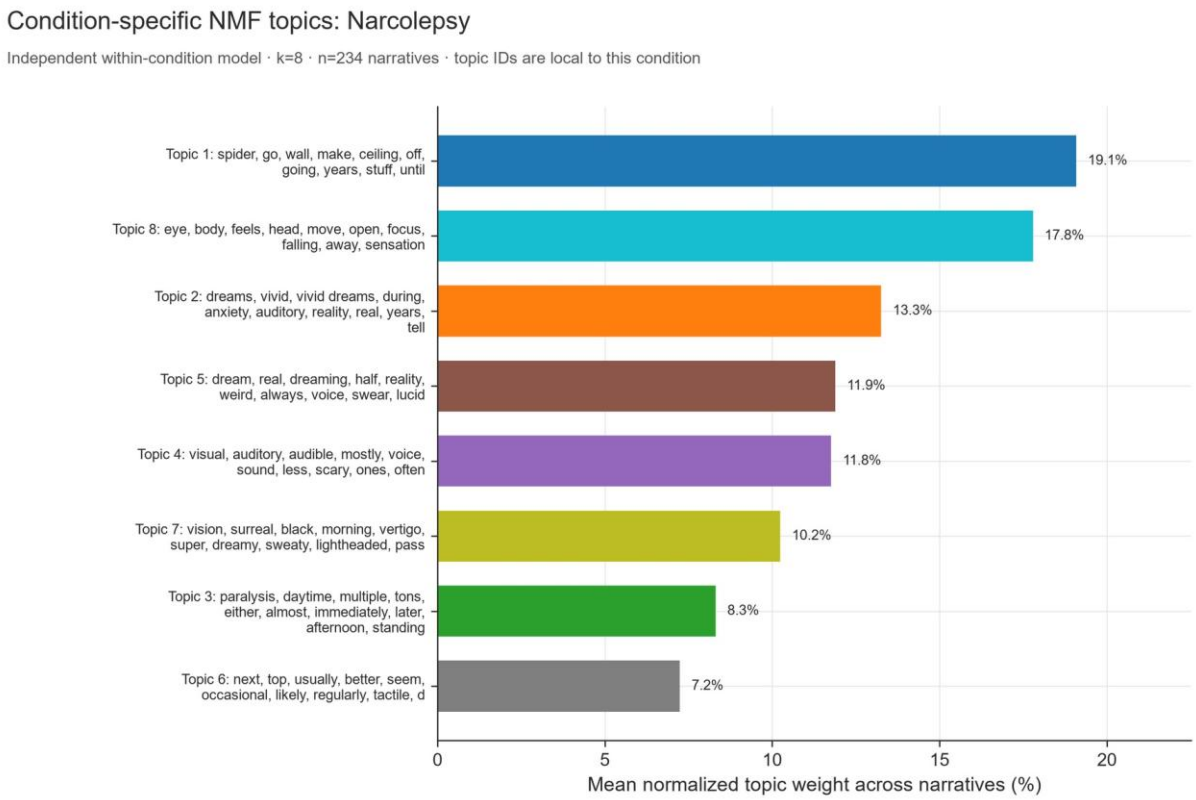
Supplementary Figure S14R



Supplementary Figure S14R. *Condition-specific NMF topic summary for Migraine*

Eight-topic NMF model fitted independently within Migraine. Topic numbers are local to this condition and are not directly comparable across conditions.

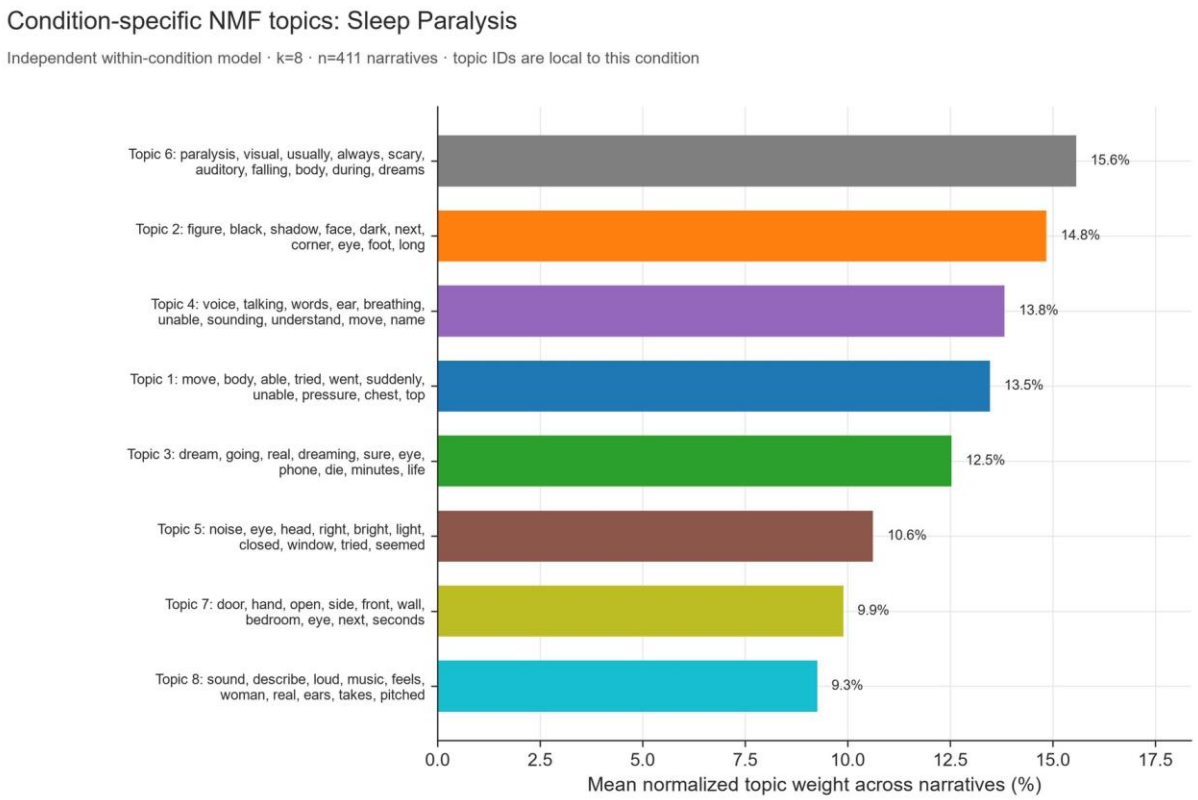
Supplementary Figure S14S



Supplementary Figure S14S. *Condition-specific NMF topic summary for Narcolepsy*

Eight-topic NMF model fitted independently within Narcolepsy. Topic numbers are local to this condition and are not directly comparable across conditions.

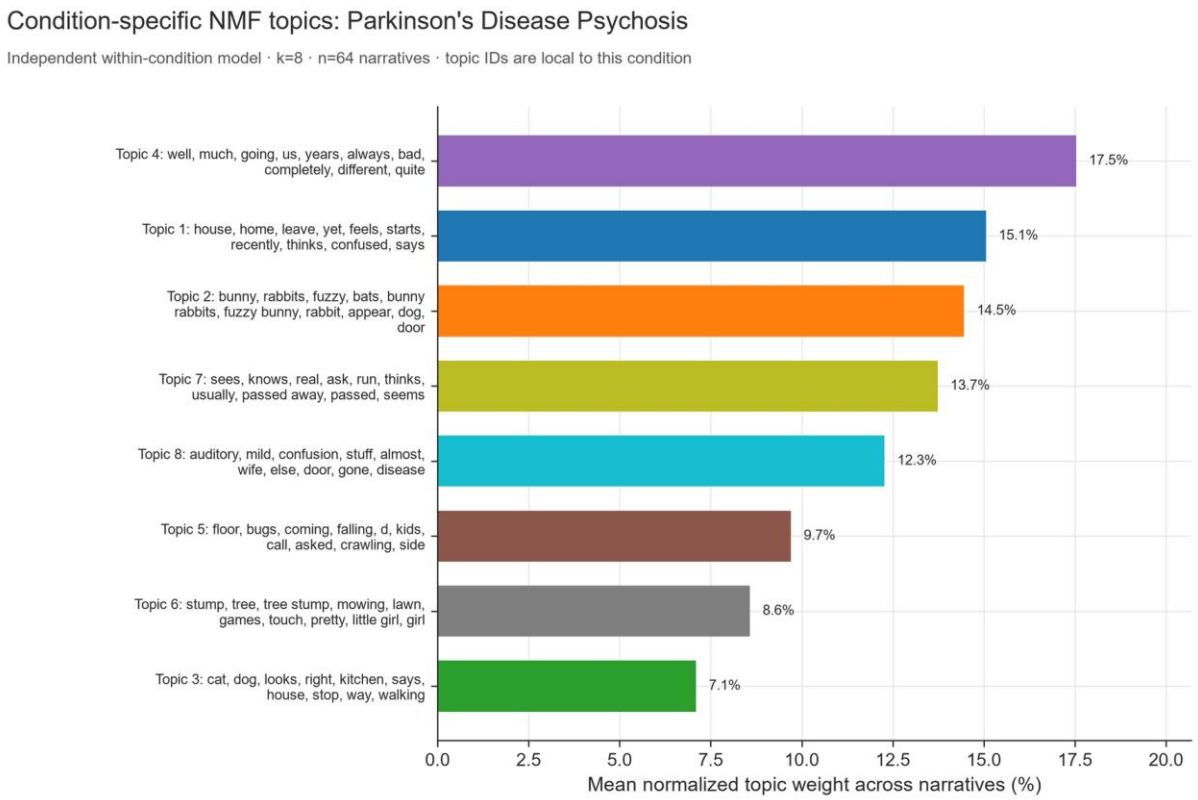
Supplementary Figure S14T



Supplementary Figure S14T. *Condition-specific NMF topic summary for Sleep paralysis*

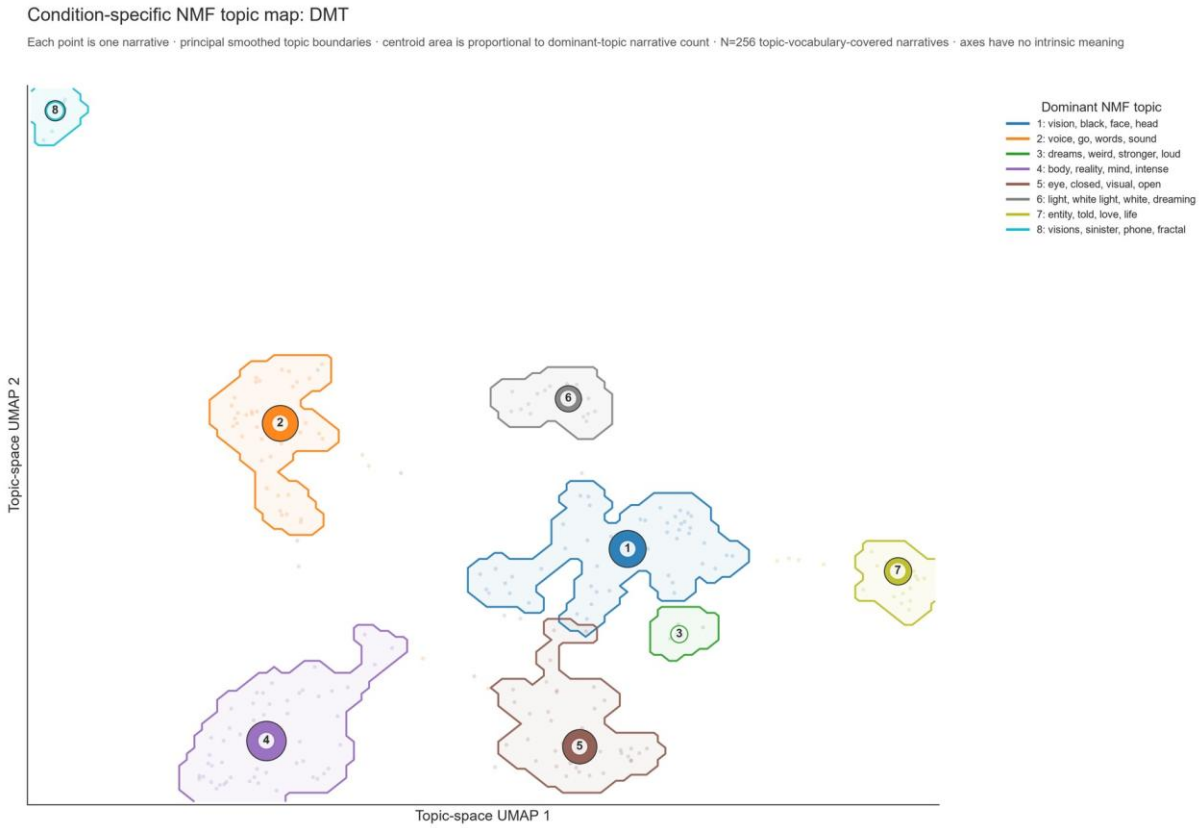
Eight-topic NMF model fitted independently within Sleep paralysis. Topic numbers are local to this condition and are not directly comparable across conditions.

Supplementary Figure S14U



Supplementary Figure S14U. *Condition-specific NMF topic summary for Parkinson's disease psychosis*
Eight-topic NMF model fitted independently within Parkinson's disease psychosis. Topic numbers are local to this condition and are not directly comparable across conditions.

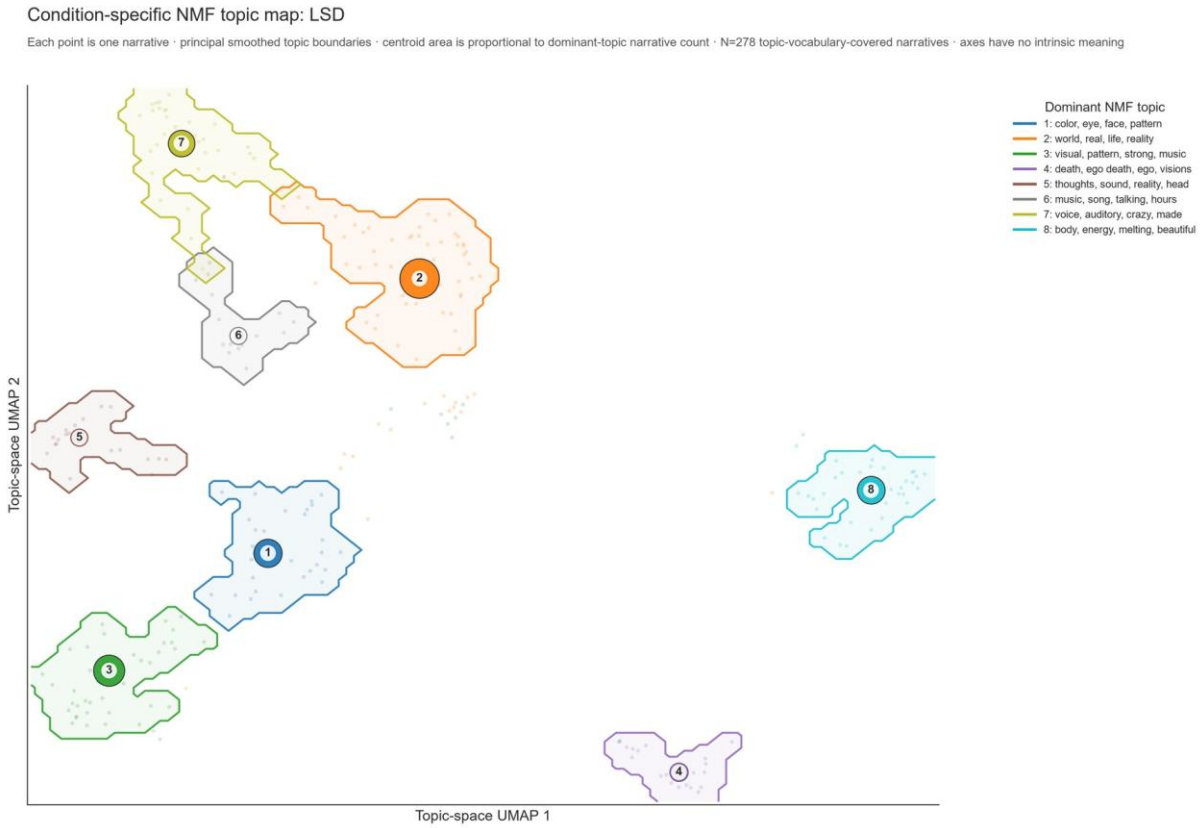
Supplementary Figure S15A



Supplementary Figure S15A. *Condition-specific NMF topic map for DMT*

UMAP of narrative topic weights from the independent eight-topic DMT model. Each point is one narrative; axes have no intrinsic meaning.

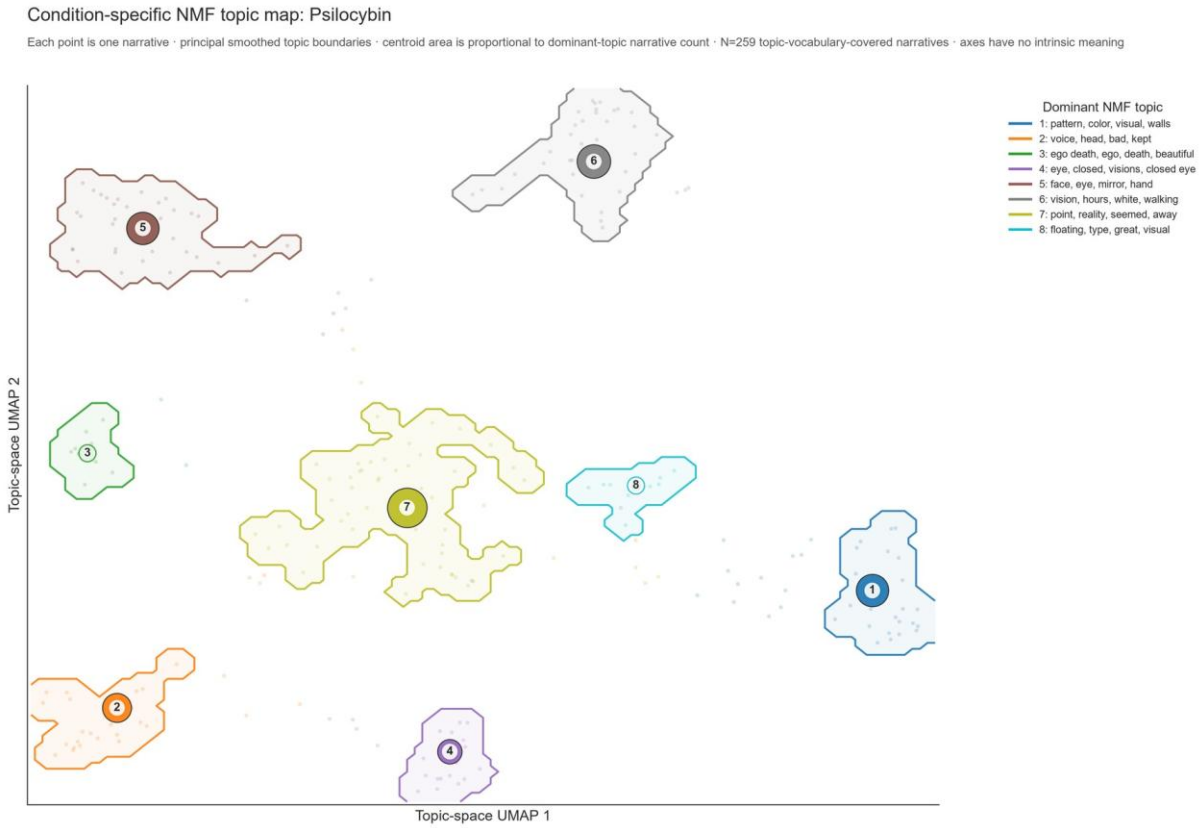
Supplementary Figure S15B



Supplementary Figure S15B. *Condition-specific NMF topic map for LSD*

UMAP of narrative topic weights from the independent eight-topic LSD model. Each point is one narrative; axes have no intrinsic meaning.

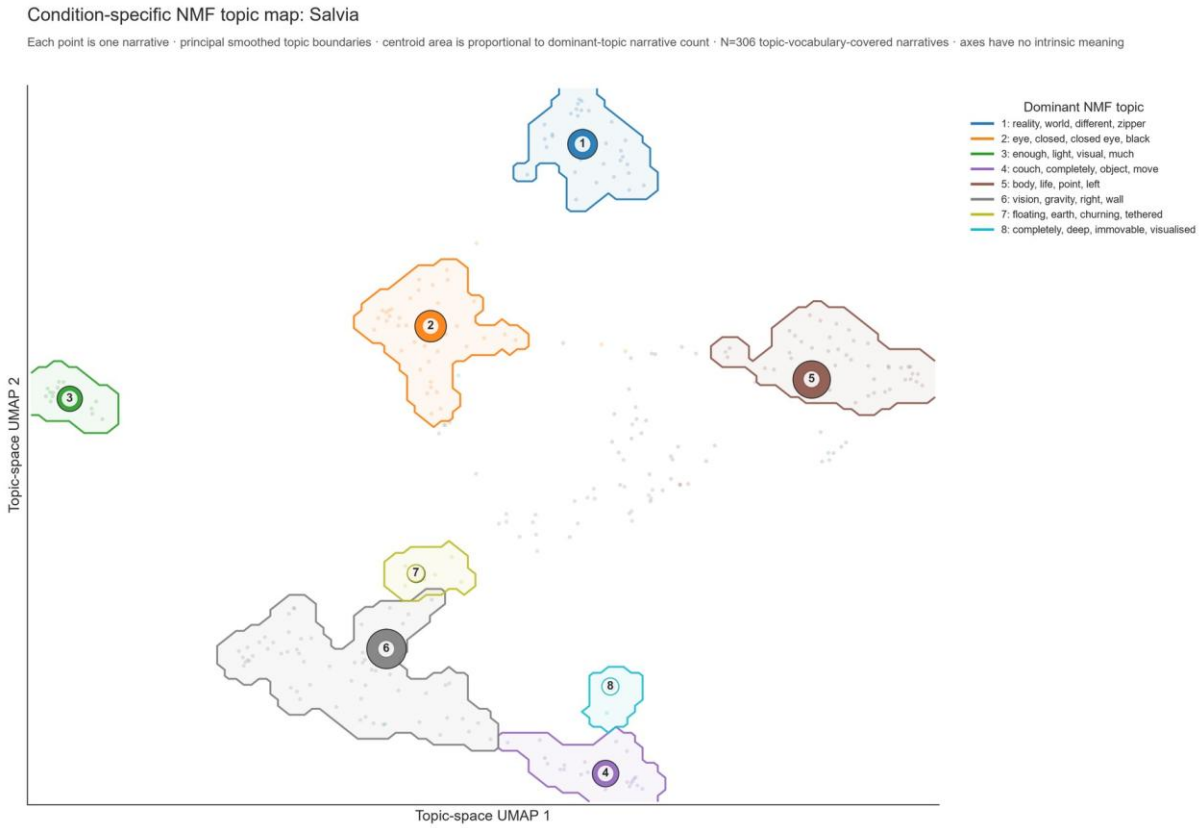
Supplementary Figure S15C



Supplementary Figure S15C. *Condition-specific NMF topic map for Psilocybin*

UMAP of narrative topic weights from the independent eight-topic Psilocybin model. Each point is one narrative; axes have no intrinsic meaning.

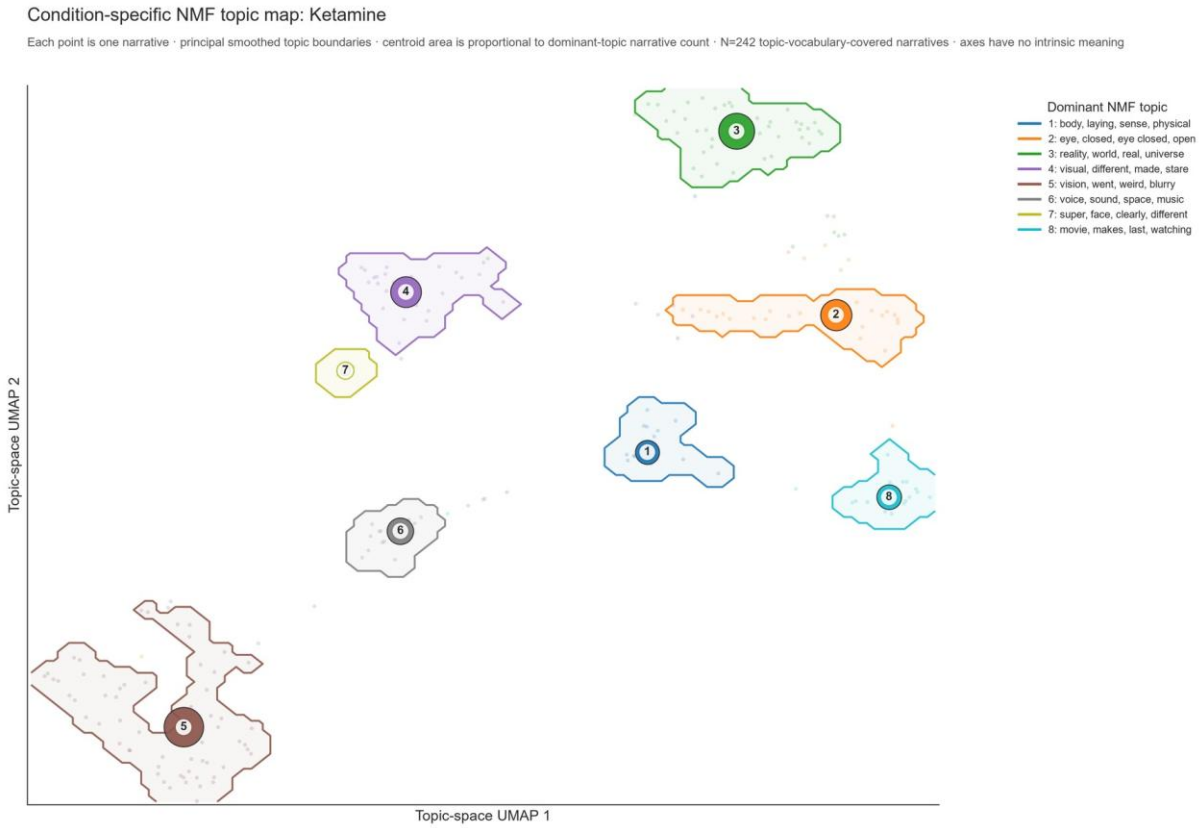
Supplementary Figure S15D



Supplementary Figure S15D. *Condition-specific NMF topic map for Salvia*

UMAP of narrative topic weights from the independent eight-topic Salvia model. Each point is one narrative; axes have no intrinsic meaning.

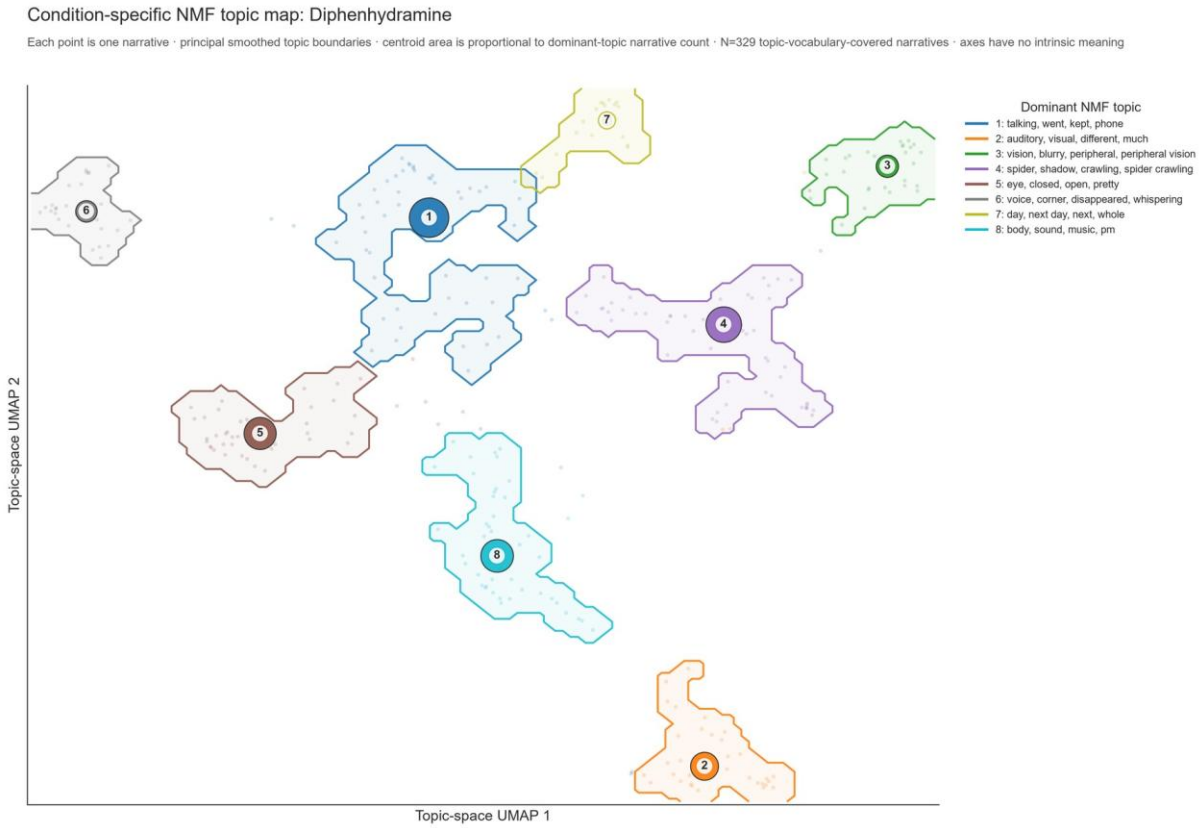
Supplementary Figure S15E



Supplementary Figure S15E. *Condition-specific NMF topic map for Ketamine*

UMAP of narrative topic weights from the independent eight-topic Ketamine model. Each point is one narrative; axes have no intrinsic meaning.

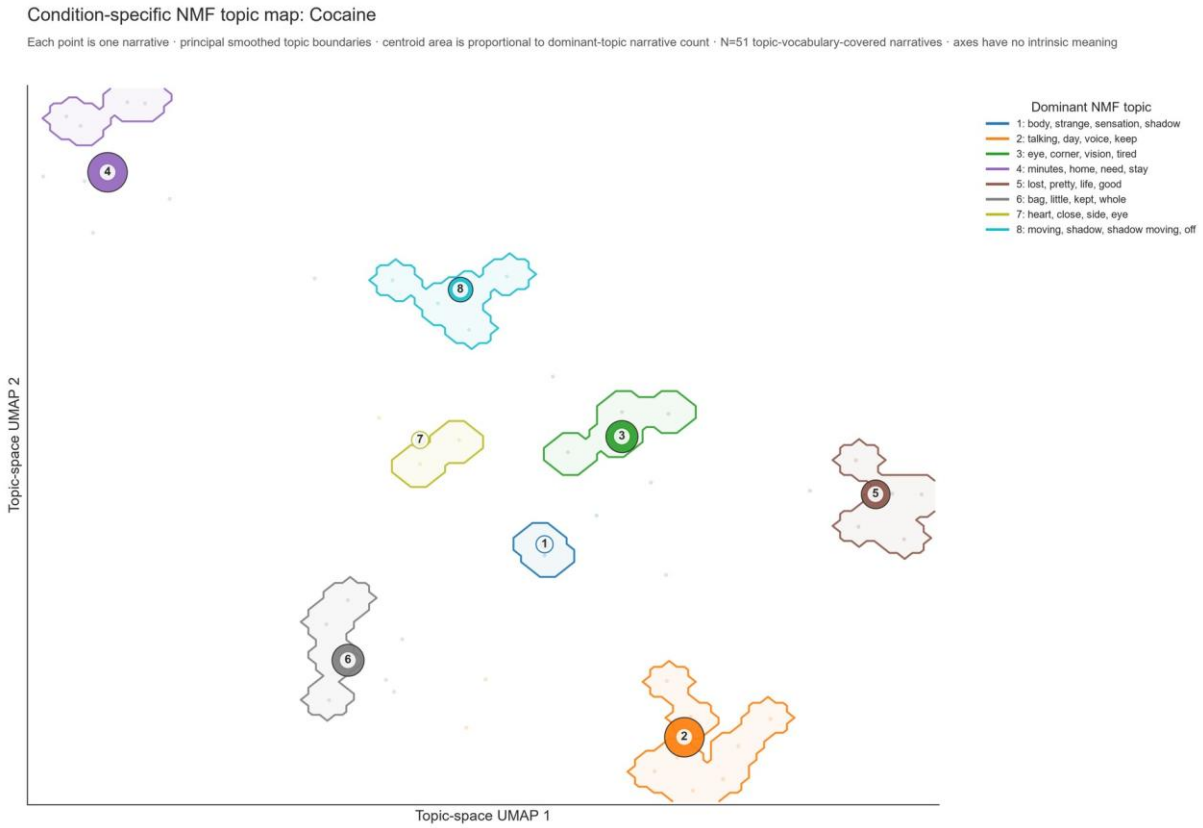
Supplementary Figure S15F



Supplementary Figure S15F. *Condition-specific NMF topic map for Diphenhydramine*

UMAP of narrative topic weights from the independent eight-topic Diphenhydramine model. Each point is one narrative; axes have no intrinsic meaning.

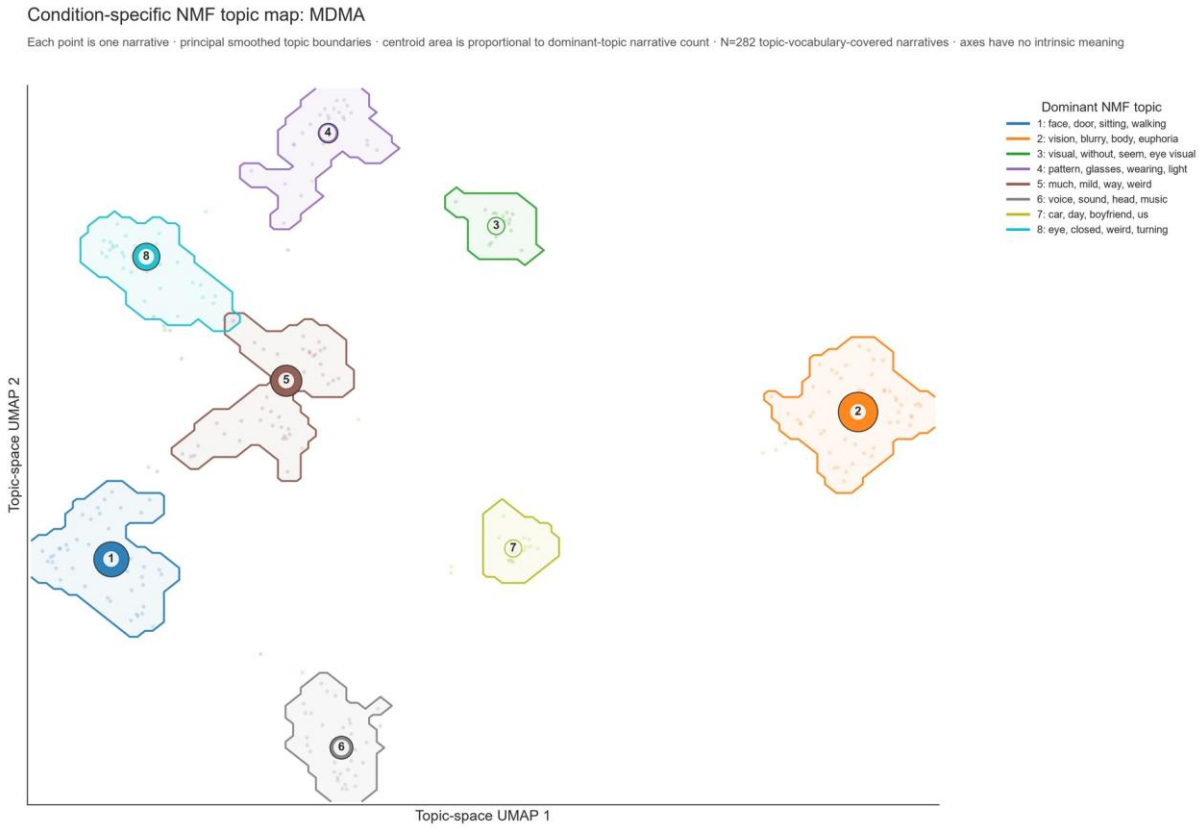
Supplementary Figure S15G



Supplementary Figure S15G. *Condition-specific NMF topic map for Cocaine*

UMAP of narrative topic weights from the independent eight-topic Cocaine model. Each point is one narrative; axes have no intrinsic meaning.

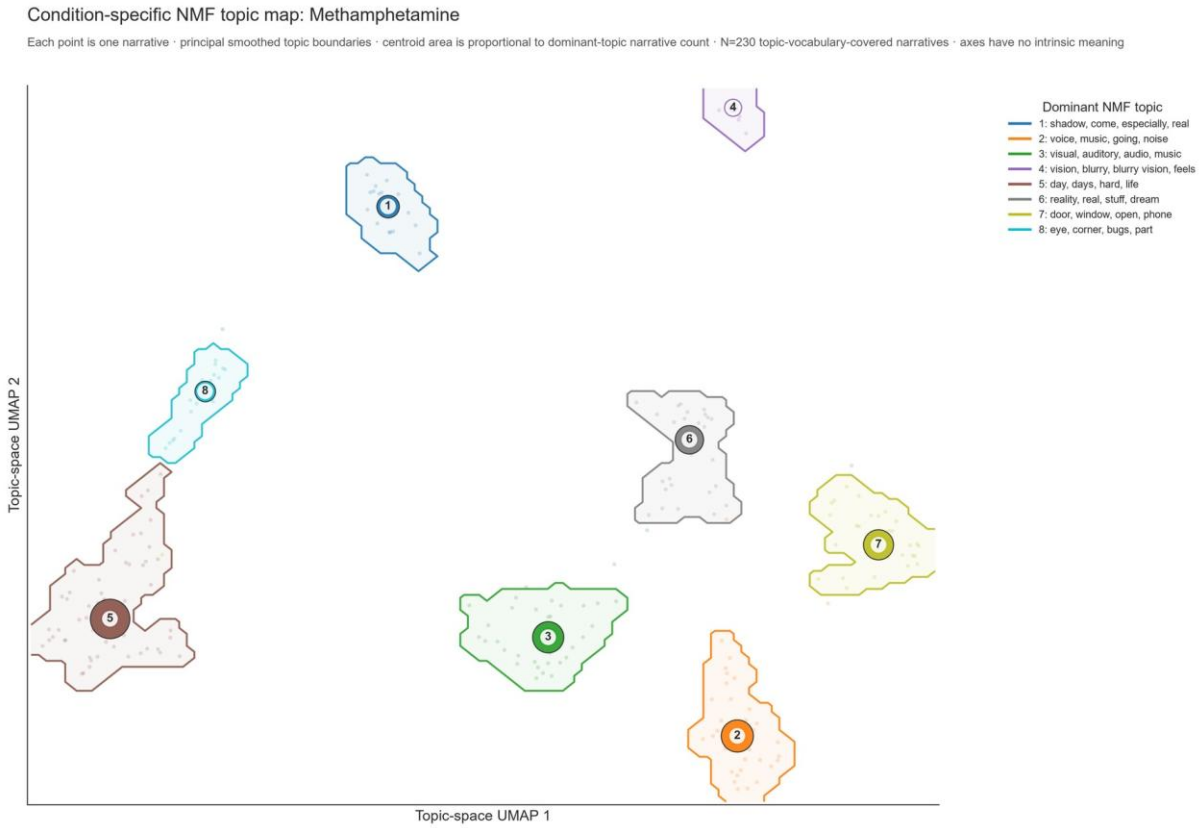
Supplementary Figure S15H



Supplementary Figure S15H. *Condition-specific NMF topic map for MDMA*

UMAP of narrative topic weights from the independent eight-topic MDMA model. Each point is one narrative; axes have no intrinsic meaning.

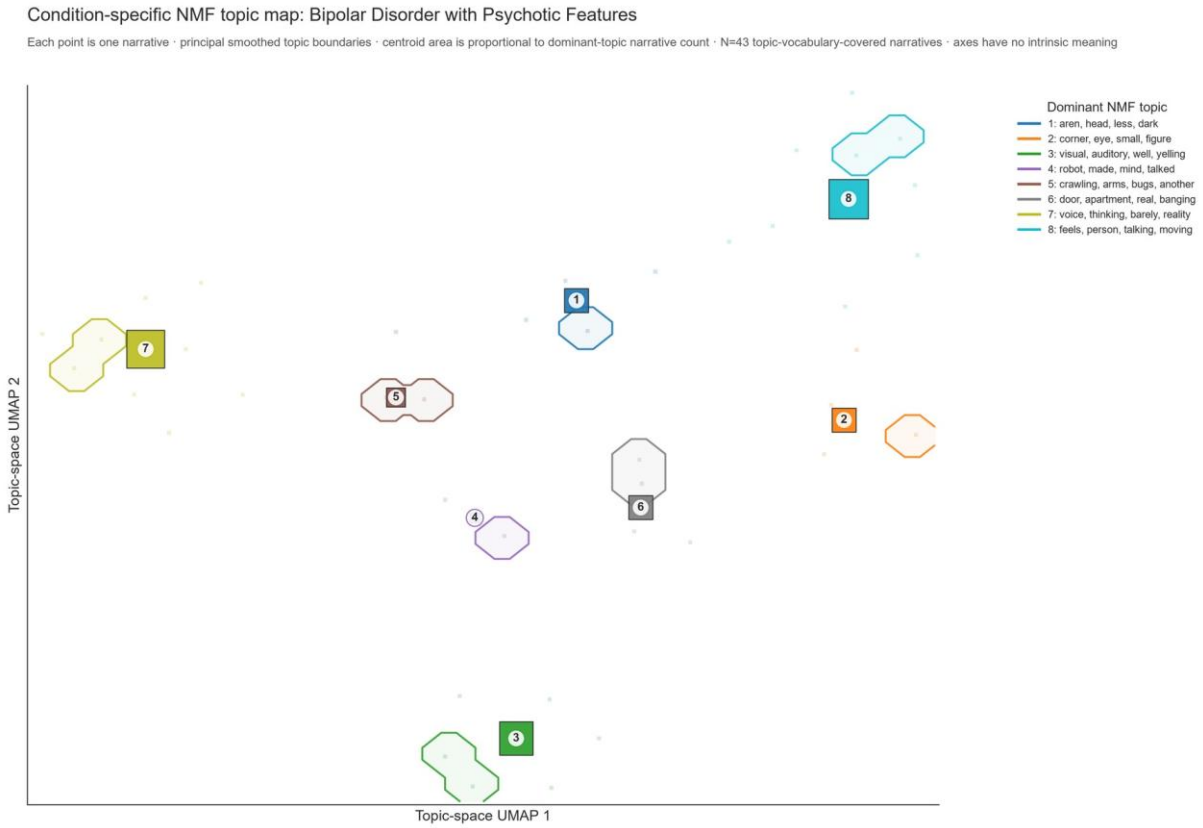
Supplementary Figure S15I



Supplementary Figure S15I. *Condition-specific NMF topic map for Methamphetamine*

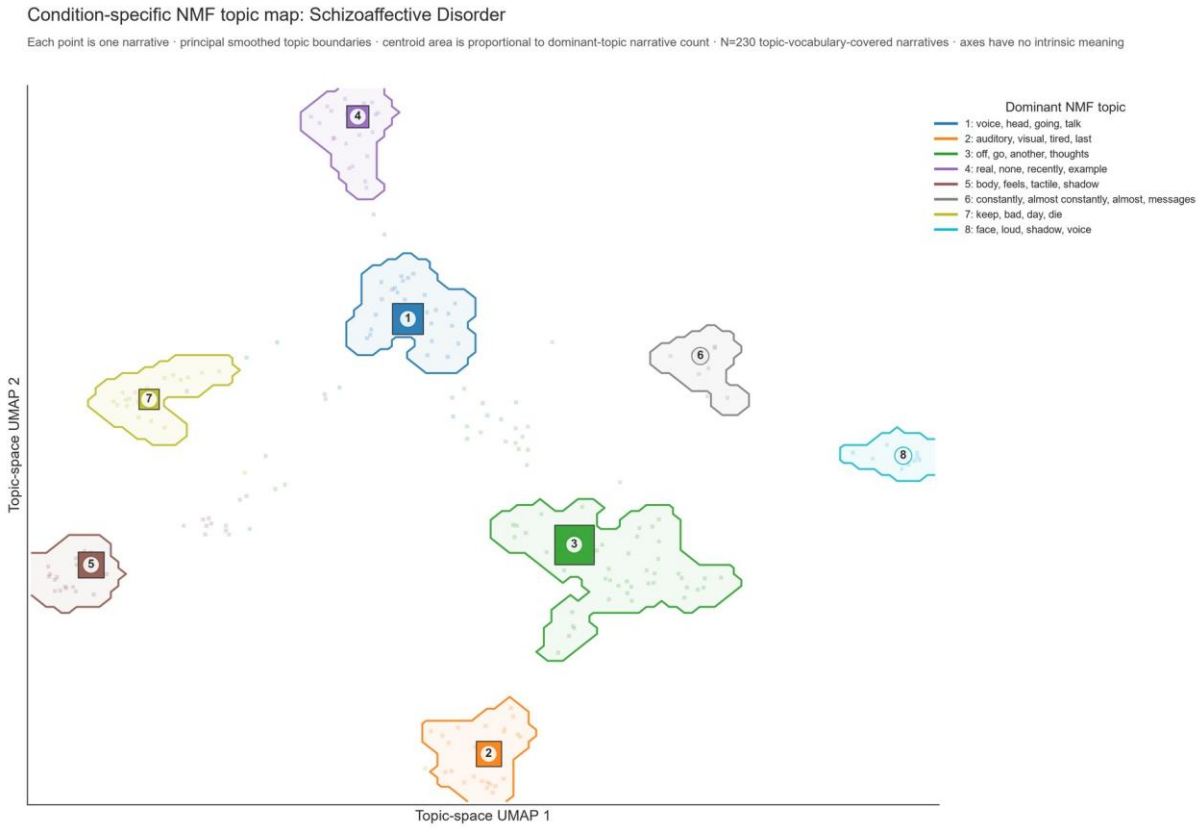
UMAP of narrative topic weights from the independent eight-topic Methamphetamine model. Each point is one narrative; axes have no intrinsic meaning.

Supplementary Figure S15J



Supplementary Figure S15J. *Condition-specific NMF topic map for Bipolar disorder with psychotic features*
UMAP of narrative topic weights from the independent eight-topic Bipolar disorder with psychotic features model. Each point is one narrative; axes have no intrinsic meaning.

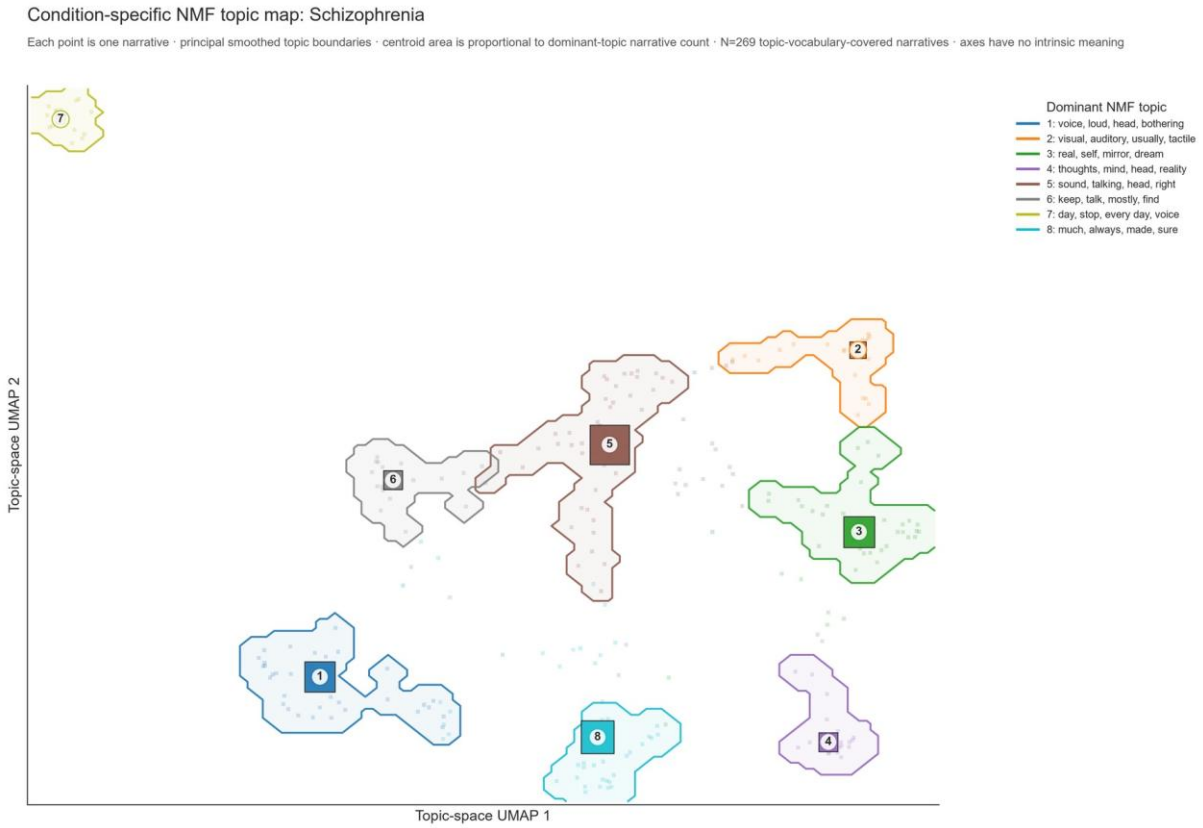
Supplementary Figure S15K



Supplementary Figure S15K. *Condition-specific NMF topic map for Schizoaffective disorder*

UMAP of narrative topic weights from the independent eight-topic Schizoaffective disorder model. Each point is one narrative; axes have no intrinsic meaning.

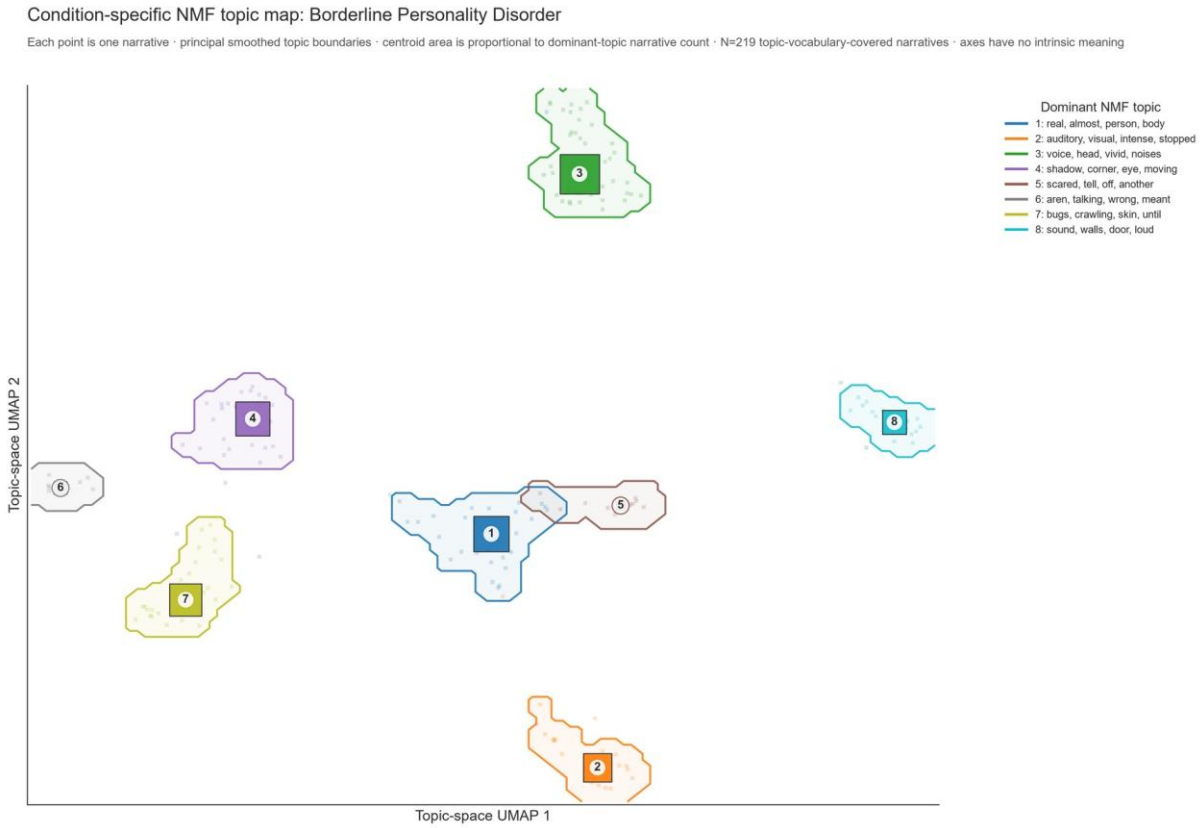
Supplementary Figure S15L



Supplementary Figure S15L. *Condition-specific NMF topic map for Schizophrenia*

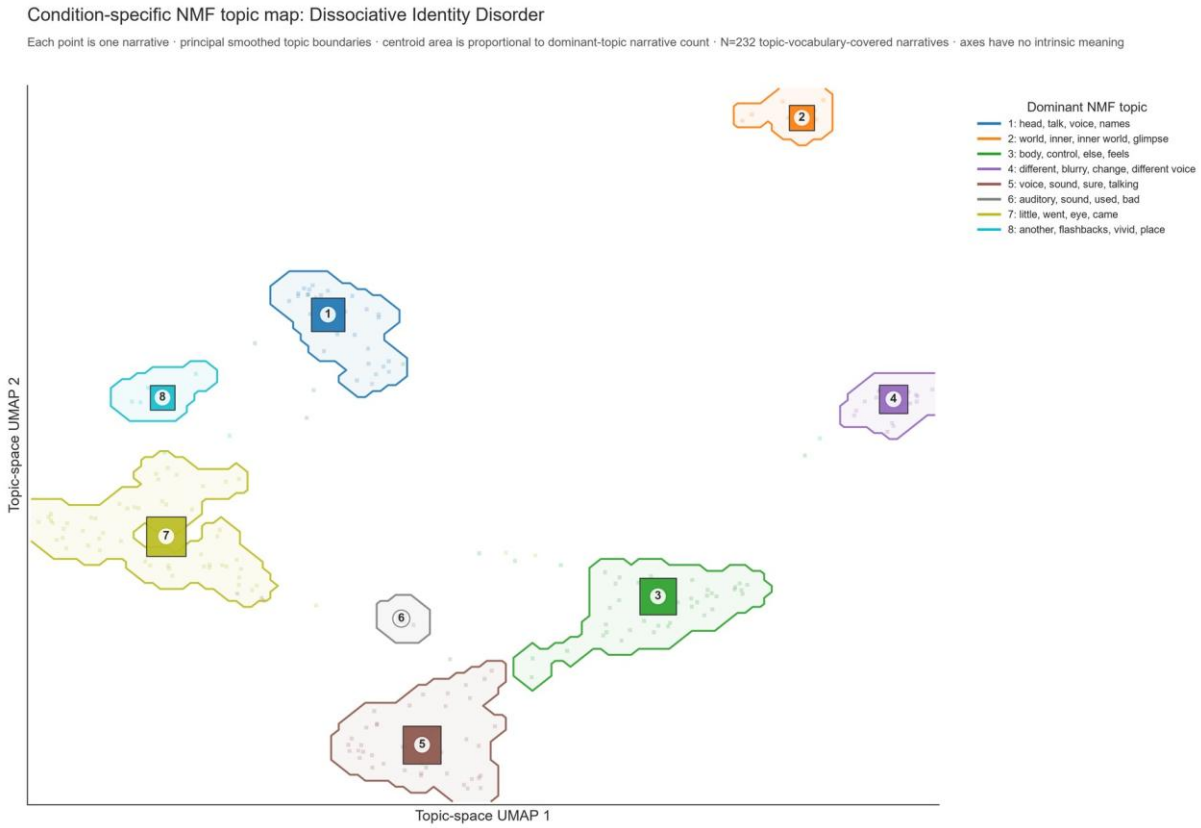
UMAP of narrative topic weights from the independent eight-topic Schizophrenia model. Each point is one narrative; axes have no intrinsic meaning.

Supplementary Figure S15M



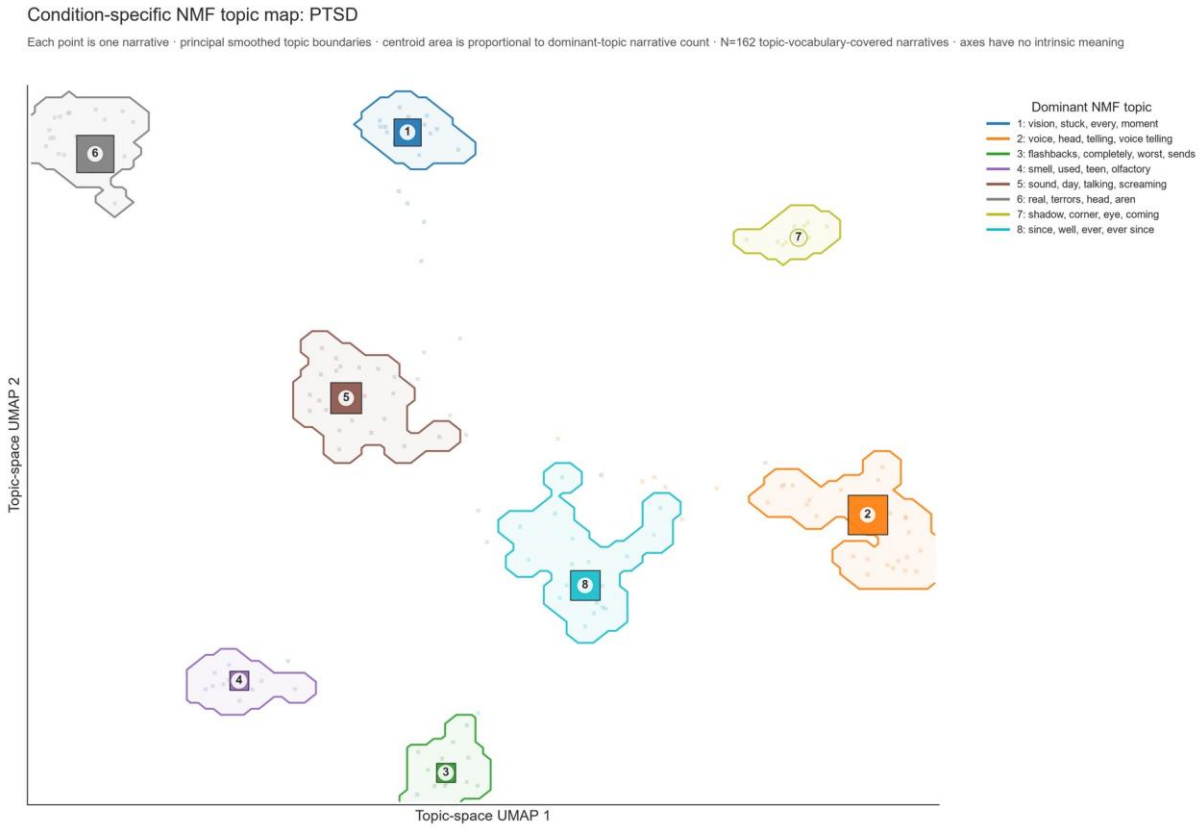
Supplementary Figure S15M. *Condition-specific NMF topic map for Borderline personality disorder*
UMAP of narrative topic weights from the independent eight-topic Borderline personality disorder model. Each point is one narrative; axes have no intrinsic meaning.

Supplementary Figure S15N



Supplementary Figure S15N. *Condition-specific NMF topic map for Dissociative identity disorder*
UMAP of narrative topic weights from the independent eight-topic Dissociative identity disorder model. Each point is one narrative; axes have no intrinsic meaning.

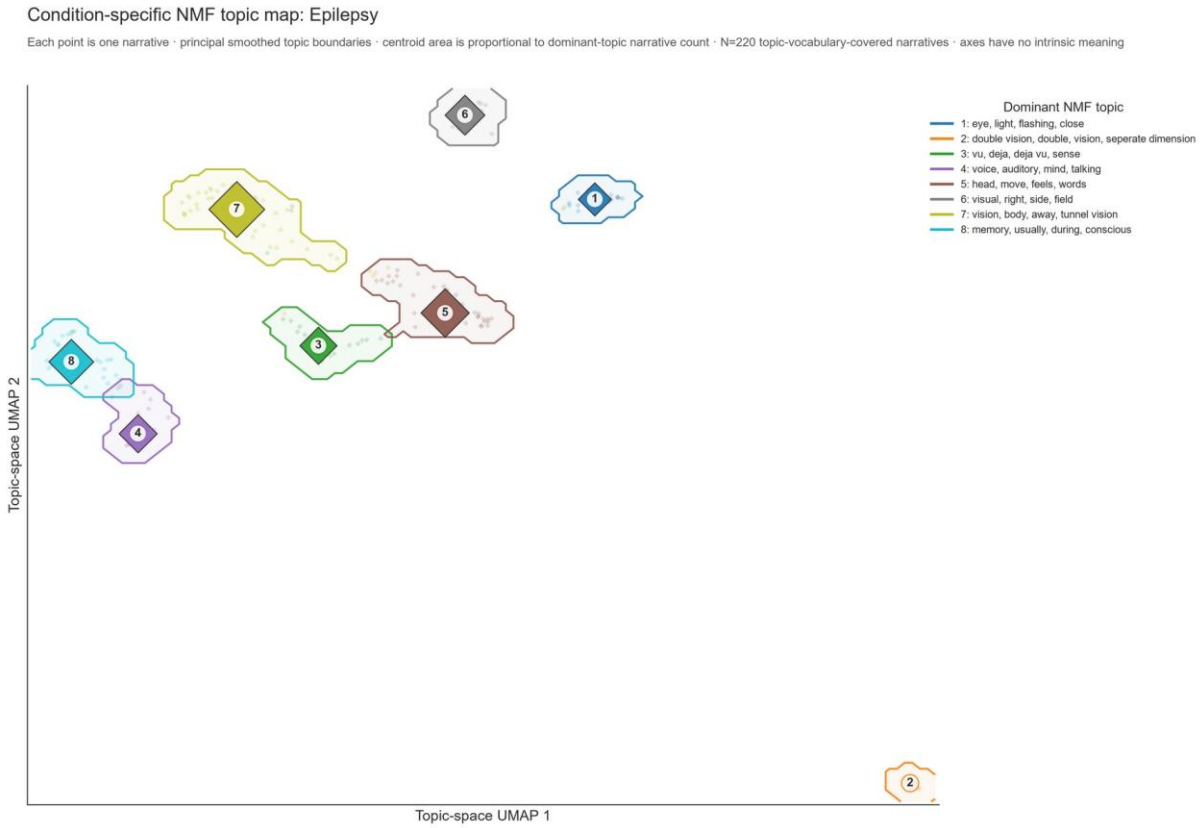
Supplementary Figure S15O



Supplementary Figure S15O. *Condition-specific NMF topic map for PTSD*

UMAP of narrative topic weights from the independent eight-topic PTSD model. Each point is one narrative; axes have no intrinsic meaning.

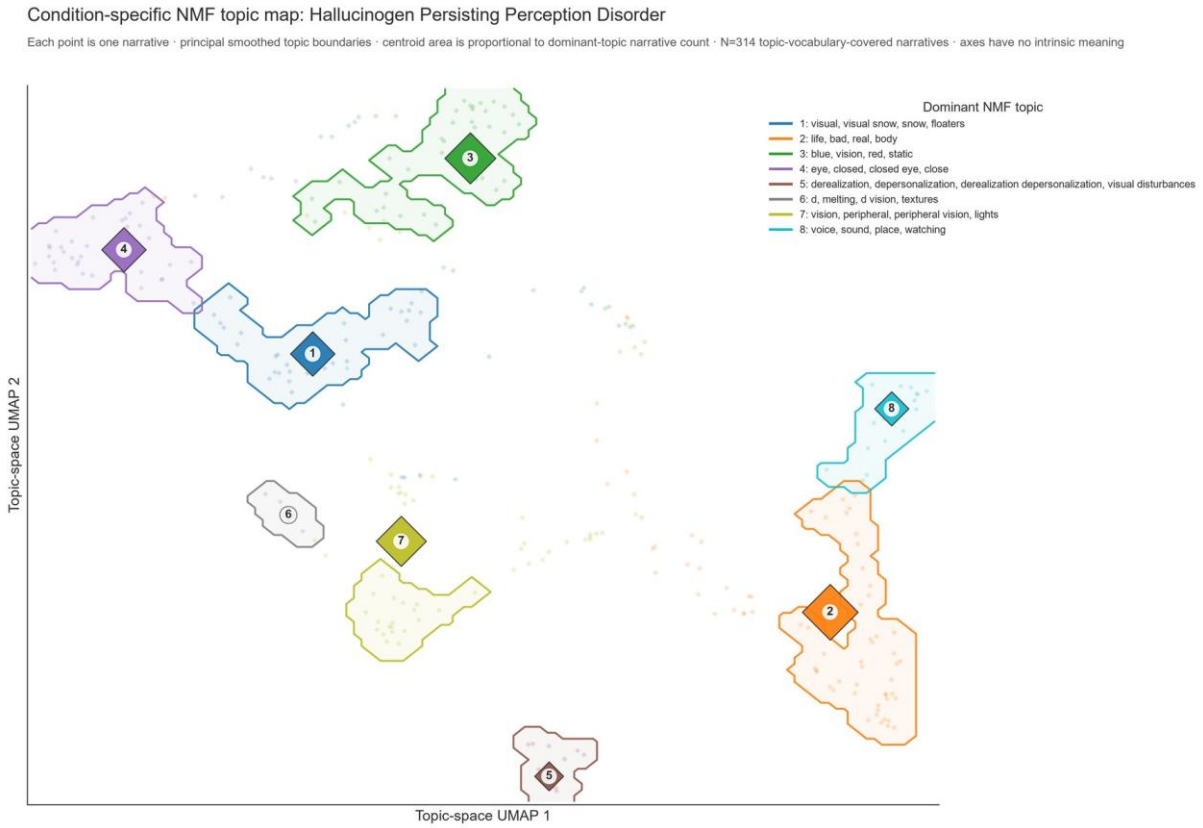
Supplementary Figure S15P



Supplementary Figure S15P. *Condition-specific NMF topic map for Epilepsy*

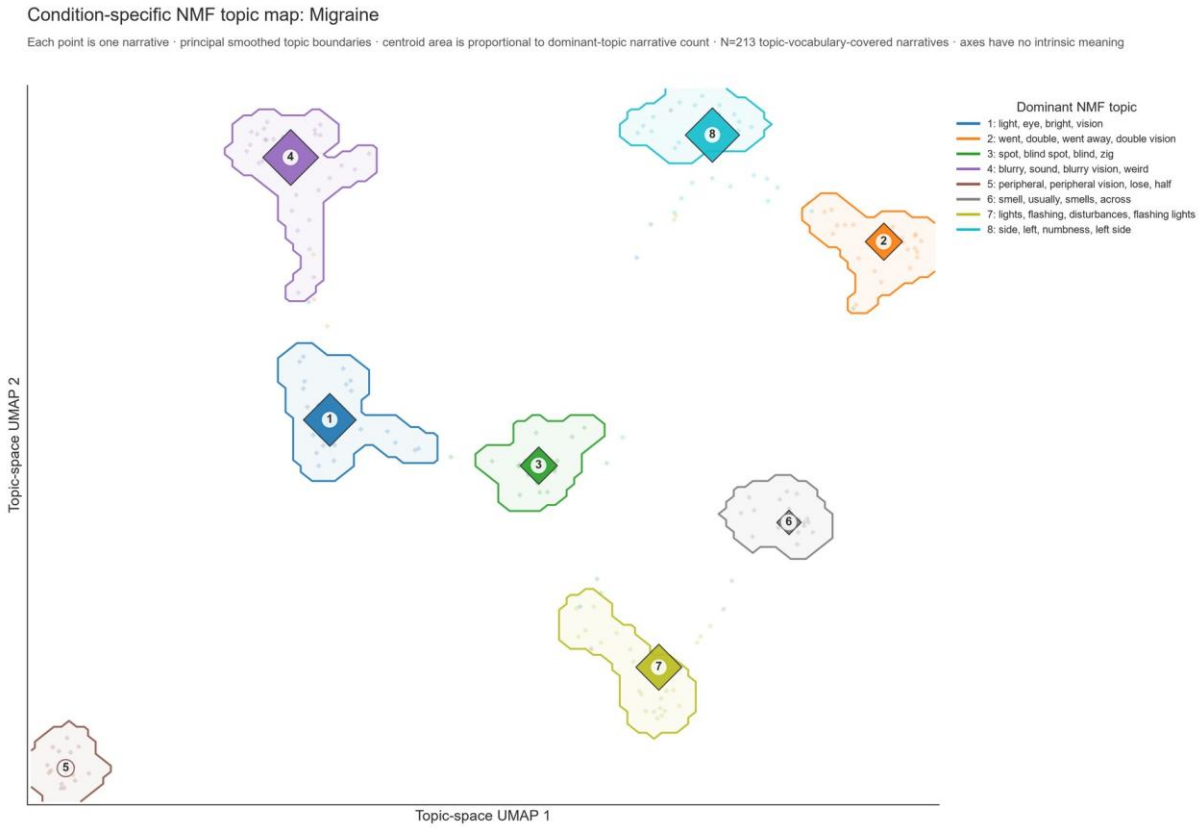
UMAP of narrative topic weights from the independent eight-topic Epilepsy model. Each point is one narrative; axes have no intrinsic meaning.

Supplementary Figure S15Q



Supplementary Figure S15Q. *Condition-specific NMF topic map for Hallucinogen persisting perception disorder*
UMAP of narrative topic weights from the independent eight-topic Hallucinogen persisting perception disorder model. Each point is one narrative; axes have no intrinsic meaning.

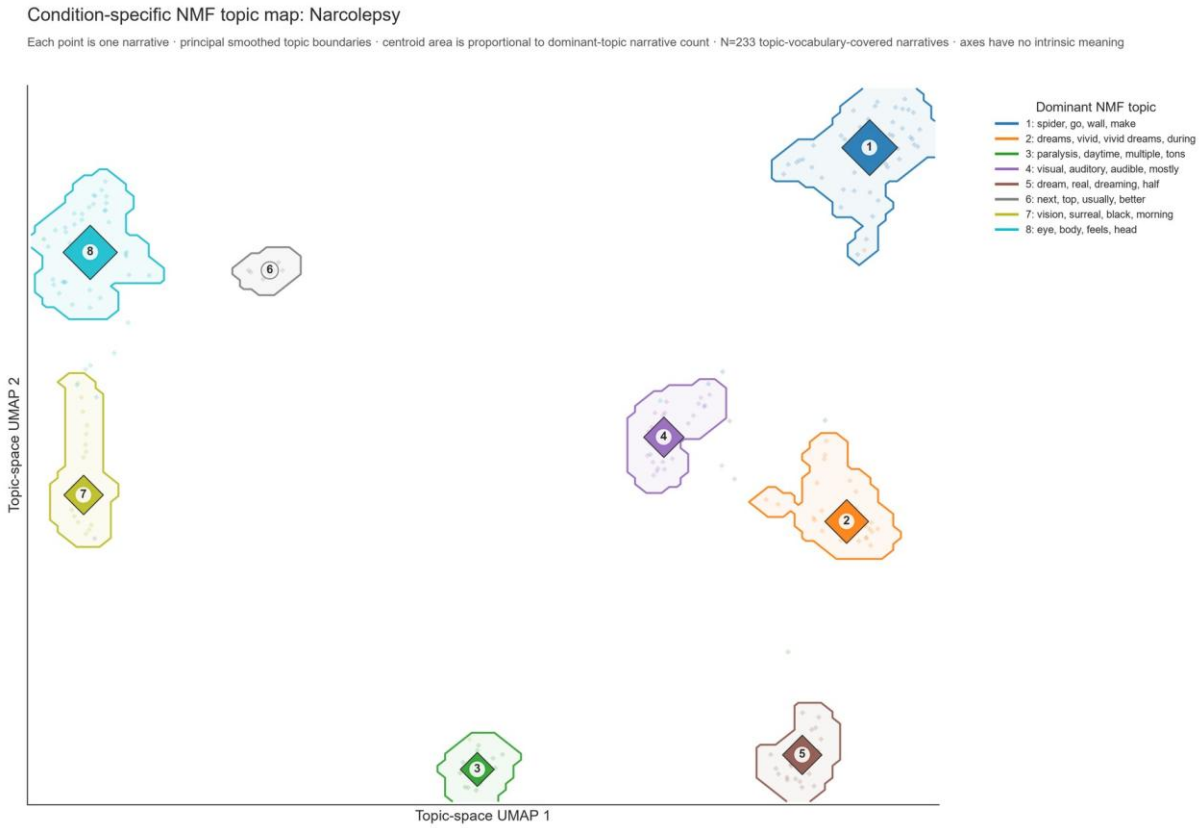
Supplementary Figure S15R



Supplementary Figure S15R. *Condition-specific NMF topic map for Migraine*

UMAP of narrative topic weights from the independent eight-topic Migraine model. Each point is one narrative; axes have no intrinsic meaning.

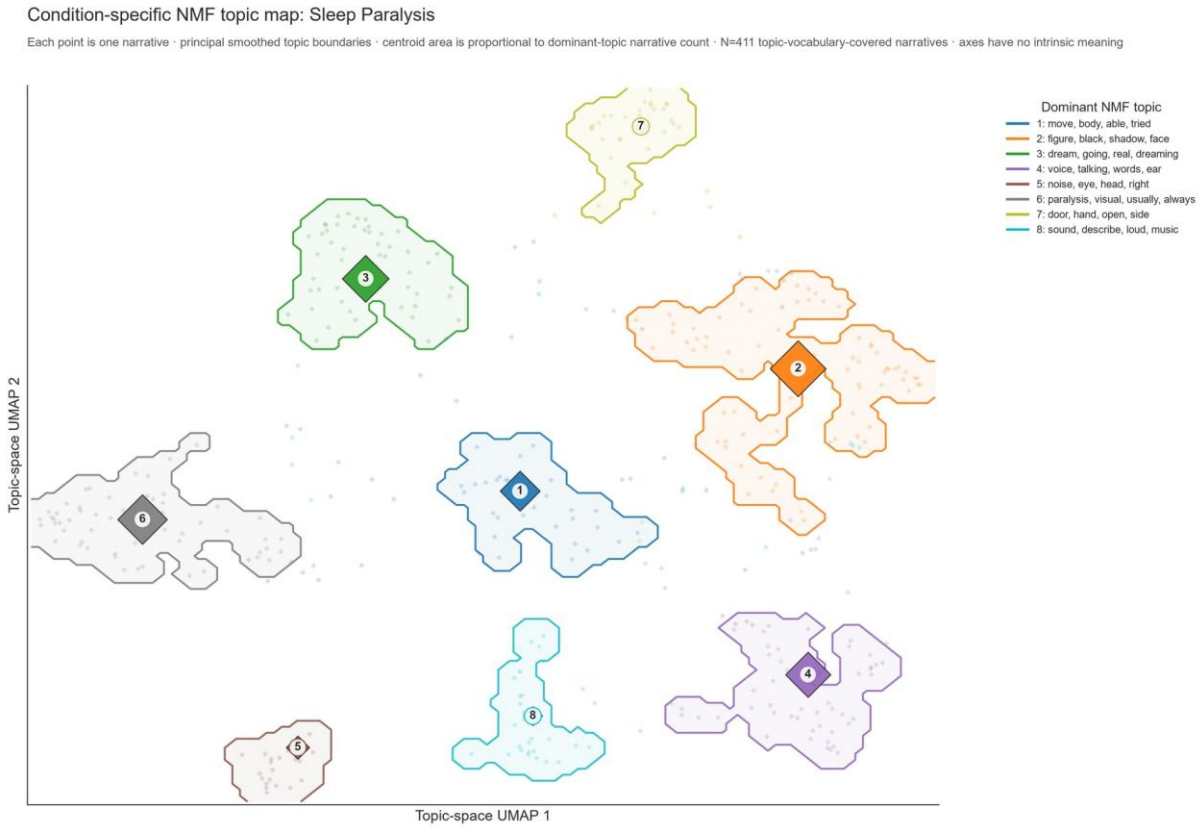
Supplementary Figure S15S



Supplementary Figure S15S. *Condition-specific NMF topic map for Narcolepsy*

UMAP of narrative topic weights from the independent eight-topic Narcolepsy model. Each point is one narrative; axes have no intrinsic meaning.

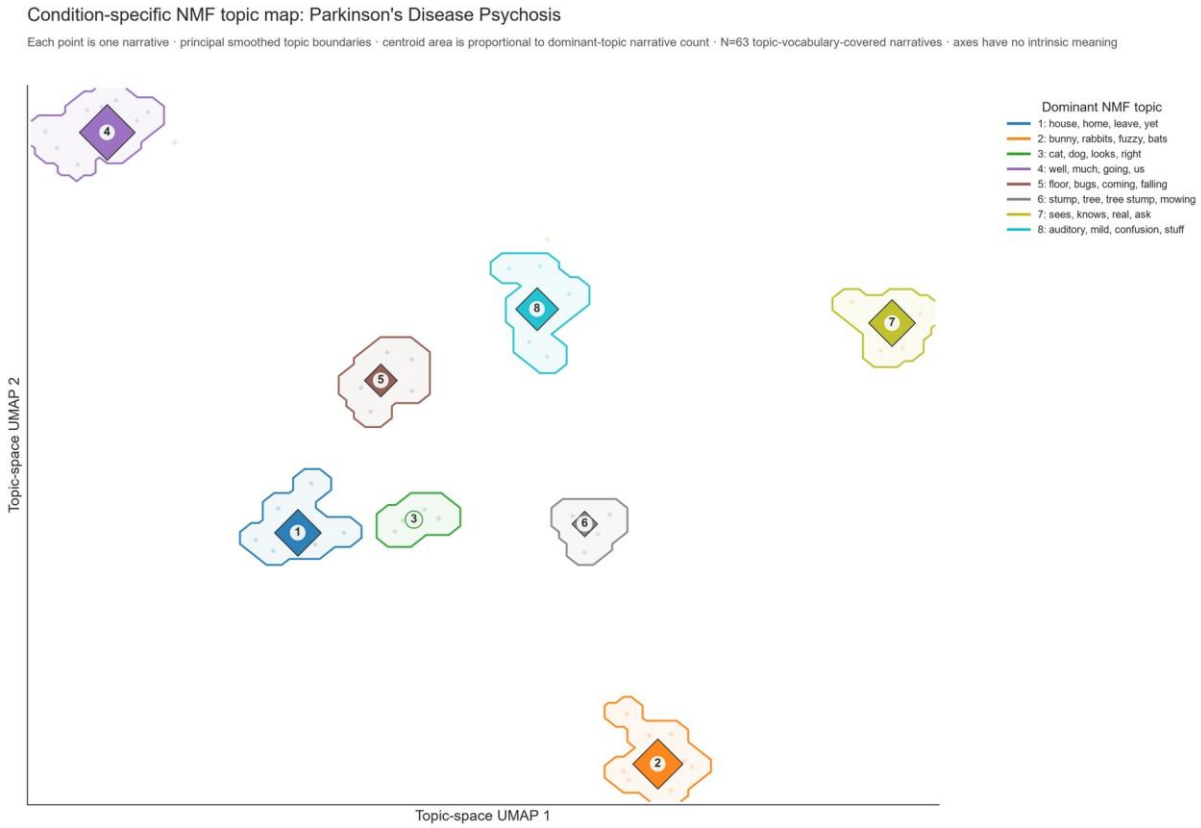
Supplementary Figure S15T



Supplementary Figure S15T. *Condition-specific NMF topic map for Sleep paralysis*

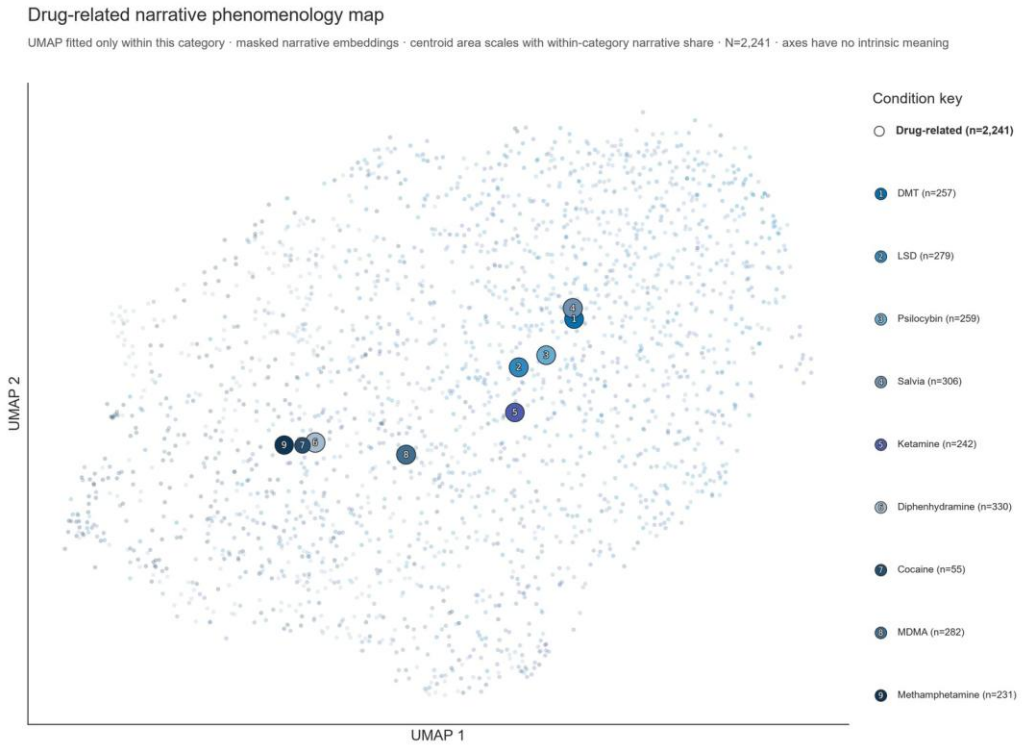
UMAP of narrative topic weights from the independent eight-topic Sleep paralysis model. Each point is one narrative; axes have no intrinsic meaning.

Supplementary Figure S15U



Supplementary Figure S15U. *Condition-specific NMF topic map for Parkinson's disease psychosis*
UMAP of narrative topic weights from the independent eight-topic Parkinson's disease psychosis model. Each point is one narrative; axes have no intrinsic meaning.

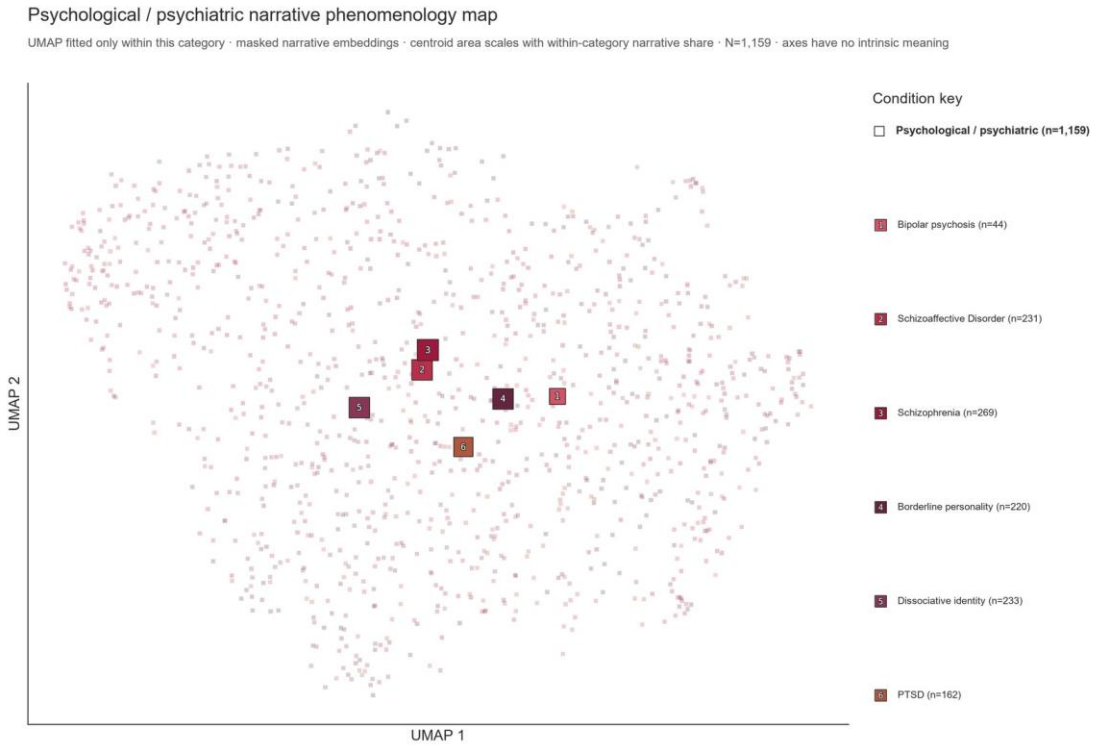
Supplementary Figure S16A



Supplementary Figure S16A. *Drug-related narrative UMAP*

Narrative-level UMAP restricted to drug-related conditions. Colours distinguish the predeclared drug families; axes have no intrinsic meaning.

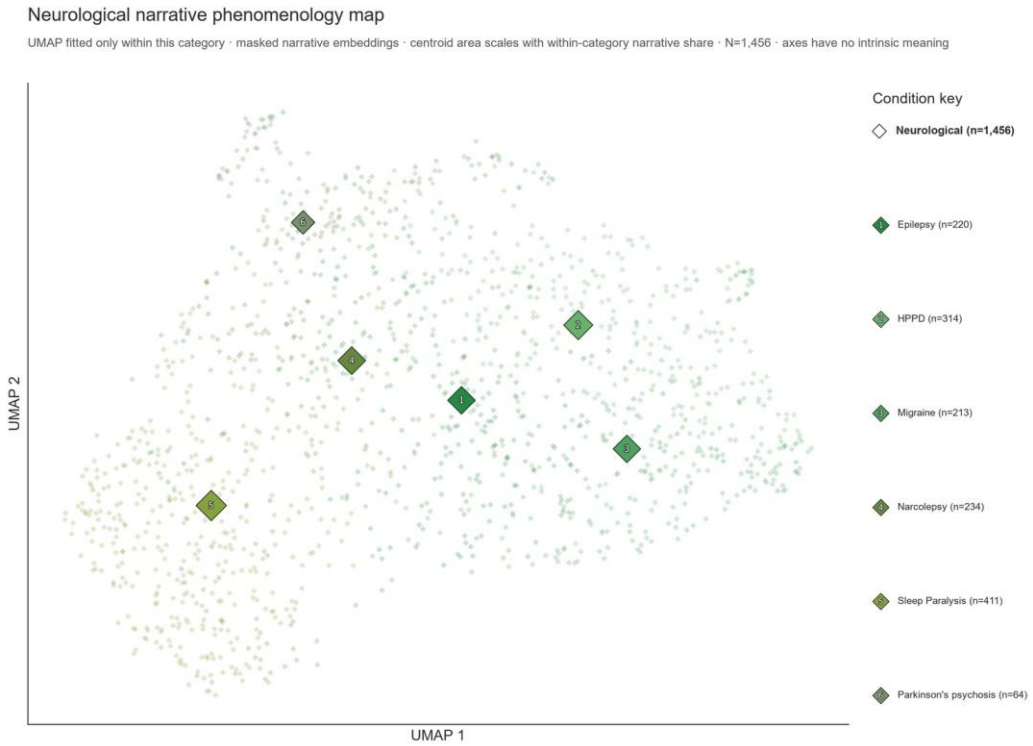
Supplementary Figure S16B



Supplementary Figure S16B. *Psychiatric narrative UMAP*

Narrative-level UMAP restricted to psychiatric conditions. Colours distinguish the predeclared psychiatric families; axes have no intrinsic meaning.

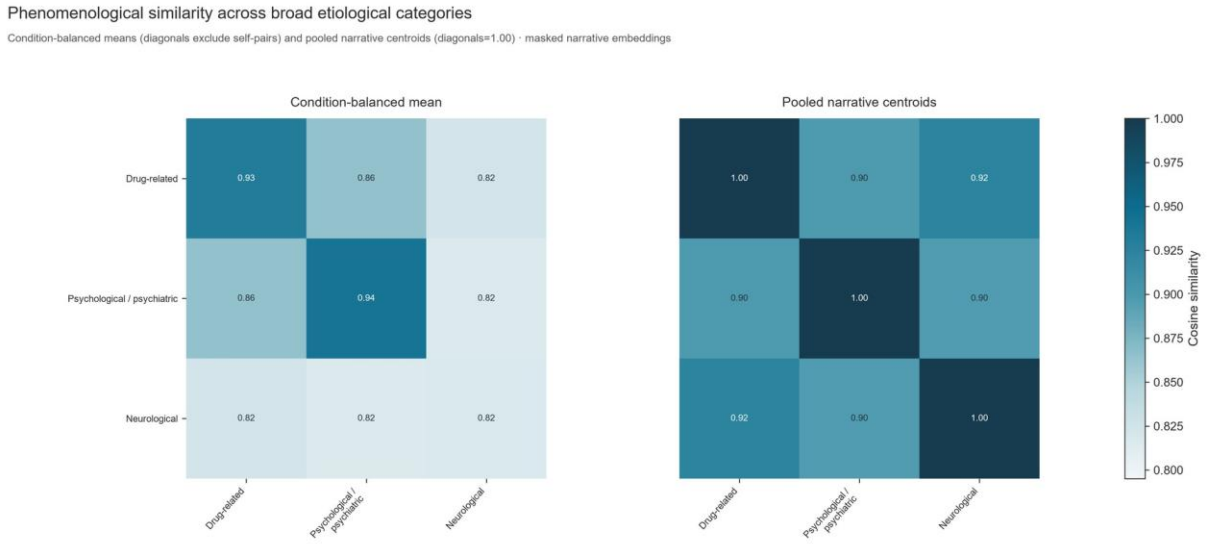
Supplementary Figure S16C



Supplementary Figure S16C. *Neurological narrative UMAP*

Narrative-level UMAP restricted to neurological conditions. Colours distinguish the predeclared neurological families; axes have no intrinsic meaning.

Supplementary Figure S17



Supplementary Figure S17. *Broad-category cosine similarity*
Condition-balanced cosine similarity is summarized for the three broad-category pairings in the primary and sensitivity corpora.

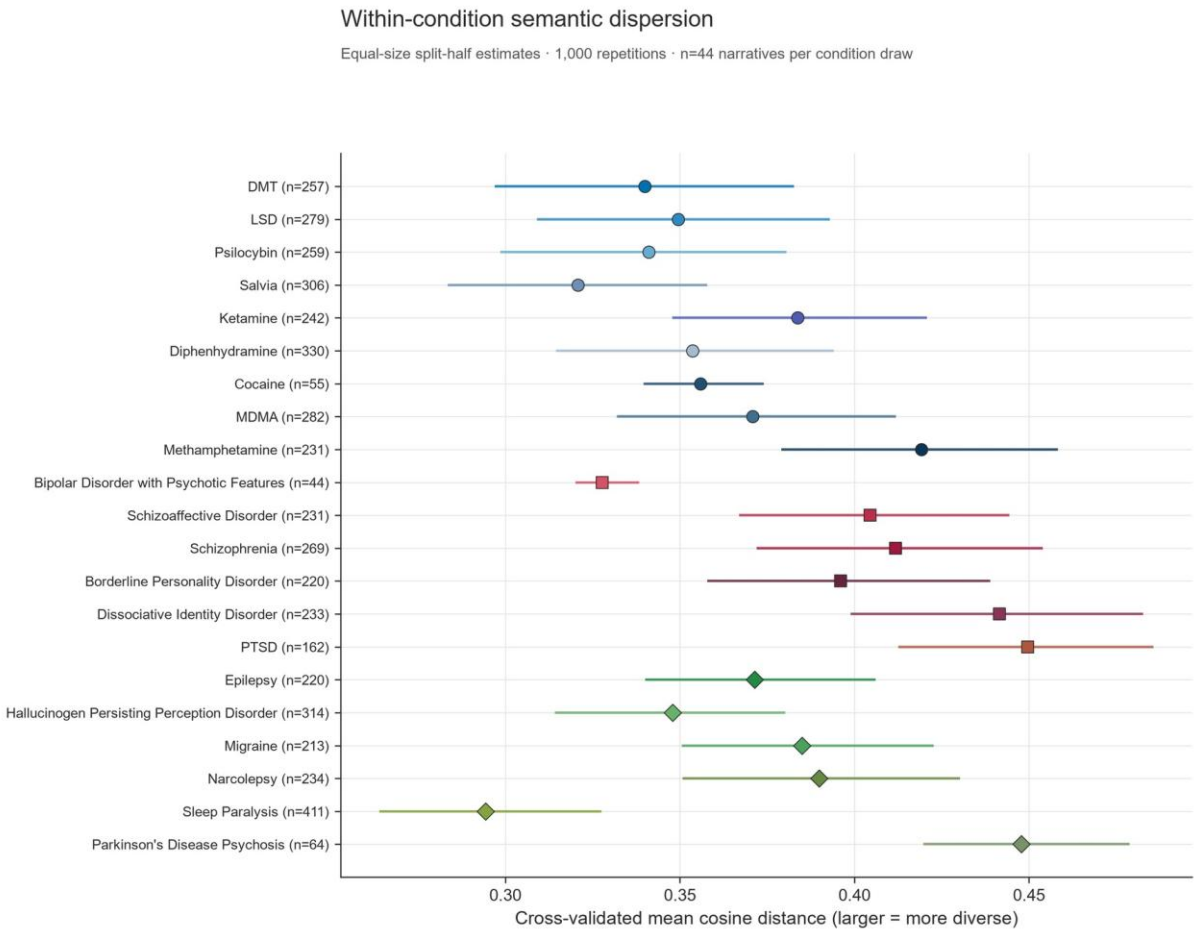
Supplementary Figure S18



Supplementary Figure S18. *Condition-family cosine similarity*

Condition-balanced cosine similarity is summarized for the predeclared condition-family pairings. Estimates describe the analysed corpus and do not establish biological equivalence.

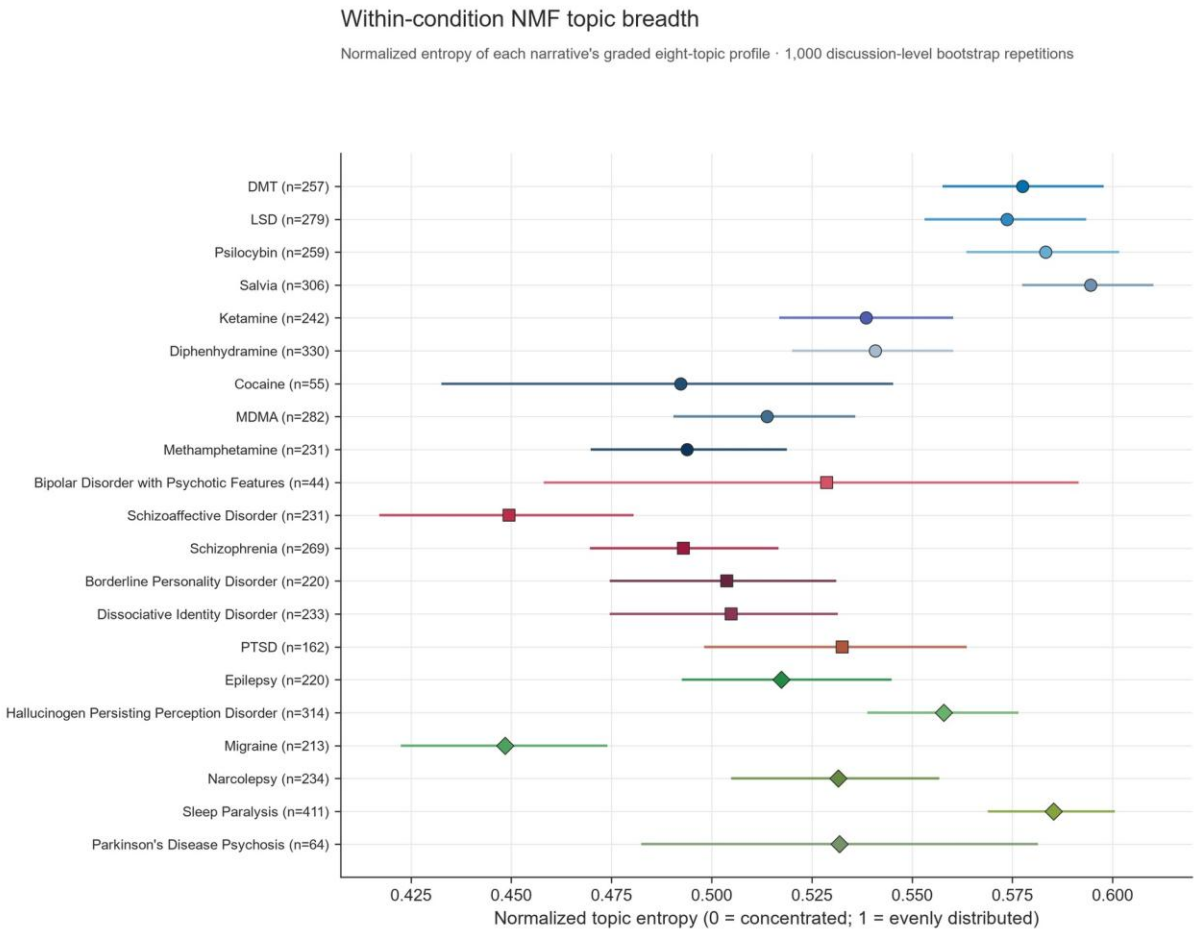
Supplementary Figure S19



Supplementary Figure S19. *Within-condition semantic dispersion*

Equal-size split-half cosine distances summarize within-condition narrative diversity. Larger values indicate greater semantic dispersion; intervals show resampling stability.

Supplementary Figure S20



Supplementary Figure S20. *Within-condition NMF topic breadth*

Normalized entropy summarizes the breadth of each condition's graded eight-topic narrative profiles. Larger values indicate more even topic mixtures, not emotional valence.

Supplementary Tables

Each entry below uses the table number cited in the thesis. The exact complete tables are included as CSV files in the accompanying tables folder. Display extracts preserve the original values but may omit additional rows or diagnostic columns for readability.

Supplementary Table S1

Supplementary Table S1. *Primary broad-category discriminative terms*

Positive masked TF-IDF coefficients from the descriptive full-corpus classifier; larger values indicate stronger association with that category, not causal importance.

Complete machine-readable table: tables/table_s01_primary_broad_category_discriminative_terms.csv (75 rows; 6 columns).

Category Name	Category n	Rank	Term	Coefficient
Drug-related	2241	1	visuals	2.56
Drug-related	2241	2	reality	2.069
Drug-related	2241	3	felt	1.886
Drug-related	2241	4	music	1.712
Drug-related	2241	5	looked	1.629
Drug-related	2241	6	space	1.282
Drug-related	2241	7	hours	1.277
Drug-related	2241	8	spiders	1.206
Drug-related	2241	9	began	1.195
Drug-related	2241	10	amazing	1.18

Display extract shown above; 65 additional rows are retained in the complete accompanying CSV.

Supplementary Table S2

Supplementary Table S2. *Sensitivity broad-category discriminative terms*

Positive masked TF-IDF coefficients from the descriptive full-corpus classifier; larger values indicate stronger association with that category, not causal importance.

Complete machine-readable table: tables/table_s02_sensitivity_broad_category_discriminative_terms.csv (75 rows; 6 columns).

Category Name	Category n	Rank	Term	Coefficient
Drug-related	3159	1	visuals	2.891
Drug-related	3159	2	reality	2.196
Drug-related	3159	3	felt	1.931
Drug-related	3159	4	looked	1.773
Drug-related	3159	5	music	1.658
Drug-related	3159	6	space	1.515
Drug-related	3159	7	hours	1.426
Drug-related	3159	8	shadow people	1.33
Drug-related	3159	9	amazing	1.328
Drug-related	3159	10	ego	1.285

Display extract shown above; 65 additional rows are retained in the complete accompanying CSV.

Supplementary Table S3

Supplementary Table S3. *Primary condition-level lexical feature summary*

A readable top-term summary across mean TF-IDF, smoothed log-odds and chi-square rankings.

Complete machine-readable table: tables/table_s03_primary_condition_feature_summary.csv (21 rows; 5 columns).

Condition	Post Count	Top TF-IDF Terms	Top Log Odds Terms	Top Chi Square Terms
Bipolar Disorder with Psychotic Features	44	voices, not, hear, corner of, i hear, corner, aren t, aren, head, etc, hear voices, see	person who, convinced me, illusions, lightly, rocking, worse than, s normal, s moving, am currently, someones, palms, crawling up	illusions, lightly, person who, convinced me, robot, numbers, worse than, s normal, s moving, someones, am currently, rocking
Borderline Personality Disorder	220	not, hear, hearing, voices, auditory, feel, sometimes, auditory hallucinations, do not, see, seeing, i hear	emptiness, feel bugs, child i, of dissociation, amd, true i, stressed, dissociating, voices at, the name, another voice, recognise	hearing things, stressed, seeing things, dissociating, emptiness, keep seeing, hearing, child i, feel bugs, of dissociation, amd, auditory
Cocaine	55	not, hearing, thought, seeing, hearing things, i thought, talking, numb, seeing things, hear, too long, door	wardrobe, garbage, plate, was said, bag, and stay, audible hallucinations, heard was, for days, or seeing, twice and, leave the	bag, wardrobe, garbage, was said, plate, for days, y, hiding, pretty sure, heard was, and stay, audible hallucinations
DMT	257	felt, not, i felt, eyes, body, felt like, vision, see, saw, no, time, my eyes	geometries, crown, stay with, these beings, swirled, loaded, hyperspace, toke, jester, these shapes, of love, observe	entity, entities, fractals, hyperspace, shapes, felt, energy, this entity, these beings, jester, i felt, the entity
Diphenhydramine	330	not, spiders, see, saw, auditory, time, seeing, even, talking, auditory hallucinations, feel, music	hatman, and spiders, spiders were, of spiders, to pee, seeing spiders, dry mouth, spiders crawling, spiders but, see spiders, piss, spiders and	spiders, pm, heavy, spider, spiders and, my wall, crawling, the spiders, dry, spiders crawling, phone, hatman
Dissociative Identity Disorder	233	not, voices, voice, hear, my head, head, feel, do not, time, sometimes, felt, body	jade, inner world, dissociated and, am tired, the inner, lt, do hear, named, their voice, not shake, hate it, british	inner world, jade, the inner, i hate, voice, inner, i want, the body, hate, dissociated, am tired, dissociated and
Epilepsy	220	vision, not, my vision, feeling, vu, sometimes, feel, deja, deja vu, body, head, my head	dejavu, j vu, d j, vu, deja, tingling in, deja vu, the memory, of dread, of deja, involuntarily, convulsing	vu, deja, deja vu, dejavu, of dread, tingling, j vu, d j, of deja, followed by, memory, vision
Hallucinogen Persisting Perception Disorder	314	vision, snow, visual, visual snow, not, my vision, see, visuals, time, sometimes, floaters, anxiety	around lights, ghosting, afterimages, halos around, halos, after images, snow and, visual snow, mold, floaters, glare, snow is	visual snow, snow, floaters, visual, halos, after images, tinnitus, afterimages, tracers, anxiety, vision, snow and
Ketamine	242	not, felt, feel, vision, body, felt like, visuals, time, hole, eyes, one, feeling	hole it, cupboard, snorting, wobbly, the hole, wonky, points of, lazy, universes, feel kind, slowing down, miles away	hole, the hole, holes, hole it, wonky, wobbly, snorting, cupboard, points, landscapes, of body, universes
LSD	279	felt, not, visuals, time, see, vision, felt like, i	face started, her skin, tinted, the phrase, about	visuals, colors, felt, face started, cats, looked,

Condition	Post Count	Top TF-IDF Terms	Top Log Odds Terms	Top Chi Square Terms
		felt, body, saw, looked, feel	seeing, visuals had, melting into, wires, switched to, smashed, around us, had gotten	began to, stars, began, geometry, felt like, stars and
MDMA	282	not, felt, saw, see, vision, seeing, looked, time, weird, i saw, eyes, hallucinating	the euphoria, my mates, smooth and, mates, tingly and, carry, the crowd, people standing, then turned, cone, clenching, remember getting	euphoria, the euphoria, my mates, mates, sweating, hallucinating, was hallucinating, everytime i, glasses, bush, everytime, s faces
Methamphetamine	231	not, shadow, hear, voices, see, shadow people, day, hearing, seeing, real, even, still	shadow ppl, piles, hear music, lyrics, of day, right outside, neighbors, am already, points in, laundry, dart, deprivation	shadow, shadow people, shadow ppl, the shadow, slept, hear music, hear, neighbors, day, lyrics, piles, days
Migraine	213	vision, my vision, not, light, spots, numbness, visual, pain, vision and, see, blind, vision in	zig, blind spots, zag, zig zag, scintillating, numbness in, both eyes, blind spot, went numb, shimmering, spots in, numbness	spots, vision, numbness, blind, zig, vision in, my vision, numbness in, blind spots, blind spot, spots in, zig zag
Narcolepsy	234	not, dreams, feel, dream, paralysis, sometimes, falling, time, real, often, even, auditory	dreams of, dreams are, vivid dreams, meow, rage, collapse, hanging from, am dreaming, the dreams, cause me, fluorescent, sweaty	dreams, vivid dreams, dream, dreams of, falling, dreams are, paralysis, my dreams, dreams and, fog, often, dreaming
PTSD	162	not, flashbacks, voice, feel, sometimes, i feel, hear, hearing, see, time, voices, can not	flashbacks and, disassociating, mindset, feel what, months of, no no, terrors, flashbacks, non stop, paranormal, nightmares and, a corner	flashbacks, no no, nightmares, flashbacks and, smell, terrors, flashback, feel what, disassociating, mindset, months of, paranormal
Parkinson's Disease Psychosis	64	house, the house, not, bunny, seeing, thinks, bats, rabbits, rabbit, sees, knows, fuzzy	our house, rabbits, bunny, he thinks, bats, rabbit, he sees, thinks, she says, thump, he knows, tail	thinks, bunny, rabbits, the house, our house, bats, rabbit, he thinks, he sees, sees, house, he knows
Psilocybin	259	felt, not, felt like, visuals, i felt, looked, saw, time, vision, i saw, eyes, feel	expanded, oneness, the said, be described, aztec, of oneness, connect, my conscious, the dimensions, dissolution, a warm, the moon	felt, felt like, patterns, looked, i felt, visuals, beautiful, i saw, seemed, saw, colors, dimensions
Salvia	306	felt, body, not, reality, felt like, i felt, my body, time, feeling, feel, vision, one	of x, x and, if one, a page, gravity and, zipped, crunch, the zipper, zipper, sensed, book and, of gravity	reality, felt, body, book, zipper, gravity, wheel, i felt, my body, felt like, a book, space
Schizoaffective Disorder	231	voices, the voices, not, voice, hear, do not, auditory, sometimes, thoughts, feel, head, one	commentary, lucifer, narrator, voices do, are often, thoughts or, train of, peoples, voices are, keep telling, singer, home in	voices, the voices, commentary, my voices, voices are, lucifer, peoples, tired of, narrator, voices that, voices have, echo
Schizophrenia	269	voices, not, the voices, hear, voice, hearing, sometimes, head, one, my head, feel, talking	internal voices, bothering, bothering me, voices have, voices all, they seem, has made, easier, might hear, consistently, harm,	voices, the voices, internal voices, my voices, voices have, voices i, hear them, been hearing, bothering me, bothering, internal, kill

Condition	Post Count	Top TF-IDF Terms	Top Log Odds Terms	Top Chi Square Terms
			voices do	
Sleep Paralysis	411	move, not, paralysis, could not, felt, not move, eyes, body, to move, see, my eyes, hear	had paralysis, having paralysis, scream but, in paralysis, experiencing paralysis, move again, hag, the visitor, not open, the paralysis, paralysed, to scream	paralysis, move, not move, to move, could not, scream, move my, move i, tried, paralyzed, paralysis i, to scream

Supplementary Table S4

Supplementary Table S4. *Primary condition-level mean TF-IDF terms*

Terms with the largest mean TF-IDF within each condition; characteristic does not necessarily mean uniquely discriminative.

Complete machine-readable table: tables/table_s04_primary_condition_mean_tfidf_terms.csv (1,050 rows; 5 columns).

Condition	Rank	Term	Mean TF-IDF	Post Count
Bipolar Disorder with Psychotic Features	1	voices	0.0348	44
Bipolar Disorder with Psychotic Features	2	not	0.0269	44
Bipolar Disorder with Psychotic Features	3	hear	0.026	44
Bipolar Disorder with Psychotic Features	4	corner of	0.0213	44
Bipolar Disorder with Psychotic Features	5	i hear	0.021	44
Bipolar Disorder with Psychotic Features	6	corner	0.0203	44
Bipolar Disorder with Psychotic Features	7	aren t	0.0201	44
Bipolar Disorder with Psychotic Features	8	aren	0.0201	44
Bipolar Disorder with Psychotic Features	9	head	0.0185	44
Bipolar Disorder with Psychotic Features	10	etc	0.0184	44

Display extract shown above; 1,040 additional rows are retained in the complete accompanying CSV.

Supplementary Table S5

Supplementary Table S5. *Primary condition-level overrepresented terms*

Smoothed log-odds terms enriched in each condition relative to all other conditions.

Complete machine-readable table: tables/table_s05_primary_condition_overrepresented_log_odds_terms.csv (1,050 rows; 9 columns).

Condition	Rank	Term	Log Odds	Condition Count	Rest Count
Bipolar Disorder with Psychotic Features	1	person who	4.259	2	5
Bipolar Disorder with Psychotic Features	2	convinced me	4.259	2	5
Bipolar Disorder with Psychotic Features	3	illusions	4.258	4	10
Bipolar Disorder with Psychotic Features	4	lightly	4.194	3	8
Bipolar Disorder with Psychotic Features	5	rocking	4.077	2	6
Bipolar Disorder with Psychotic Features	6	worse than	4.077	2	6
Bipolar Disorder with Psychotic Features	7	s normal	4.077	2	6
Bipolar Disorder with Psychotic Features	8	s moving	4.077	2	6
Bipolar Disorder with Psychotic Features	9	am currently	4.077	2	6
Bipolar Disorder with Psychotic Features	10	someones	4.077	2	6

Display extract shown above; 1,040 additional rows are retained in the complete accompanying CSV.

Supplementary Table S6

Supplementary Table S6. *Primary condition-level chi-square terms*

Terms whose document-frequency pattern differs most for one condition versus the remainder; exploratory rather than confirmatory inference.

Complete machine-readable table: tables/table_s06_primary_condition_chi_square_terms.csv (1,050 rows; 10 columns).

Condition	Rank	Term	Chi2	P Value	Condition Rate	Rest Rate
Bipolar Disorder with Psychotic Features	1	illusions	119.3	0	0.002	0
Bipolar Disorder with Psychotic Features	2	lightly	85.169	0	0.0015	0
Bipolar Disorder with Psychotic Features	3	person who	59.669	0	0.001	0
Bipolar Disorder with Psychotic Features	4	convinced me	59.669	0	0.001	0
Bipolar Disorder with Psychotic Features	5	robot	58.172	0	0.002	0.0001
Bipolar Disorder with Psychotic Features	6	numbers	56.739	0	0.0015	0
Bipolar Disorder with Psychotic Features	7	worse than	51.723	0	0.001	0
Bipolar Disorder with Psychotic Features	8	s normal	51.723	0	0.001	0
Bipolar Disorder with Psychotic Features	9	s moving	51.723	0	0.001	0
Bipolar Disorder with Psychotic Features	10	someones	51.723	0	0.001	0

Display extract shown above; 1,040 additional rows are retained in the complete accompanying CSV.

Supplementary Table S7

Supplementary Table S7. *Primary residual contextual-term audit*

Known contextual or label-related vocabulary still appearing among high-ranked features and requiring interpretation or masking review.

Complete machine-readable table: tables/table_s07_primary_residual_contextual_term_audit.csv (8 rows; 7 columns).

Condition	Rank	Term	Mean TF-IDF	Matched Confound Terms
DMT	34	did not	0.0125	did
Diphenhydramine	22	did not	0.0129	did
Dissociative Identity Disorder	49	did not	0.0087	did
Ketamine	49	did not	0.0089	did
LSD	43	did not	0.011	did
MDMA	19	did not	0.012	did
Psilocybin	22	did not	0.0143	did
Salvia	22	did not	0.0139	did

Supplementary Table S8

Supplementary Table S8. *Primary narrative-level NMF topic definitions*

Ranked words defining each graded NMF topic. Labels are descriptive top-word summaries, not manually validated clinical names.

Complete machine-readable table: tables/table_s08_primary_narrative_nmf_topic_definitions.csv (160 rows; 6 columns).

Topic	Label	Ranked terms
1	NMF topic 1: face / went / kept / sound / off	face, went, kept, sound, off, go, going, music, little, talking, color, another, point, right, away, weird, made, way, every, next
2	NMF topic 2: voice / head / talk / talking / telling	voice, head, talk, talking, telling, sound, thoughts, day, loud, name, kill, told, keep, aren, voice telling, always, bad, make, want, hard
3	NMF topic 3: visual / auditory / visual snow / snow / intense	visual, auditory, visual snow, snow, intense, pattern, music, much, mild, made, little, intense visual, mostly, anxiety, since, pretty, tactile, audio, stuff, slight
4	NMF topic 4: vision / blurry / spot / light / peripheral	vision, blurry, spot, light, peripheral, peripheral vision, weird, snow, feels, double, focus, static, double vision, visual snow, lights, blurry vision, noticed, day, blind, field
5	NMF topic 5: shadow / spider / crawling / bugs / figure	shadow, spider, crawling, bugs, figure, corner, door, day, walls, wall, moving, stuff, usually, always, spider crawling, window, small, shadow figure, name, walking
6	NMF topic 6: eye / closed / open / corner / closed eye	eye, closed, open, corner, closed eye, close, eye closed, light, eye visual, opened, black, pattern, white, face, visions, color, focus, blue, bright, mind
7	NMF topic 7: body / move / paralysis / tried / feels	body, move, paralysis, tried, feels, able, head, unable, control, suddenly, sensation, scream, presence, try, heavy, dream, moving, legs, chest, shaking
8	NMF topic 8: real / reality / life / world / dream	real, reality, life, world, dream, dreams, mind, different, sense, thoughts, feels, vivid, completely, though, almost, real life, happening, way, person, ever

Supplementary Table S9

Supplementary Table S9. *Primary dominant narrative-level NMF topic by condition*

The highest average NMF topic weight for each condition; topics remain graded and conditions can load on several topics.

Complete machine-readable table: tables/table_s09_primary_condition_dominant_narrative_topics.csv (21 rows; 6 columns).

Condition	Post Count	Top NMF Topic	Top NMF Topic Label	Top NMF Topic Mean Weight	Top NMF Topic Terms
DMT	257	1	NMF topic 1: face / went / kept / sound / off	0.0245	face, went, kept, sound, off, go, going, music
LSD	279	1	NMF topic 1: face / went / kept / sound / off	0.0275	face, went, kept, sound, off, go, going, music
MDMA	282	1	NMF topic 1: face / went / kept / sound / off	0.0259	face, went, kept, sound, off, go, going, music
Parkinson's Disease Psychosis	64	1	NMF topic 1: face / went / kept / sound / off	0.0166	face, went, kept, sound, off, go, going, music
Psilocybin	259	1	NMF topic 1: face / went / kept / sound / off	0.0292	face, went, kept, sound, off, go, going, music
Bipolar Disorder with Psychotic Features	44	2	NMF topic 2: voice / head / talk / talking / telling	0.0284	voice, head, talk, talking, telling, sound, thoughts, day
Borderline Personality Disorder	220	2	NMF topic 2: voice / head / talk / talking / telling	0.0231	voice, head, talk, talking, telling, sound, thoughts, day
Dissociative Identity Disorder	233	2	NMF topic 2: voice / head / talk / talking / telling	0.0414	voice, head, talk, talking, telling, sound, thoughts, day
PTSD	162	2	NMF topic 2: voice / head / talk / talking / telling	0.0251	voice, head, talk, talking, telling, sound, thoughts, day
Schizoaffective Disorder	231	2	NMF topic 2: voice / head / talk / talking / telling	0.0497	voice, head, talk, talking, telling, sound, thoughts, day
Schizophrenia	269	2	NMF topic 2: voice / head / talk / talking / telling	0.0465	voice, head, talk, talking, telling, sound, thoughts, day
Epilepsy	220	4	NMF topic 4: vision / blurry / spot / light / peripheral	0.0357	vision, blurry, spot, light, peripheral, peripheral vision, weird, snow
Hallucinogen Persisting Perception Disorder	314	4	NMF topic 4: vision / blurry / spot / light / peripheral	0.0445	vision, blurry, spot, light, peripheral, peripheral vision, weird, snow
Migraine	213	4	NMF topic 4: vision / blurry / spot / light / peripheral	0.0686	vision, blurry, spot, light, peripheral, peripheral vision, weird, snow

Condition	Post Count	Top NMF Topic	Top NMF Topic Label	Top NMF Topic Mean Weight	Top NMF Topic Terms
Cocaine	55	5	NMF topic 5: shadow / spider / crawling / bugs / figure	0.0197	shadow, spider, crawling, bugs, figure, corner, door, day
Diphenhydramine	330	5	NMF topic 5: shadow / spider / crawling / bugs / figure	0.0373	shadow, spider, crawling, bugs, figure, corner, door, day
Methamphetamine	231	5	NMF topic 5: shadow / spider / crawling / bugs / figure	0.0321	shadow, spider, crawling, bugs, figure, corner, door, day
Sleep Paralysis	411	7	NMF topic 7: body / move / paralysis / tried / feels	0.054	body, move, paralysis, tried, feels, able, head, unable
Ketamine	242	8	NMF topic 8: real / reality / life / world / dream	0.0222	real, reality, life, world, dream, dreams, mind, different
Narcolepsy	234	8	NMF topic 8: real / reality / life / world / dream	0.0241	real, reality, life, world, dream, dreams, mind, different
Salvia	306	8	NMF topic 8: real / reality / life / world / dream	0.031	real, reality, life, world, dream, dreams, mind, different

Supplementary Table S10

Supplementary Table S10. *Primary mean narrative-level NMF topic weights by condition*

Average graded topic weights for every condition, with top-word descriptions embedded in the column names.

Complete machine-readable table: tables/table_s10_primary_condition_mean_nmf_topic_weights.csv (21 rows; 10 columns).

Condition	NMF Topic 1 Face / Went / Kept / Sound / Off	NMF Topic 2 Voice / Head / Talk / Talking / Telling	NMF Topic 3 Visual / Auditory / Visual Snow / Snow / Intense	NMF Topic 4 Vision / Blurry / Spot / Light / Peripheral	NMF Topic 5 Shadow / Spider / Crawling / Bugs / Figure	NMF Topic 6 Eye / Closed / Open / Corner / Closed Eye
Bipolar Disorder with Psychotic Features	0.0102	0.0284	0.0138	0.0071	0.0223	0.0113
Borderline Personality Disorder	0.0128	0.0231	0.0199	0.0045	0.0208	0.006
Cocaine	0.0193	0.0111	0.0091	0.0113	0.0197	0.0118
DMT	0.0245	0.0121	0.0101	0.0146	0.0027	0.0242
Diphenhydramine	0.0225	0.0093	0.0186	0.0122	0.0373	0.013
Dissociative Identity Disorder	0.0143	0.0414	0.0032	0.0071	0.0039	0.0047
Epilepsy	0.0135	0.0109	0.0076	0.0357	0.004	0.0091
Hallucinogen Persisting Perception Disorder	0.0128	0.0047	0.0421	0.0445	0.0083	0.0176
Ketamine	0.0208	0.0068	0.0143	0.0198	0.0023	0.0199
LSD	0.0275	0.0077	0.0216	0.0153	0.0031	0.0168
MDMA	0.0259	0.0064	0.0114	0.0165	0.0097	0.0152
Methamphetamine	0.0173	0.0173	0.0141	0.0088	0.0321	0.0077
Migraine	0.0114	0.0034	0.0114	0.0686	0.0034	0.0171
Narcolepsy	0.0123	0.0067	0.0124	0.009	0.0161	0.0113
PTSD	0.0118	0.0251	0.0081	0.0115	0.0114	0.0069
Parkinson's Disease Psychosis	0.0166	0.0036	0.0088	0.0048	0.011	0.0031
Psilocybin	0.0292	0.0085	0.0176	0.0151	0.0041	0.0226
Salvia	0.0259	0.0107	0.0078	0.013	0.0026	0.0136
Schizoaffective Disorder	0.0109	0.0497	0.0117	0.0043	0.0101	0.0029
Schizophrenia	0.014	0.0465	0.0103	0.0035	0.0114	0.0057
Sleep Paralysis	0.0217	0.0163	0.0047	0.0073	0.0168	0.0217

Supplementary Table S11

Supplementary Table S11. *Primary narrative-level LSA component terms*

Signed terms defining latent semantic axes. Positive and negative directions are opposing ends of an axis, not separate topics.

Complete machine-readable table: tables/table_s11_primary_narrative_lsa_component_terms.csv (240 rows; 8 columns).

Component ID	Component Label	Rank	Term	Signed Weight	Direction
1	LSA component 1: voice / eye / vision / visual / body	1	voice	0.2319	positive
1	LSA component 1: voice / eye / vision / visual / body	2	eye	0.1939	positive
1	LSA component 1: voice / eye / vision / visual / body	3	vision	0.1844	positive
1	LSA component 1: voice / eye / vision / visual / body	4	visual	0.1761	positive
1	LSA component 1: voice / eye / vision / visual / body	5	body	0.1683	positive
1	LSA component 1: voice / eye / vision / visual / body	6	head	0.1321	positive
1	LSA component 1: voice / eye / vision / visual / body	7	real	0.1079	positive
1	LSA component 1: voice / eye / vision / visual / body	8	face	0.1031	positive
1	LSA component 1: voice / eye / vision / visual / body	9	move	0.1	positive
1	LSA component 1: voice / eye / vision / visual / body	10	reality	0.0972	positive

Display extract shown above; 230 additional rows are retained in the complete accompanying CSV.

Supplementary Table S12

Supplementary Table S12. *Primary condition-document NMF topic definitions*

Topics learned after combining narratives within each condition; these differ from narrative-level NMF topics.

Complete machine-readable table: tables/table_s12_primary_condition_document_nmf_topic_definitions.csv (200 rows; 7 columns).

Condition n	Post n	Topic ID	Topic Label	Rank	Term	Weight
21	4856	1	Condition NMF topic 1: pattern / music / entity / fractal / universe	1	pattern	0.3802
21	4856	1	Condition NMF topic 1: pattern / music / entity / fractal / universe	2	music	0.2807
21	4856	1	Condition NMF topic 1: pattern / music / entity / fractal / universe	3	entity	0.2502
21	4856	1	Condition NMF topic 1: pattern / music / entity / fractal / universe	4	fractal	0.2395
21	4856	1	Condition NMF topic 1: pattern / music / entity / fractal / universe	5	universe	0.2307
21	4856	1	Condition NMF topic 1: pattern / music / entity / fractal / universe	6	ego	0.2036
21	4856	1	Condition NMF topic 1: pattern / music / entity / fractal / universe	7	sky	0.1751
21	4856	1	Condition NMF topic 1: pattern / music / entity / fractal / universe	8	beautiful	0.164
21	4856	1	Condition NMF topic 1: pattern / music / entity / fractal / universe	9	energy	0.1613
21	4856	1	Condition NMF topic 1: pattern / music / entity / fractal / universe	10	amazing	0.1562

Display extract shown above; 190 additional rows are retained in the complete accompanying CSV.

Supplementary Table S13

Supplementary Table S13. *Primary condition-document NMF topic weights*

NMF weights for condition-level aggregate documents, including a top-word label for the largest component.

Complete machine-readable table: tables/table_s13_primary_condition_document_nmf_topic_weights.csv (21 rows; 13 columns).

Condition	Post Count	Combined Word Count	Dominant NMF Topic	Dominant NMF Topic Label
Bipolar Disorder with Psychotic Features	44	3520	7	Condition NMF topic 7: illusions / tactile / shadow / lightly / external world
Borderline Personality Disorder	220	17587	4	Condition NMF topic 4: shadow / spider / music / bugs / blurry
Cocaine	55	5676	8	Condition NMF topic 8: little person / bag / garbage bag / shadow / wardrobe
DMT	257	50240	1	Condition NMF topic 1: pattern / music / entity / fractal / universe
Diphenhydramine	330	46531	4	Condition NMF topic 4: shadow / spider / music / bugs / blurry
Dissociative Identity Disorder	233	23881	2	Condition NMF topic 2: flashbacks / inner world / inner / memories / dissociating
Epilepsy	220	18778	3	Condition NMF topic 3: visual snow / snow / floaters / numbness / blind spot
Hallucinogen Persisting Perception Disorder	314	36382	3	Condition NMF topic 3: visual snow / snow / floaters / numbness / blind spot
Ketamine	242	33083	1	Condition NMF topic 1: pattern / music / entity / fractal / universe
LSD	279	55281	1	Condition NMF topic 1: pattern / music / entity / fractal / universe
MDMA	282	31805	4	Condition NMF topic 4: shadow / spider / music / bugs / blurry
Methamphetamine	231	23494	4	Condition NMF topic 4: shadow / spider / music / bugs / blurry
Migraine	213	16940	3	Condition NMF topic 3: visual snow / snow / floaters / numbness / blind spot
Narcolepsy	234	21201	5	Condition NMF topic 5: paralysis / dreaming / spider / shadow / scream
PTSD	162	11642	2	Condition NMF topic 2: flashbacks / inner world / inner / memories /

Condition	Post Count	Combined Word Count	Dominant NMF Topic	Dominant NMF Topic Label
				dissociating
Parkinson's Disease Psychosis	64	4849	6	Condition NMF topic 6: stump / bunny / thinks / fuzzy bunny / tree stump
Psilocybin	259	54097	1	Condition NMF topic 1: pattern / music / entity / fractal / universe
Salvia	306	70780	1	Condition NMF topic 1: pattern / music / entity / fractal / universe
Schizoaffective Disorder	231	17943	4	Condition NMF topic 4: shadow / spider / music / bugs / blurry
Schizophrenia	269	26201	2	Condition NMF topic 2: flashbacks / inner world / inner / memories / dissociating
Sleep Paralysis	411	69370	5	Condition NMF topic 5: paralysis / dreaming / spider / shadow / scream

Supplementary Table S14

Supplementary Table S14. *Primary condition-document LSA component terms*

Signed terms defining latent semantic axes among condition-level aggregate documents.

Complete machine-readable table: tables/table_s14_primary_condition_document_lsa_component_terms.csv (200 rows; 9 columns).

Condition n	Post n	Component ID	Component Label	Rank	Term	Signed Weight	Direction
21	4856	1	Condition LSA component 1: shadow / music / pattern / spider / paralysis	1	shadow	0.1879	positive
21	4856	1	Condition LSA component 1: shadow / music / pattern / spider / paralysis	2	music	0.1787	positive
21	4856	1	Condition LSA component 1: shadow / music / pattern / spider / paralysis	3	pattern	0.142	positive
21	4856	1	Condition LSA component 1: shadow / music / pattern / spider / paralysis	4	spider	0.1051	positive
21	4856	1	Condition LSA component 1: shadow / music / pattern / spider / paralysis	5	paralysis	0.091	positive
21	4856	1	Condition LSA component 1: shadow / music / pattern / spider / paralysis	6	entity	0.0799	positive
21	4856	1	Condition LSA component 1: shadow / music / pattern / spider / paralysis	7	blurry	0.0768	positive

Condition n	Post n	Component ID	Component Label	Rank	Term	Signed Weight	Direction
21	4856	1	Condition LSA component 1: shadow / music / pattern / spider / paralysis	8	floating	0.0696	positive
21	4856	1	Condition LSA component 1: shadow / music / pattern / spider / paralysis	9	universe	0.0682	positive
21	4856	1	Condition LSA component 1: shadow / music / pattern / spider / paralysis	10	fractal	0.0666	positive

Display extract shown above; 190 additional rows are retained in the complete accompanying CSV.

Supplementary Table S15

Supplementary Table S15. *Primary condition-level MDS coordinates*

Plot coordinates and point stress for TF-IDF, LSA and NMF distance representations; axes have no intrinsic substantive meaning.

Complete machine-readable table: tables/table_s15_primary_condition_mds_coordinates.csv (63 rows; 7 columns).

Representation Name	Condition	Mds X	Mds Y	Point Stress	Post Count
TF-IDF lexical distance	Bipolar Disorder with Psychotic Features	0.7394	-1.187	0.4966	44
TF-IDF lexical distance	Borderline Personality Disorder	-0.1113	-0.2636	0.1205	220
TF-IDF lexical distance	Cocaine	-0.5571	-0.9093	0.539	55
TF-IDF lexical distance	DMT	0.3226	0.0724	0.0945	257
TF-IDF lexical distance	Diphenhydramine	0.0501	-0.2189	0.3079	330
TF-IDF lexical distance	Dissociative Identity Disorder	-0.3523	0.3086	0.2433	233
TF-IDF lexical distance	Epilepsy	0.0652	0.6326	0.1105	220
TF-IDF lexical distance	Hallucinogen Persisting Perception Disorder	0.7477	0.5242	0.3836	314
TF-IDF lexical distance	Ketamine	0.2044	0.1632	0.0438	242
TF-IDF lexical distance	LSD	0.2256	0.0904	0.0946	279
TF-IDF lexical distance	MDMA	0.1139	0.084	0.1988	282
TF-IDF lexical distance	Methamphetamine	-0.0114	-0.3798	0.1031	231
TF-IDF lexical distance	Migraine	0.425	0.8752	0.222	213
TF-IDF lexical distance	Narcolepsy	-0.2004	0.0881	0.2597	234
TF-IDF lexical distance	PTSD	-0.5394	-0.0285	0.3524	162
TF-IDF lexical distance	Parkinson's Disease Psychosis	-1.546	0.7426	0.4283	64
TF-IDF lexical distance	Psilocybin	0.2487	0.0885	0.0635	259
TF-IDF lexical distance	Salvia	0.264	0.0041	0.1288	306
TF-IDF lexical distance	Schizoaffective Disorder	-0.2331	-0.2279	0.2541	231
TF-IDF lexical distance	Schizophrenia	-0.1005	-0.1295	0.1559	269
TF-IDF lexical distance	Sleep Paralysis	0.2452	-0.3295	0.449	411

Representation Name	Condition	Mds X	Mds Y	Point Stress	Post Count
LSA semantic-component distance	Bipolar Disorder with Psychotic Features	-1.008	-0.7324	1.062	44
LSA semantic-component distance	Borderline Personality Disorder	-0.3358	-0.0488	0.1874	220
LSA semantic-component distance	Cocaine	-0.0295	-1.056	0.9788	55
LSA semantic-component distance	DMT	-0.065	0.59	0.1922	257
LSA semantic-component distance	Diphenhydramine	0.054	-0.0506	0.3638	330
LSA semantic-component distance	Dissociative Identity Disorder	-0.5693	0.2304	0.7678	233
LSA semantic-component distance	Epilepsy	0.5689	-0.0987	0.756	220
LSA semantic-component distance	Hallucinogen Persisting Perception Disorder	0.7108	0.535	0.6232	314
LSA semantic-component distance	Ketamine	0.1364	0.3079	0.1667	242
LSA semantic-component distance	LSD	0.1507	0.402	0.0947	279
LSA semantic-component distance	MDMA	0.1613	0.1151	0.3044	282
LSA semantic-component distance	Methamphetamine	-0.3931	-0.1973	0.5128	231
LSA semantic-component distance	Migraine	0.903	0.07	0.4209	213
LSA semantic-component distance	Narcolepsy	0.1612	-0.2686	0.8441	234
LSA semantic-component distance	PTSD	-0.23	-0.3686	0.7378	162
LSA semantic-component distance	Parkinson's Disease Psychosis	0.9397	-0.9968	1.086	64
LSA semantic-component distance	Psilocybin	0.1105	0.4664	0.1169	259
LSA semantic-component distance	Salvia	-0.0517	0.4768	0.1798	306
LSA semantic-component distance	Schizoaffective Disorder	-0.476	-0.045	0.2643	231
LSA semantic-component distance	Schizophrenia	-0.2481	0.1197	0.17	269
LSA semantic-component distance	Sleep Paralysis	-0.4901	0.55	0.7969	411
NMF topic-profile distance	Bipolar Disorder with Psychotic Features	-0.8405	-0.3043	1.268	44
NMF topic-profile distance	Borderline Personality Disorder	0.0411	-0.5865	0.183	220
NMF topic-profile	Cocaine	-0.8457	0.0442	1.233	55

Representation Name	Condition	Mds X	Mds Y	Point Stress	Post Count
distance					
NMF topic-profile distance	DMT	0.6305	0.1492	0.4674	257
NMF topic-profile distance	Diphenhydramine	0.2373	-0.4752	0.6643	330
NMF topic-profile distance	Dissociative Identity Disorder	-0.3708	-0.5375	0.7221	233
NMF topic-profile distance	Epilepsy	0.0283	0.5282	0.439	220
NMF topic-profile distance	Hallucinogen Persisting Perception Disorder	0.2198	0.7014	0.5314	314
NMF topic-profile distance	Ketamine	0.5542	0.0722	0.2369	242
NMF topic-profile distance	LSD	0.6023	0.0809	0.2473	279
NMF topic-profile distance	MDMA	0.4042	-0.1727	0.1796	282
NMF topic-profile distance	Methamphetamine	0.2701	-0.6467	0.6071	231
NMF topic-profile distance	Migraine	0.1539	0.772	0.7482	213
NMF topic-profile distance	Narcolepsy	-0.4475	0.639	0.8291	234
NMF topic-profile distance	PTSD	-0.4039	-0.536	0.6407	162
NMF topic-profile distance	Parkinson's Disease Psychosis	-0.7839	0.381	1.258	64
NMF topic-profile distance	Psilocybin	0.6309	0.0778	0.3312	259
NMF topic-profile distance	Salvia	0.5234	0.127	0.4385	306
NMF topic-profile distance	Schizoaffective Disorder	-0.0756	-0.5325	0.1085	231
NMF topic-profile distance	Schizophrenia	-0.0894	-0.422	0.0955	269
NMF topic-profile distance	Sleep Paralysis	-0.4388	0.6406	0.8561	411

Supplementary Table S16

Supplementary Table S16. *Primary category composition within dominant narrative-level NMF topics*

Counts and within-topic percentages for drug-related, psychological/psychiatric and neurological narratives assigned to each highest-weight NMF topic; the overall-corpus row is the baseline.

Complete machine-readable table: tables/table_s16_primary_narrative_nmf_topic_category_composition.csv (27 rows; 8 columns).

Topic ID	Topic Label	Topic n	Category Label	Narrative n	Within Topic %	Row Type
1	Topic 1: face, went, kept, sound	897	Drug-related	571	63.657	topic
1	Topic 1: face, went, kept, sound	897	Psychological / psychiatric	156	17.391	topic
1	Topic 1: face, went, kept, sound	897	Neurological	170	18.952	topic
2	Topic 2: voice, head, talk, talking	699	Drug-related	155	22.175	topic
2	Topic 2: voice, head, talk, talking	699	Psychological / psychiatric	465	66.524	topic
2	Topic 2: voice, head, talk, talking	699	Neurological	79	11.302	topic
3	Topic 3: visual, auditory, visual snow, snow	482	Drug-related	233	48.34	topic
3	Topic 3: visual, auditory, visual snow, snow	482	Psychological / psychiatric	92	19.087	topic
3	Topic 3: visual, auditory, visual snow, snow	482	Neurological	157	32.573	topic
4	Topic 4: vision, blurry, spot, light	664	Drug-related	236	35.542	topic
4	Topic 4: vision, blurry, spot, light	664	Psychological / psychiatric	40	6.024	topic
4	Topic 4: vision, blurry, spot, light	664	Neurological	388	58.434	topic
5	Topic 5: shadow, spider, crawling, bugs	420	Drug-related	207	49.286	topic
5	Topic 5: shadow, spider, crawling, bugs	420	Psychological / psychiatric	104	24.762	topic
5	Topic 5: shadow, spider, crawling, bugs	420	Neurological	109	25.952	topic
6	Topic 6: eye, closed, open, corner	426	Drug-related	257	60.329	topic

Topic ID	Topic Label	Topic n	Category Label	Narrative n	Within Topic %	Row Type
6	Topic 6: eye, closed, open, corner	426	Psychological / psychiatric	46	10.798	topic
6	Topic 6: eye, closed, open, corner	426	Neurological	123	28.873	topic
7	Topic 7: body, move, paralysis, tried	563	Drug-related	217	38.544	topic
7	Topic 7: body, move, paralysis, tried	563	Psychological / psychiatric	65	11.545	topic
7	Topic 7: body, move, paralysis, tried	563	Neurological	281	49.911	topic
8	Topic 8: real, reality, life, world	705	Drug-related	365	51.773	topic
8	Topic 8: real, reality, life, world	705	Psychological / psychiatric	191	27.092	topic
8	Topic 8: real, reality, life, world	705	Neurological	149	21.135	topic
0	Overall corpus	4856	Drug-related	2241	46.149	overall
0	Overall corpus	4856	Psychological / psychiatric	1159	23.867	overall
0	Overall corpus	4856	Neurological	1456	29.984	overall

Supplementary Table S17

Supplementary Table S17. *Curated condition biology metadata*

Source-linked exploratory coding of category, condition family, neurotransmitter systems, molecular targets, brain regions/networks and mechanisms. Blank fields mean insufficiently established coding, not biological absence.

Complete machine-readable table: tables/table_s17_condition_biology_metadata.csv (21 rows; 19 columns).

Cocaine

Category	drug
Condition family	stimulant
Neurotransmitter systems	dopamine|norepinephrine|serotonin
Molecular targets	DAT|NET|SERT
Directional molecular effects	dopamine:increase|norepinephrine:increase|serotonin:increase|DAT:inhibit|NET:inhibit|SERT:inhibit
Brain regions or networks	mesolimbic_reward|striatum_basal_ganglia|prefrontal_cortex|limbic_medial_temporal
Evidence scope	direct drug pharmacology and psychosis literature
Hallucination specificity	moderate
Caveat	Transporter inhibition and increased synaptic monoamine availability are direct pharmacology; regional coding summarizes stimulant reward and psychosis circuits rather than a co...
Source 1	Roque Bravo R, et al. (2022). Cocaine: An updated overview on chemistry, detection, biokinetics, and pharmacotoxicological aspects including abuse pattern. Toxins, 14(4), 278. h...
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/35448887/
Source 2	Hearing MC, Zink AN, & Wickman K. (2012). Cocaine-induced adaptations in metabotropic inhibitory signaling in the mesocorticolimbic system. Reviews in the Neurosciences, 23(4),...
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/22944653/

Diphenhydramine

Category	drug
Condition family	antihistamine_anticholinergic_deliriant
Neurotransmitter systems	acetylcholine|histamine
Molecular targets	M1_M5|H1
Directional molecular effects	acetylcholine:decrease|histamine:decrease|M1_M5:antagonize|H1:antagonize
Brain regions or networks	prefrontal_cortex|limbic_medial_temporal|salience_cingulo_insular
Evidence scope	direct_drug_pharmacology_plus_delirium_mechanism
Hallucination specificity	moderate
Caveat	Central antimuscarinic toxicity is the principal hallucination-relevant model; named regions summarize delirium and attention-memory networks and are less drug-specific than the...
Source 1	Dawson AH & Buckley NA. (2016). Pharmacological management of anticholinergic delirium: Theory, evidence and practice. British Journal of Clinical Pharmacology, 81(3), 516-524....
Source 1 URL	https://pmc.ncbi.nlm.nih.gov/articles/PMC4767198/
Source 2	Nerush MO, et al. (2024). Classics in chemical neuroscience: Deliriant antihistaminic drugs. ACS Chemical Neuroscience, 15(21), 3848-3862. https://doi.org/10.1021/acschemneuro.4...
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/39404616/

DMT

Category	drug
Condition family	classic_psychedelic
Neurotransmitter systems	serotonin
Molecular targets	5-HT2A|5-HT1A
Directional molecular effects	5-HT2A:agonize|5-HT1A:agonize
Brain regions or networks	visual_cortex|thalamus|prefrontal_cortex|default_mode_network|limbic_medial_temporal
Evidence scope	direct_drug_pharmacology_plus_human_neuroimaging
Hallucination specificity	high
Caveat	5-HT2A agonism is the principal class mechanism; 5-HT1A and regional effects are secondary and the source of individual phenomenological features remains uncertain.
Source 1	Cummins BR, Billac GB, Nichols DE, & Nichols CD. (2025). 5-HT2A receptors: Pharmacology and functional selectivity. Pharmacological Reviews, 77(4), 100059. https://doi.org/10.10...
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/40418878/
Source 2	Castelhano J, et al. (2021). The effects of tryptamine psychedelics in the brain: A meta-analysis of functional and review of molecular imaging studies. Frontiers in Pharmacolog...
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/34658876/

Ketamine

Category	drug
Condition family	dissociative_anesthetic
Neurotransmitter systems	glutamate
Molecular targets	NMDA|HCN1
Directional molecular effects	NMDA:antagonize|HCN1:inhibit
Brain regions or networks	prefrontal_cortex|limbic_medial_temporal|default_mode_network|thalamus
Evidence scope	direct_drug_pharmacology_plus_human_neurophysiology
Hallucination specificity	high
Caveat	NMDA antagonism is established; HCN1 is an additional action and should not be interpreted as the sole cause of ketamine phenomenology.
Source 1	Mion G & Villevieille T. (2013). Ketamine pharmacology: An update. CNS Neuroscience & Therapeutics, 19(6), 370-380. https://doi.org/10.1111/cns.12099
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/23575437/
Source 2	Tian F, et al. (2023). Characterizing brain dynamics during ketamine-induced dissociation and subsequent interactions with propofol using human intracranial neurophysiology. Nat...
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/36991011/

LSD

Category	drug
Condition family	classic_psychedelic
Neurotransmitter systems	serotonin
Molecular targets	5-HT2A
Directional molecular effects	5-HT2A:agonize

Category	drug
Brain regions or networks	visual_cortex|thalamus|prefrontal_cortex|default_mode_network
Evidence scope	direct_drug_pharmacology_plus_human_neuroimaging
Hallucination specificity	high
Caveat	LSD has secondary receptor actions, but 5-HT2A agonism is retained as the primary hallucination-relevant action to keep the coding parsimonious.
Source 1	Preller KH, et al. (2018). Changes in global and thalamic brain connectivity in LSD-induced altered states of consciousness are attributable to the 5-HT2A receptor. eLife, 7, e3...
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/30355445/
Source 2	Becker AM, et al. (2023). Ketanserin reverses the acute response to LSD in a randomized double-blind placebo-controlled crossover study. International Journal of Neuropsychophar...
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/36342343/

MDMA

Category	drug
Condition family	empathogen_stimulant
Neurotransmitter systems	serotonin|dopamine|norepinephrine
Molecular targets	SERT|DAT|NET|VMAT2
Directional molecular effects	serotonin:increase|dopamine:increase|norepinephrine:increase|SERT:reverse_transport|DAT:reverse_transport|NET:reverse_transport
Brain regions or networks	prefrontal_cortex|limbic_medial_temporal|thalamus|mesolimbic_reward
Evidence scope	direct_drug_pharmacology_plus_human_neuroimaging
Hallucination specificity	low
Caveat	Monoamine release is well established but is not hallucination-specific; phenomenology can also reflect dose, setting, co-use, and secondary psychedelic-like effects.
Source 1	de la Torre R, et al. (2004). Human pharmacology of MDMA: Pharmacokinetics, metabolism, and disposition. Therapeutic Drug Monitoring, 26(2), 137-144. https://doi.org/10.1097/000...
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/15228154/
Source 2	Vollenweider FX. (2001). Brain mechanisms of hallucinogens and entactogens. Dialogues in Clinical Neuroscience, 3(4), 265-279. https://doi.org/10.31887/DCNS.2001.3.4/fxvollenweider
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/22033605/

Methamphetamine

Category	drug
Condition family	stimulant
Neurotransmitter systems	dopamine|norepinephrine|glutamate|GABA
Molecular targets	DAT|NET|VMAT2|TAAR1
Directional	dopamine:increase|norepinephrine:increase|DAT:reverse transport|NET:reverse transport|VMAT2:inhibit|TAAR1:agoni

Category	drug
molecular effects	ze|glutamate:mixed|GABA:mixed
Brain regions or networks	striatum_basal_ganglia|mesolimbic_reward|prefrontal_cortex|limbic_medial_temporal
Evidence scope	direct_drug_pharmacology_and_psychosis_literature
Hallucination specificity	moderate
Caveat	Monoamine actions are direct; glutamate, GABA, and regional codes summarize downstream psychosis models and are not specific to every hallucination.
Source 1	Ramli FF, et al. (2025). A mechanistic review on toxicity effects of methamphetamine. International Journal of Medical Sciences, 22(3), 482-507. https://doi.org/10.7150/ijms.99159
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/39898237/
Source 2	Chen C, Hsu FC, Li CW, & Huang MC. (2019). Structural, functional, and neurochemical neuroimaging of methamphetamine-associated psychosis: A systematic review. Psychiatry Resear...
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/31476712/

Psilocybin

Category	drug
Condition family	classic_psychedelic
Neurotransmitter systems	serotonin
Molecular targets	5-HT2A
Directional molecular effects	5-HT2A:agonize
Brain regions or networks	visual_cortex|thalamus|prefrontal_cortex|default_mode_network
Evidence scope	direct_drug_pharmacology_plus_human_neuroimaging
Hallucination specificity	high
Caveat	5-HT2A agonism is strong at the drug-class level; regional labels summarize recurring imaging findings rather than a one-to-one map of subjective features.
Source 1	Cummins BR, Billac GB, Nichols DE, & Nichols CD. (2025). 5-HT2A receptors: Pharmacology and functional selectivity. Pharmacological Reviews, 77(4), 100059. https://doi.org/10.10...
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/40418878/
Source 2	Quednow BB, Kometer M, Geyer MA, & Vollenweider FX. (2012). Psilocybin-induced deficits in automatic and controlled inhibition are attenuated by ketanserin in healthy human volu...
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/21956447/

Salvia

Category	drug
Condition family	kappa_opioid_hallucinogen
Neurotransmitter systems	kappa_opioid
Molecular targets	KOR
Directional molecular effects	KOR:agonize
Brain regions or networks	limbic_medial_temporal|thalamus|prefrontal_cortex|sensorimotor_cortex

Category	drug
Evidence scope	direct_drug_pharmacology_plus_small_human_imaging_studies
Hallucination specificity	high
Caveat	KOR agonism is direct and distinctive; the regional evidence is based on comparatively small human studies and broad perfusion or electrophysiological changes.
Source 1	Cunningham CW, Rothman RB, & Prisinzano TE. (2011). Neuropharmacology of the naturally occurring kappa-opioid hallucinogen salvinorin A. Pharmacological Reviews, 63(2), 316-347....
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/21444610/
Source 2	Ona G, et al. (2022). The kappa opioid receptor and the sleep of reason: Cortico-subcortical imbalance following salvinorin-A. International Journal of Neuropsychopharmacology,...
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/34537829/

Bipolar Disorder with Psychotic Features

Category	psychological
Condition family	mood_disorder_with_psychosis
Neurotransmitter systems	dopamine|glutamate
Molecular targets	D2_D3|NMDA
Directional molecular effects	dopamine:mixed|glutamate:mixed|D2_D3:mixed|NMDA:mixed
Brain regions or networks	prefrontal_cortex|salience_cingulo_insular|limbic_medial_temporal
Evidence scope	condition pathophysiology with limited hallucination specificity
Hallucination specificity	low
Caveat	Psychotic bipolar biomarkers are sparse and incompletely replicated; mixed directional codes are retained but excluded from concordance and opposition scores.
Source 1	Buoli M, et al. (2016). Biological aspects and candidate biomarkers for psychotic bipolar disorder: A systematic review. Psychiatry and Clinical Neurosciences, 70(6), 227-244. h...
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/26969211/
Source 2	Chabert J, et al. (2022). Glutamatergic and N-acetylaspartate metabolites in bipolar disorder: A systematic review and meta-analysis of proton magnetic resonance spectroscopy st...
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/36012234/

Borderline Personality Disorder

Category	psychological
Condition family	personality_disorder
Neurotransmitter systems	serotonin|glutamate|GABA
Directional molecular effects	serotonin:mixed|glutamate:mixed|GABA:mixed
Brain regions or networks	prefrontal_cortex|limbic_medial_temporal|salience_cingulo_insular|default_mode_network
Evidence scope	condition_pathophysiology_not_hallucination_specific
Hallucination specificity	low
Caveat	The cited biology concerns BPD generally; no settled receptor-level mechanism explains hallucinations, so the target field is intentionally blank and directional claims are not...
Source 1	Krause-Utz A, Winter D, Niedtfeld I, & Schmahl C. (2014). The latest neuroimaging findings in borderline personality disorder. Current Psychiatry Reports, 16(3), 438.

Category	psychological
	https://do...
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/24492919/
Source 2	O'Neill A & Frodl T. (2012). Brain structure and function in borderline personality disorder. Brain Structure and Function, 217(4), 767-782. https://doi.org/10.1007/s00429-012-0...
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/22252376/

Dissociative Identity Disorder

Category	psychological
Condition family	dissociative_disorder
Brain regions or networks	prefrontal_cortex|salience_cingulo_insular|striatum_basal_ganglia|frontoparietal_attention|temporal_auditory_language
Evidence scope	small_condition_neuroimaging_literature_not_hallucination_specific
Hallucination specificity	low
Caveat	Neurochemical mechanisms are not established and the imaging literature is small and heterogeneous; blank transmitter, target, and directional fields are deliberate.
Source 1	Modesti MN, Rapisarda L, Capriotti G, & Del Casale A. (2022). Functional neuroimaging in dissociative disorders: A systematic review. Journal of Personalized Medicine, 12(9), 14...
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/36143190/

PTSD

Category	psychological
Condition family	trauma_related_disorder
Neurotransmitter systems	norepinephrine|serotonin|glutamate|GABA
Molecular targets	alpha2_adrenergic|5-HT1B|NMDA|GABA_A
Directional molecular effects	norepinephrine:increase|serotonin:mixed|glutamate:mixed|GABA:mixed
Brain regions or networks	limbic_medial_temporal|prefrontal_cortex|salience_cingulo_insular|default_mode_network|thalamus
Evidence scope	condition_pathophysiology_plus_cross_diagnostic_voice_literature
Hallucination specificity	low
Caveat	Most evidence is PTSD-general rather than hallucination-specific; only elevated noradrenergic signaling receives a directional sign and other systems remain mixed.
Source 1	Hendrickson RC & Raskind MA. (2016). Noradrenergic dysregulation in the pathophysiology of PTSD. Experimental Neurology, 284(B), 181-195. https://doi.org/10.1016/j.expneurol.201...
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/27222130/
Source 2	Averill LA, et al. (2017). Glutamate dysregulation and glutamatergic therapeutics for PTSD: Evidence from human studies. Neuroscience Letters, 649, 147-155. https://doi.org/10.1...
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/27916636/

Schizoaffective Disorder

Category	psychological
Condition family	schizophrenia_spectrum
Neurotransmitter systems	dopamine|glutamate|GABA

Category	psychological
Molecular targets	D2_D3|NMDA
Directional molecular effects	dopamine:increase|glutamate:decrease|GABA:decrease|D2_D3:increase|NMDA:decrease
Brain regions or networks	temporal_auditory_language|salience_cingulo_insular|prefrontal_cortex|limbic_medial_temporal|striatum_basal_ganglia
Evidence scope	partly extrapolated from schizophrenia spectrum biology
Hallucination specificity	moderate
Caveat	Condition-specific mechanistic evidence is limited; transmitter and target directions are explicit extrapolations from schizophrenia-spectrum models and require cautious interpr...
Source 1	Madre M, et al. (2016). Neuropsychological and neuroimaging underpinnings of schizoaffective disorder: A systematic review. Acta Psychiatrica Scandinavica, 134(1), 16-30. https:...
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/27028168/
Source 2	Mo F, et al. (2024). Network localization of state and trait of auditory verbal hallucinations in schizophrenia. Schizophrenia Bulletin, 50(6), 1326-1336. https://doi.org/10.109...
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/38401526/

Schizophrenia

Category	psychological
Condition family	schizophrenia_spectrum
Neurotransmitter systems	dopamine|glutamate|GABA|serotonin
Molecular targets	D2_D3|NMDA|5-HT2A
Directional molecular effects	dopamine:increase|glutamate:decrease|GABA:decrease|D2_D3:increase|NMDA:decrease|5-HT2A:mixed
Brain regions or networks	temporal_auditory_language|salience_cingulo_insular|limbic_medial_temporal|striatum_basal_ganglia|thalamus|prefrontal_cortex
Evidence scope	psychosis pathophysiology plus hallucination specific imaging
Hallucination specificity	high
Caveat	No single mechanism explains schizophrenia hallucinations; dopamine, glutamate-GABA, serotonin, and network findings represent complementary and partly heterogeneous levels of e...
Source 1	Howes OD & Kapur S. (2009). The dopamine hypothesis of schizophrenia: Version III. Schizophrenia Bulletin, 35(3), 549-562. https://doi.org/10.1093/schbul/sbp006
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/19325164/
Source 2	Mo F, et al. (2024). Network localization of state and trait of auditory verbal hallucinations in schizophrenia. Schizophrenia Bulletin, 50(6), 1326-1336. https://doi.org/10.109...
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/38401526/

Epilepsy

Category	neurological
Condition family	seizure_disorder
Neurotransmitter systems	glutamate|GABA
Molecular targets	NMDA|GABA_A

Category	neurological
Directional molecular effects	glutamate:increase|GABA:decrease|NMDA:increase|GABA_A:decrease
Brain regions or networks	visual_cortex|temporal_auditory_language|limbic_medial_temporal|salience_cingulo_insular
Evidence scope	general epilepsy neurochemistry plus hallucination localization
Hallucination specificity	moderate
Caveat	Hallucination modality and complexity depend strongly on seizure focus; directional transmitter coding represents the broad excitation-inhibition model rather than every epileps...
Source 1	Sherwin AL. (1999). Neuroactive amino acids in focally epileptic human brain: A review. Neurochemical Research, 24(11), 1387-1395. https://doi.org/10.1023/A:1022580506443
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/10555779/
Source 2	Elliott B, Joyce E, & Shorvon S. (2009). Delusions, illusions and hallucinations in epilepsy: 1. Elementary phenomena. Epilepsy Research, 85(2-3), 162-171. https://doi.org/10.10...
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/19423297/

Hallucinogen Persisting Perception Disorder

Category	neurological
Condition family	persistent_perceptual_disorder
Neurotransmitter systems	serotonin|GABA
Molecular targets	5-HT2A
Directional molecular effects	serotonin:mixed|GABA:decrease|5-HT2A:mixed
Brain regions or networks	visual_cortex|visual_association_cortex|thalamus
Evidence scope	proposed pathophysiology from reviews and case series
Hallucination specificity	moderate
Caveat	Pathophysiology is not established. Serotonergic and GABAergic visual-disinhibition models are hypotheses; mixed effects are excluded from directional scoring.
Source 1	Litjens RP, Brunt TM, Alderliefste GJ, & Westerink RH. (2014). Hallucinogen persisting perception disorder and the serotonergic system: A comprehensive review including new MDMA...
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/24933532/
Source 2	Doyle MA, et al. (2022). Hallucinogen persisting perceptual disorder: A scoping review covering frequency, risk factors, prevention, and treatment. Expert Opinion on Drug Safety...
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/35426769/

Migraine

Category	neurological
Condition family	migraine_spectrum
Neurotransmitter systems	glutamate
Molecular targets	cortical_spreading_depolarization|CGRP_pathway
Directional molecular effects	glutamate:increase|cortical_spreading_depolarization:increase|CGRP_pathway:increase
Brain regions or networks	visual_cortex|visual_association_cortex|sensorimotor_cortex|brainstem_sensory_modulation
Evidence scope	aura_specific_pathophysiology_plus_human_imaging
Hallucination specificity	high
Caveat	Coding is most applicable to migraine aura and complex perceptual symptoms; not every migraine narrative is generated by cortical spreading depolarization.

Category	neurological
Source 1	O'Hare L, Asher JM, & Hibbard PB. (2021). Migraine visual aura and cortical spreading depression: Linking mathematical models to empirical evidence. Vision, 5(2), 30. https://do...
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/34200625/
Source 2	Pikor D, et al. (2025). fMRI insights into visual cortex dysfunction as a biomarker for migraine with aura. Neurology International, 17(2), 15. https://doi.org/10.3390/neurolint...
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/39997646/

Narcolepsy

Category	neurological
Condition family	central_hypersomnolence
Neurotransmitter systems	orexin
Molecular targets	OX1_OX2
Directional molecular effects	orexin:decrease|OX1_OX2:decrease
Brain regions or networks	hypothalamus|brainstem_REM_arousal|thalamus|salience_cingulo_insular|temporal_auditory_language
Evidence scope	direct_condition_biology_plus_sleep_imaging
Hallucination specificity	high
Caveat	Orexin deficiency is well established in narcolepsy type 1; the neural correlates of hypnagogic and hypnopompic hallucination content remain less certain.
Source 1	Bassetti CLA, et al. (2019). Narcolepsy: Clinical spectrum, aetiopathophysiology, diagnosis and treatment. Nature Reviews Neurology, 15(9), 519-539. https://doi.org/10.1038/s415...
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/31324898/
Source 2	Dang-Vu TT, Desseilles M, Schwartz S, & Maquet P. (2009). Neuroimaging of narcolepsy. CNS and Neurological Disorders Drug Targets, 8(4), 254-263. https://doi.org/10.2174/1871527...
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/19689307/

Parkinson's Disease Psychosis

Category	neurological
Condition family	parkinsonian_psychosis
Neurotransmitter systems	acetylcholine|serotonin|dopamine
Molecular targets	5-HT2A|D2_D3
Directional molecular effects	acetylcholine:decrease|serotonin:mixed|dopamine:mixed|5-HT2A:mixed|D2_D3:mixed
Brain regions or networks	visual_cortex|visual_association_cortex|limbic_medial_temporal|frontoparietal_attention|default_mode_network|brainstem_REM_arousal
Evidence scope	condition_pathophysiology_plus_visual_hallucination_literature
Hallucination specificity	high
Caveat	Hallucinations are multifactorial and medication exposure can contribute; only cholinergic loss receives a directional sign and other transmitter effects remain mixed.
Source 1	Pagonabarraga J, Bejr-Kasem H, Martinez-Horta S, & Kulisevsky J. (2024). Parkinson disease psychosis: From phenomenology to neurobiological mechanisms. Nature Reviews Neurology,...
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/38225264/

Category	neurological
Source 2	Liu J, et al. (2026). Cholinergic neurotransmission underlying visual hallucinations in Parkinson's disease: Integration of multimodal evidence and translational approaches. Neu...
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/41672204/

Sleep Paralysis

Category	neurological
Condition family	REM_parasomnia
Neurotransmitter systems	GABA|glycine|acetylcholine
Molecular targets	GABA_A|GABA_B|glycine_receptor
Directional molecular effects	GABA:increase|glycine:increase|acetylcholine:increase
Brain regions or networks	brainstem_REM_arousal|thalamus|limbic_medial_temporal|sensorimotor_cortex
Evidence scope	REM_sleep_mechanism_with_partial_hallucination_specificity
Hallucination specificity	high
Caveat	GABAergic and glycinergic activity explains REM atonia more directly than hallucination content; hallucinations are modeled as REM imagery intruding into wakefulness with body-s...
Source 1	Brooks PL & Peever JH. (2012). Identification of the transmitter and receptor mechanisms responsible for REM sleep paralysis. Journal of Neuroscience, 32(29), 9785-9795. https:/...
Source 1 URL	https://pubmed.ncbi.nlm.nih.gov/22815493/
Source 2	Fraigne JJ, Torontali ZA, Snow MB, & Peever JH. (2015). REM sleep at its core: Circuits, neurotransmitters, and pathophysiology. Frontiers in Neurology, 6, 123. https://doi.org/...
Source 2 URL	https://pubmed.ncbi.nlm.nih.gov/26074874/

Supplementary Table S18

Supplementary Table S18. *Condition-biology univariate QAP associations*

Primary and sensitivity total pairwise associations between each curated metadata domain and masked embedding-centroid similarity, tested by condition-label permutations.

Complete machine-readable table: tables/table_s18_condition_biology_univariate_qap.csv (24 rows; 20 columns).

Corpus	Analysis Set	Predictor Label	Observed QAP Correlation	Permutation P Two Sided	BH Q Value	n Pairs
primary	all_conditions	Same broad category	0.4118	0.0002	0.0006	210
primary	all_conditions	Same condition family	0.2382	0.0008	0.0012	210
primary	all_conditions	Neurotransmitter-system overlap	0.169	0.1472	0.1766	210
primary	all_conditions	Receptor / molecular-target overlap	0.0971	0.3302	0.3302	210
primary	all_conditions	Brain-region / network overlap	0.5572	0.0002	0.0006	210
primary	all_conditions	Mechanism-keyword overlap	0.2492	0.0008	0.0012	210
primary	moderate_high_evidence	Same broad category	0.416	0.0006	0.0018	120
primary	moderate_high_evidence	Same condition family	0.2612	0.014	0.021	120
primary	moderate_high_evidence	Neurotransmitter-system overlap	0.1429	0.2794	0.28	120
primary	moderate_high_evidence	Receptor / molecular-target overlap	0.1194	0.28	0.28	120
primary	moderate_high_evidence	Brain-region / network overlap	0.5313	0.0002	0.0012	120
primary	moderate_high_evidence	Mechanism-keyword overlap	0.2699	0.0096	0.0192	120
sensitivity	all_conditions	Same broad category	0.4278	0.0002	0.0006	210
sensitivity	all_conditions	Same condition family	0.237	0.0014	0.0021	210
sensitivity	all_conditions	Neurotransmitter-system overlap	0.1817	0.1142	0.137	210
sensitivity	all_conditions	Receptor / molecular-target overlap	0.1084	0.2706	0.2706	210
sensitivity	all_conditions	Brain-region / network overlap	0.5535	0.0002	0.0006	210
sensitivity	all_conditions	Mechanism-keyword overlap	0.2476	0.001	0.002	210
sensitivity	moderate_high_evidence	Same broad category	0.4418	0.0004	0.0012	120
sensitivity	moderate_high_evidence	Same condition family	0.2593	0.0116	0.0174	120
sensitivity	moderate_high_evidence	Neurotransmitter-system overlap	0.1461	0.2756	0.2756	120

Corpus	Analysis Set	Predictor Label	Observed QAP Correlation	Permutation P Two Sided	BH Q Value	n Pairs
sensitivity	moderate_high_evidence	Receptor / molecular-target overlap	0.1235	0.2716	0.2756	120
sensitivity	moderate_high_evidence	Brain-region / network overlap	0.5353	0.0002	0.0012	120
sensitivity	moderate_high_evidence	Mechanism-keyword overlap	0.2687	0.0096	0.0174	120

Supplementary Table S19

Supplementary Table S19. *Condition-biology multivariable MRQAP*

Primary and sensitivity standardized unique associations from a joint residual-permutation MRQAP. Results are exploratory associations and do not establish biological causation.

Complete machine-readable table: tables/table_s19_condition_biology_mrqap.csv (24 rows; 18 columns).

Corpus	Analysis Set	Predictor Label	Standardized Beta	Drop One Delta R Squared	Freedman Lane QAP P Two Sided	BH Q Value	Variance Inflation Factor
primary	all_conditions	Same broad category	0.2383	0.0482	0.0002	0.0006	1.177
primary	all_conditions	Same condition family	-0.0515	0.0009	0.507	0.5772	3.102
primary	all_conditions	Neurotransmitter-system overlap	0.068	0.0028	0.5772	0.5772	1.627
primary	all_conditions	Receptor / molecular-target overlap	-0.128	0.0105	0.1672	0.3344	1.561
primary	all_conditions	Brain-region / network overlap	0.4876	0.1675	0.0002	0.0006	1.42
primary	all_conditions	Mechanism-keyword overlap	0.0609	0.0012	0.5166	0.5772	3.182
primary	moderate_high_evidence	Same broad category	0.2522	0.0539	0.0086	0.0258	1.181
primary	moderate_high_evidence	Same condition family	-0.1167	0.0033	0.4202	0.5042	4.133
primary	moderate_high_evidence	Neurotransmitter-system overlap	0.0524	0.0014	0.7292	0.7292	1.928
primary	moderate_high_evidence	Receptor / molecular-target overlap	-0.1317	0.009	0.1692	0.3384	1.933
primary	moderate_high_evidence	Brain-region / network overlap	0.4753	0.1465	0.0072	0.0258	1.542
primary	moderate_high_evidence	Mechanism-keyword overlap	0.1353	0.0041	0.4012	0.5042	4.421
sensitivity	all_conditions	Same broad category	0.2603	0.0575	0.0002	0.0006	1.177
sensitivity	all_conditions	Same condition family	-0.0534	0.0009	0.4902	0.5914	3.102
sensitivity	all_conditions	Neurotransmitter-system overlap	0.0837	0.0043	0.5098	0.5914	1.627
sensitivity	all_conditions	Receptor / molecular-target overlap	-0.1179	0.0089	0.196	0.392	1.561
sensitivity	all_conditions	Brain-region / network overlap	0.4728	0.1575	0.0002	0.0006	1.42
sensitivity	all_conditions	Mechanism-keyword overlap	0.0502	0.0008	0.5914	0.5914	3.182
sensitivity	moderate_high_evidence	Same broad category	0.2824	0.0675	0.0038	0.0228	1.181
sensitivity	moderate_high_evidence	Same condition family	-0.1241	0.0037	0.39	0.504	4.133

Corpus	Analysis Set	Predictor Label	Standardized Beta	Drop One Delta R Squared	Freedman Lane QAP P Two Sided	BH Q Value	Variance Inflation Factor
sensitivity	moderate_high_evidence	Neurotransmitter-system overlap	0.0597	0.0019	0.6682	0.6682	1.928
sensitivity	moderate_high_evidence	Receptor / molecular-target overlap	-0.1239	0.0079	0.2008	0.4016	1.933
sensitivity	moderate_high_evidence	Brain-region / network overlap	0.4711	0.1439	0.0106	0.0318	1.542
sensitivity	moderate_high_evidence	Mechanism-keyword overlap	0.1268	0.0036	0.42	0.504	4.421

Supplementary Table S20

Supplementary Table S20. *Condition-biology QAP/MRQAP model summary*

Primary and sensitivity condition/pair counts, descriptive model R-squared, global QAP tests and assumptions for the full and moderate/high-evidence sets.

Complete machine-readable table: tables/table_s20_condition_biology_model_summary.csv (4 rows; 18 columns).

Corpus	Analysis Set	n Conditions	n Pairs	Predictor n	Full Model R Squared Descriptive	Global QAP P	Permutations
primary	all_conditions	21	210	6	0.3718	0.0002	4999
primary	moderate high evidence	16	120	6	0.3552	0.0008	4999
sensitivity	all_conditions	21	210	6	0.3752	0.0002	4999
sensitivity	moderate high evidence	16	120	6	0.3723	0.0004	4999

Supplementary Table S21

Supplementary Table S21. *Broad-category pair similarity summary*

Primary and sensitivity descriptive masked centroid similarity for each within- and between-category condition pairing.

Complete machine-readable table: tables/table_s21_condition_category_pair_similarity.csv (12 rows; 10 columns).

Corpus	Category 1	Category 2	Pair n	Mean	Median	Sd	Same Category
primary	drug	drug	36	0.9317	0.9439	0.0418	True
primary	drug	psychological	54	0.8647	0.8579	0.045	False
primary	drug	neurological	54	0.8227	0.8455	0.0775	False
primary	psychological	psychological	15	0.9412	0.9345	0.0268	True
primary	psychological	neurological	36	0.8152	0.8333	0.086	False
primary	neurological	neurological	15	0.8162	0.8441	0.0964	True
sensitivity	drug	drug	36	0.9369	0.9525	0.041	True
sensitivity	drug	psychological	54	0.867	0.8618	0.0447	False
sensitivity	drug	neurological	54	0.8232	0.8456	0.0769	False
sensitivity	psychological	psychological	15	0.9454	0.9396	0.0278	True
sensitivity	psychological	neurological	36	0.8232	0.8425	0.083	False
sensitivity	neurological	neurological	15	0.8215	0.8464	0.0926	True

Supplementary Table S22

Supplementary Table S22. *Condition-level linguistic-form summary*

Primary and sensitivity discussion-weighted medians, quartiles and bootstrap intervals for narrative amount, clause complexity, lexical richness, coherence, readability and style.

Complete machine-readable table: tables/table_s22_condition_linguistic_form_summary.csv (780 rows; 16 columns).

Corpus	Label	Metric Label	Narrative n Nonmissing	Discussion Weighted Median	Discussion Weighted Q1	Discussion Weighted Q3	Median CI Low	Median CI High
primary	PTSD	Accepted phenomenology words	162	49	30.25	77.5	44.5	57.5
primary	PTSD	Accepted clause count	162	2.75	1.25	4	2	3
primary	PTSD	Mean words per accepted clause	162	18.65	14	25	16.833	20.125
primary	PTSD	Share of clauses with at least 20 words	162	0.5	0	0.9583	0.3333	0.5
primary	PTSD	Subordination markers per 100 words	162	2.282	0	4.348	2	2.857
primary	PTSD	Moving-average type-token ratio (window 25)	132	0.87	0.8411	0.9031	0.86	0.8769
primary	PTSD	Measure of textual lexical diversity	132	65.543	55.539	78.701	61.116	68.43
primary	PTSD	Non-stopword token share	162	0.3876	0.3495	0.4373	0.3778	0.4
primary	PTSD	Adjacent-clause content-token overlap	120	0.0091	0	0.037	0	0.0199
primary	PTSD	Adjacent-clause TF-IDF coherence	120	0.0106	0	0.0556	0	0.0259

Display extract shown above; 770 additional rows are retained in the complete accompanying CSV.

Supplementary Table S23

Supplementary Table S23. *Condition differences in linguistic form*

Discussion-level omnibus rank tests with condition-label permutations, epsilon-squared effect sizes and Benjamini-Hochberg correction across metrics. Emphasize effect sizes over p-values.

Complete machine-readable table: tables/table_s23_condition_linguistic_form_tests.csv (30 rows; 16 columns).

Corpus	Metric Label	Discussion n	Condition n	Epsilon Squared	Permutation P	BH Q Value	Effect Size Rank
primary	Accepted phenomenology words	4123	21	0.1703	0.0002	0.0002	1
primary	Accepted clause count	4123	21	0.167	0.0002	0.0002	2
primary	First-person pronouns per 100 words	4123	21	0.0537	0.0002	0.0002	3
primary	Clause-based Flesch reading ease	4123	21	0.0479	0.0002	0.0002	4
primary	Non-stopword token share	4123	21	0.0379	0.0002	0.0002	5
primary	Discourse connectives per 100 words	4123	21	0.0331	0.0002	0.0002	6
primary	Subordination markers per 100 words	4123	21	0.0287	0.0002	0.0002	7
primary	Mean words per accepted clause	4123	21	0.0219	0.0002	0.0002	8
primary	Negations per 100 words	4123	21	0.0169	0.0002	0.0002	9
primary	Share of clauses with at least 20 words	4123	21	0.0167	0.0002	0.0002	10
primary	Internal punctuation marks per 100 words	4123	21	0.016	0.0002	0.0002	11
primary	Moving-average type-token ratio (window 25)	3830	21	0.012	0.0002	0.0002	12
primary	Measure of textual lexical diversity	3830	21	0.0106	0.0002	0.0002	13
primary	Adjacent-clause content-token overlap	3619	21	0.0075	0.0006	0.0006	14
primary	Adjacent-clause TF-IDF coherence	3618	21	0.0064	0.0034	0.0034	15
sensitivity	Accepted phenomenology words	5150	21	0.1905	0.0002	0.0002	1
sensitivity	Accepted	5150	21	0.1885	0.0002	0.0002	2

Corpus	Metric Label	Discussion n	Condition n	Epsilon Squared	Permutation P	BH Q Value	Effect Size Rank
	clause count						
sensitivity	First-person pronouns per 100 words	5150	21	0.0493	0.0002	0.0002	3
sensitivity	Clause-based Flesch reading ease	5150	21	0.0463	0.0002	0.0002	4
sensitivity	Non-stopword token share	5150	21	0.0404	0.0002	0.0002	5
sensitivity	Discourse connectives per 100 words	5150	21	0.0306	0.0002	0.0002	6
sensitivity	Subordination markers per 100 words	5150	21	0.0285	0.0002	0.0002	7
sensitivity	Mean words per accepted clause	5150	21	0.0207	0.0002	0.0002	8
sensitivity	Share of clauses with at least 20 words	5150	21	0.0156	0.0002	0.0002	9
sensitivity	Negations per 100 words	5150	21	0.0131	0.0002	0.0002	10
sensitivity	Internal punctuation marks per 100 words	5150	21	0.0128	0.0002	0.0002	11
sensitivity	Measure of textual lexical diversity	4825	21	0.0119	0.0002	0.0002	12
sensitivity	Moving-average type-token ratio (window 25)	4825	21	0.011	0.0002	0.0002	13
sensitivity	Adjacent-clause TF-IDF coherence	4587	21	0.0068	0.0002	0.0002	14
sensitivity	Adjacent-clause content-token overlap	4588	21	0.0063	0.0008	0.0008	15

Supplementary Table S24

Supplementary Table S24. *Broad-category linguistic-form summary*

Primary and sensitivity discussion-weighted descriptive summaries for the three broad narrative categories; these are descriptive rather than causal contrasts.

Complete machine-readable table: tables/table_s24_category_linguistic_form_summary.csv (90 rows; 16 columns).

Corpus	Label	Metric Label	Narrative n Nonmissing	Discussion Weighted Median	Discussion Weighted Q1	Discussion Weighted Q3
primary	psychological	Accepted phenomenology words	1169	60	36	97
primary	psychological	Accepted clause count	1169	3	2	5
primary	psychological	Mean words per accepted clause	1169	18.6	14.071	26
primary	psychological	Share of clauses with at least 20 words	1169	0.4444	0.1325	0.75
primary	psychological	Subordination markers per 100 words	1169	2.565	0.8706	4
primary	psychological	Moving-average type-token ratio (window 25)	1010	0.8722	0.8431	0.8994
primary	psychological	Measure of textual lexical diversity	1010	65.154	51.343	80.864
primary	psychological	Non-stopword token share	1169	0.392	0.3543	0.4348
primary	psychological	Adjacent-clause content-token overlap	942	0.0242	0	0.0512
primary	psychological	Adjacent-clause TF-IDF coherence	941	0.0294	0	0.0768

Display extract shown above; 80 additional rows are retained in the complete accompanying CSV.

Supplementary Table S25

Supplementary Table S25. *Linguistic-form metric definitions and coverage*

Definitions, analysis coverage and clause-recovery diagnostics needed to interpret missingness and the limits of each non-semantic proxy.

Complete machine-readable table: tables/table_s25_linguistic_metric_codebook_and_coverage.csv (30 rows; 11 columns).

Corpus	Metric Label	Metric Family	Description	Narrative Nonmissing n	Narrative Nonmissing %
primary	Accepted phenomenology words	amount	Number of alphabetic word tokens across accepted clauses before masking.	4927	100.0
primary	Accepted clause count	amount	Number of Gemini-accepted phenomenological clauses.	4927	100.0
primary	Mean words per accepted clause	complexity	Mean alphabetic word-token count per accepted clause; a surface complexity proxy.	4927	100.0
primary	Share of clauses with at least 20 words	complexity	Fraction of accepted clauses containing at least 20 alphabetic word tokens.	4927	100.0
primary	Subordination markers per 100 words	complexity	Rate of common subordinating conjunctions and relative markers; not a full syntactic parse.	4927	100.0
primary	Moving-average type-token ratio (window 25)	lexical_richness	Mean type-token ratio across moving 25-token windows after contextual masking; undefined below 25 tokens.	4512	91.577
primary	Measure of textual lexical diversity	lexical_richness	Forward/reverse MTLD after contextual masking; undefined below 25 tokens.	4512	91.577
primary	Non-stopword token share	lexical_richness	Share of masked tokens not in the project's conservative function/filler-word list.	4927	100.0
primary	Adjacent-clause content-token overlap	coherence	Mean Jaccard overlap of non-stopword tokens in directly adjacent accepted clauses.	4232	85.894
primary	Adjacent-clause TF-IDF coherence	coherence	Mean corpus-fitted TF-IDF cosine similarity of directly adjacent masked clauses.	4231	85.874
primary	Discourse	coherence	Rate of common	4927	100.0

Corpus	Metric Label	Metric Family	Description	Narrative Nonmissing n	Narrative Nonmissing %
	connectives per 100 words		temporal, causal, contrastive and sequential connectives.		
primary	Clause-based Flesch reading ease	readability	Approximate Flesch score using accepted clauses as sentence units; Reddit spelling and clause segmentation limit precision.	4927	100.0
primary	First-person pronouns per 100 words	narrative_style	Rate of first-person singular and plural pronouns.	4927	100.0
primary	Negations per 100 words	narrative_style	Rate of common explicit negation tokens.	4927	100.0
primary	Internal punctuation marks per 100 words	narrative_style	Rate of commas, semicolons, colons, brackets and dashes remaining inside accepted clauses.	4927	100.0
sensitivity	Accepted phenomenology words	amount	Number of alphabetic word tokens across accepted clauses before masking.	6252	100.0
sensitivity	Accepted clause count	amount	Number of Gemini-accepted phenomenological clauses.	6252	100.0
sensitivity	Mean words per accepted clause	complexity	Mean alphabetic word-token count per accepted clause; a surface complexity proxy.	6252	100.0
sensitivity	Share of clauses with at least 20 words	complexity	Fraction of accepted clauses containing at least 20 alphabetic word tokens.	6252	100.0
sensitivity	Subordination markers per 100 words	complexity	Rate of common subordinating conjunctions and relative markers; not a full syntactic parse.	6252	100.0
sensitivity	Moving-average type-token ratio (window 25)	lexical_richness	Mean type-token ratio across moving 25-token windows after contextual masking; undefined below 25 tokens.	5762	92.163
sensitivity	Measure of textual lexical diversity	lexical_richness	Forward/reverse MTLD after contextual masking; undefined below 25 tokens.	5762	92.163
sensitivity	Non-stopword token share	lexical_richness	Share of masked tokens not in the project's	6252	100.0

Corpus	Metric Label	Metric Family	Description	Narrative Nonmissing n	Narrative Nonmissing %
			conservative function/filler-word list.		
sensitivity	Adjacent-clause content-token overlap	coherence	Mean Jaccard overlap of non-stopword tokens in directly adjacent accepted clauses.	5431	86.868
sensitivity	Adjacent-clause TF-IDF coherence	coherence	Mean corpus-fitted TF-IDF cosine similarity of directly adjacent masked clauses.	5430	86.852
sensitivity	Discourse connectives per 100 words	coherence	Rate of common temporal, causal, contrastive and sequential connectives.	6252	100.0
sensitivity	Clause-based Flesch reading ease	readability	Approximate Flesch score using accepted clauses as sentence units; Reddit spelling and clause segmentation limit precision.	6252	100.0
sensitivity	First-person pronouns per 100 words	narrative_style	Rate of first-person singular and plural pronouns.	6252	100.0
sensitivity	Negations per 100 words	narrative_style	Rate of common explicit negation tokens.	6252	100.0
sensitivity	Internal punctuation marks per 100 words	narrative_style	Rate of commas, semicolons, colons, brackets and dashes remaining inside accepted clauses.	6252	100.0

Supplementary Table S26

Supplementary Table S26. *Independent within-condition NMF topic summaries*

Eight topics fitted separately within each of the 21 eligible conditions. Topic IDs and vocabularies are local to each condition and are not directly comparable across conditions.

Complete machine-readable table: tables/table_s26_primary_condition_specific_nmf_topic_summary.csv (168 rows; 9 columns).

Condition	Category	Narrative n	Topic ID	Top Terms	Mean Topic Share %	Dominant Narrative n	Dominant Narrative %
DMT	drug	257	1	vision, black, face, head, white, dark, mouth, eye, completely, little	16.41	50	19.455
DMT	drug	257	2	voice, go, words, sound, life, let, thoughts, sure, telling, swear	15.942	45	17.51
DMT	drug	257	3	dreams, weird, stronger, loud, obvious, least, grotesque, visual, music, minutes	5.53	6	2.335
DMT	drug	257	4	body, reality, mind, intense, real, sense, physical, seemed, almost, ever	19.383	59	22.957
DMT	drug	257	5	eye, closed, visual, open, fractal, closed eye, mild, made, crazy, flame	13.846	40	15.564
DMT	drug	257	6	light, white light, white, dreaming, ball, words, state, body, lucid, pattern	9.936	21	8.171
DMT	drug	257	7	entity, told, love, life, differently, showed, energy, alone, way, powerful	10.618	23	8.949
DMT	drug	257	8	visions, sinister, phone, fractal, fractal visions, tried, dream, sinister vibe, weird eerie, eerie sinister	7.947	13	5.058
LSD	drug	279	1	color, eye, face, pattern,	14.394	38	13.62

Condition	Category	Narrative n	Topic ID	Top Terms	Mean Topic Share %	Dominant Narrative n	Dominant Narrative %
				vision, light, stars, turned, moving, sky			
LSD	drug	279	2	world, real, life, reality, away, came, going, vision, dream, car	18.708	67	24.014

Display extract shown above; 158 additional rows are retained in the complete accompanying CSV.

Supplementary Table S27

Supplementary Table S27. *Independent within-condition NMF topic terms*

Ranked terms for every condition-local topic. No narrative text is included.

Complete machine-readable table: tables/table_s27_primary_condition_specific_nmf_topic_terms.csv (1,680 rows; 7 columns).

Condition	Category	Topic ID	Rank	Term	Weight
DMT	drug	1	1	vision	0.4958
DMT	drug	1	2	black	0.4779
DMT	drug	1	3	face	0.3923
DMT	drug	1	4	head	0.328
DMT	drug	1	5	white	0.2531
DMT	drug	1	6	dark	0.2484
DMT	drug	1	7	mouth	0.2325
DMT	drug	1	8	eye	0.2137
DMT	drug	1	9	completely	0.2071
DMT	drug	1	10	little	0.1974

Display extract shown above; 1,670 additional rows are retained in the complete accompanying CSV.

Supplementary Table S28

Supplementary Table S28. *Independent within-condition NMF diagnostics*

Narrative counts, vocabulary sizes, reconstruction errors, iteration counts, and convergence diagnostics for the fixed eight-topic models.

Complete machine-readable table: tables/table_s28_primary_condition_specific_nmf_diagnostics.csv (21 rows; 14 columns).

Condition	Category	Narrative n	Topic Count	Vocabulary n	Reconstruction Error	Iterations	Converged Before Max Iter
DMT	drug	257	8	2090	15.207	160	True
LSD	drug	279	8	2400	15.859	231	True
Psilocybin	drug	259	8	2129	15.287	290	True
Salvia	drug	306	8	2842	16.677	218	True
Ketamine	drug	242	8	1528	14.743	276	True
Diphenhydramine	drug	330	8	1761	17.266	130	True
Cocaine	drug	55	8	272	6.078	75	True
MDMA	drug	282	8	1307	16.006	279	True
Methamphetamine	drug	231	8	1027	14.306	71	True
Bipolar Disorder with Psychotic Features	psychological	44	8	178	5.288	194	True
Schizoaffective Disorder	psychological	231	8	801	14.282	215	True
Schizophrenia	psychological	269	8	1114	15.589	191	True
Borderline Personality Disorder	psychological	220	8	738	13.864	62	True
Dissociative Identity Disorder	psychological	233	8	974	14.404	408	True
PTSD	psychological	162	8	551	11.85	176	True
Epilepsy	neurological	220	8	924	13.947	153	True
Hallucinogen Persisting Perception Disorder	neurological	314	8	1451	16.787	188	True
Migraine	neurological	213	8	907	13.677	397	True
Narcolepsy	neurological	234	8	919	14.378	114	True
Sleep Paralysis	neurological	411	8	2305	19.366	345	True
Parkinson's Disease Psychosis	neurological	64	8	252	6.863	80	True

Supplementary Table S29

Supplementary Table S29. *Independent within-condition NMF topic-map centres*

Topic-centre coordinates, dominant-topic narrative counts, and marker-size fields for each condition-specific topic map. Coordinates and topic IDs are local to each model.

Complete machine-readable table: tables/table_s29_primary_condition_specific_nmf_topic_map_centroids.csv (168 rows; 15 columns).

Condition	Dominant NMF Topic	Category	Centroid X	Centroid Y	n	Topic Terms	Category Share
DMT	1	drug	5.847	0.1433	49	vision, black, face, head	0.1914
DMT	2	drug	1.196	2.096	45	voice, go, words, sound	0.1758
DMT	3	drug	6.538	-1.183	6	dreams, weird, stronger, loud	0.0234
DMT	4	drug	1.011	-2.839	59	body, reality, mind, intense	0.2305
DMT	5	drug	5.202	-2.931	40	eye, closed, visual, open	0.1562
DMT	6	drug	5.053	2.481	21	light, white light, white, dreaming	0.082
DMT	7	drug	9.463	-0.2006	23	entity, told, love, life	0.0898
DMT	8	drug	-1.815	6.955	13	visions, sinister, phone, fractal	0.0508
LSD	1	drug	-0.4442	5.614	37	color, eye, face, pattern	0.1331
LSD	2	drug	1.342	9.601	67	world, real, life, reality	0.241

Display extract shown above; 158 additional rows are retained in the complete accompanying CSV.

Supplementary Table S30

Supplementary Table S30. *Broad-category phenomenological similarity*

Mean condition-centroid cosine similarity for each broad-category pairing. Within-category rows omit identical-condition self-pairs and give each condition equal status.

Complete machine-readable table: tables/table_s30_broad_category_similarity.csv (18 rows; 7 columns).

Corpus	Group A	Group B	Condition Pair n	Mean Condition Centroid Similarity	Median Condition Centroid Similarity	Sd Condition Centroid Similarity
primary	drug	drug	36	0.9317	0.9439	0.0418
primary	drug	psychological	54	0.8647	0.8579	0.045
primary	drug	neurological	54	0.8227	0.8455	0.0775
primary	psychological	drug	54	0.8647	0.8579	0.045
primary	psychological	psychological	15	0.9412	0.9345	0.0268
primary	psychological	neurological	36	0.8152	0.8333	0.086
primary	neurological	drug	54	0.8227	0.8455	0.0775
primary	neurological	psychological	36	0.8152	0.8333	0.086
primary	neurological	neurological	15	0.8162	0.8441	0.0964
sensitivity	drug	drug	36	0.9369	0.9525	0.041
sensitivity	drug	psychological	54	0.867	0.8618	0.0447
sensitivity	drug	neurological	54	0.8232	0.8456	0.0769
sensitivity	psychological	drug	54	0.867	0.8618	0.0447
sensitivity	psychological	psychological	15	0.9454	0.9396	0.0278
sensitivity	psychological	neurological	36	0.8232	0.8425	0.083
sensitivity	neurological	drug	54	0.8232	0.8456	0.0769
sensitivity	neurological	psychological	36	0.8232	0.8425	0.083
sensitivity	neurological	neurological	15	0.8215	0.8464	0.0926

Supplementary Table S31

Supplementary Table S31. *Condition-family phenomenological similarity*

Mean condition-centroid cosine similarity for the predeclared scientific families. The condition-balanced estimand prevents larger conditions from dominating.

Complete machine-readable table: tables/table_s31_condition_family_similarity.csv (128 rows; 7 columns).

Corpus	Group A	Group B	Condition Pair n	Mean Condition Centroid Similarity	Median Condition Centroid Similarity	Sd Condition Centroid Similarity
primary	Serotonergic psychedelics	Serotonergic psychedelics	3	0.9821	0.9814	0.0103
primary	Serotonergic psychedelics	Stimulants and entactogens	9	0.9001	0.894	0.0344
primary	Serotonergic psychedelics	Other psychoactive hallucinogens	9	0.9515	0.9663	0.0321
primary	Serotonergic psychedelics	Schizophrenia spectrum	6	0.818	0.8222	0.0169
primary	Serotonergic psychedelics	Other psychiatric conditions	12	0.8529	0.8494	0.0165
primary	Serotonergic psychedelics	Persistent and episodic visual conditions	6	0.7537	0.7478	0.0901
primary	Serotonergic psychedelics	Sleep-related conditions	6	0.8403	0.8476	0.018
primary	Serotonergic psychedelics	Other neurological conditions	6	0.791	0.7881	0.0716
primary	Stimulants and entactogens	Serotonergic psychedelics	9	0.9001	0.894	0.0344
primary	Stimulants and entactogens	Stimulants and entactogens	3	0.9543	0.9534	0.0161

Display extract shown above; 118 additional rows are retained in the complete accompanying CSV.

Supplementary Table S32

Supplementary Table S32. *Within-condition semantic dispersion*

Equal-size split-half cosine-distance estimates. Larger values indicate greater within-condition narrative diversity; intervals summarize resampling stability rather than a population confidence interval.

Complete machine-readable table: tables/table_s32_condition_semantic_dispersion.csv (42 rows; 14 columns).

Corpus	Condition	Category	Analysis Family	Narrative n	Raw Mean Cosine Distance	Raw Median Cosine Distance
primary	DMT	drug	Serotonergic psychedelics	257	0.3218	0.2928
primary	LSD	drug	Serotonergic psychedelics	279	0.3303	0.303
primary	Psilocybin	drug	Serotonergic psychedelics	259	0.3228	0.2922
primary	Salvia	drug	Other psychoactive hallucinogens	306	0.3032	0.2628
primary	Ketamine	drug	Other psychoactive hallucinogens	242	0.3619	0.3363
primary	Diphenhydramine	drug	Other psychoactive hallucinogens	330	0.3353	0.3048
primary	Cocaine	drug	Stimulants and entactogens	55	0.3303	0.3088
primary	MDMA	drug	Stimulants and entactogens	282	0.3505	0.3261
primary	Methamphetamine	drug	Stimulants and entactogens	231	0.396	0.3722
primary	Bipolar Disorder with Psychotic Features	psychological	Other psychiatric conditions	44	0.303	0.2799

Display extract shown above; 32 additional rows are retained in the complete accompanying CSV.

Supplementary Table S33

Supplementary Table S33. *Within-condition NMF topic breadth*

Normalized entropy across each narrative's graded eight-topic profile. Larger values indicate more even topic mixtures and are not emotional-valence scores.

Complete machine-readable table: tables/table_s33_condition_nmf_topic_entropy.csv (42 rows; 10 columns).

Corpus	Condition	Category	Narrative n	Discussion n	Mean Normalized Topic Entropy	Median Normalized Topic Entropy
primary	DMT	drug	257	232	0.5776	0.596
primary	LSD	drug	279	259	0.5737	0.5994
primary	Psilocybin	drug	259	244	0.5833	0.6082
primary	Salvia	drug	306	268	0.5945	0.6098
primary	Ketamine	drug	242	219	0.5385	0.5554
primary	Diphenhydramine	drug	330	298	0.5408	0.5732
primary	Cocaine	drug	55	49	0.4923	0.5733
primary	MDMA	drug	282	234	0.5138	0.5403
primary	Methamphetamine	drug	231	168	0.4938	0.5135
primary	Bipolar Disorder with Psychotic Features	psychological	44	35	0.5287	0.5675

Display extract shown above; 32 additional rows are retained in the complete accompanying CSV.

Supplementary Table S34

Supplementary Table S34. *Category and family similarity permutation tests*

Global condition-label permutation tests of whether within-category or within-family condition centroids are more similar than between-group centroids. Conditions, not condition pairs, are the exchangeable units.

Complete machine-readable table: tables/table_s34_group_similarity_permutation_tests.csv (4 rows; 12 columns).

Corpus	Grouping	Condition n	Group n	Within Condition Pair n	Between Condition Pair n	Mean Within Group Similarity
primary	category	21	3	66	144	0.9076
primary	analysis_family	21	8	19	191	0.9343
sensitivity	category	21	3	66	144	0.9126
sensitivity	analysis_family	21	8	19	191	0.9383

Supplementary Table S35

Supplementary Table S35. *Semantic-dispersion confound audit*

Condition-level Spearman associations between balanced dispersion and narrative sample size, phenomenology length, accepted-clause count, or submission share, with label-permutation p-values and BH correction. Strong associations limit substantive interpretation of raw dispersion rankings.

Complete machine-readable table: tables/table_s35_dispersion_confound_correlations.csv (8 rows; 8 columns).

Corpus	Predictor	Condition n	Spearman Rho With Balanced Dispersion	Permutation P Two Sided	Permutations	Random State
primary	narrative sample size	21	-0.4873	0.0274	4999	72
primary	median phenomenology word count	21	-0.6431	0.0026	4999	72
primary	median accepted-clause count	21	-0.6078	0.004	4999	72
primary	submission share	21	-0.5768	0.007	4999	72
sensitivity	narrative sample size	21	-0.5905	0.005	4999	72
sensitivity	median phenomenology word count	21	-0.7329	0.0006	4999	72
sensitivity	median accepted-clause count	21	-0.7171	0.0004	4999	72
sensitivity	submission share	21	-0.7364	0.0004	4999	72

Supplementary Table S36

Supplementary Table S36. *Global topic-label audit summary*

Text-free summary of ten high-weight representative clauses per global NMF topic, including topic dominance, clause length and category/condition coverage. The private row-level worksheet contains the clause text and remains outside the thesis package.

Complete machine-readable table: tables/table_s36_global_topic_label_audit_summary.csv (16 rows; 14 columns).

Corpus	Topic ID	Topic Label	Representative Clauses	Distinct Narratives	Represented Conditions	Represented Categories
primary	1	Faces and facial distortions	10	10	8	3
primary	2	Voices and verbal auditory experiences	10	10	5	2
primary	3	Visual and auditory hallucinations	10	10	8	3
primary	4	Vision and visual-field disturbances	10	10	8	3
primary	5	Shadows, figures, insects and crawling forms	10	10	6	3
primary	6	Eyes and closed-eye imagery	10	10	8	2
primary	7	Bodily sensations, movement and paralysis	10	10	4	3
primary	8	Altered reality and dreamlike experience	10	10	7	3
sensitivity	1	Faces and facial distortions	10	10	6	3
sensitivity	2	Voices and verbal auditory experiences	10	10	6	3
sensitivity	3	Visual and auditory hallucinations	10	10	9	3
sensitivity	4	Vision and visual-field disturbances	10	10	7	3

Display extract shown above; 4 additional rows are retained in the complete accompanying CSV.